

FINITE-USE GEOMETRY OF ORDERED BINARY QUANTUM MEASUREMENTS

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ABSTRACT. We determine the finite-use geometry of the full body of ordered binary quantum measurements with classical output and no residual quantum output, in every fixed input dimension. A regularized Sylvester form at the midpoint of two effects is uniformly comparable to their reference-assisted adaptive distance. A horizontal measurement dilation gives the upper estimate, and one- and two-dimensional nonadaptive experiments give the converse, including every spectral rank and multiplicity. In input dimension d , the covering number is comparable to $N^{d^2/2}[\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}$ for all horizons $N \geq 1$ and all sufficiently small errors δ , with constants depending only on d . Uniform volume bounds for actual operational balls reduce the entropy calculation to a spectral integral; balanced endpoint prefixes account for each logarithm. For an unknown binary qubit measurement, a common learner attains and requires order $N\delta^{-2} \log(1/\eta)$ training calls for error δ in the future- N adaptive distance with failure probability at most η , for sufficiently small δ and $0 < \eta \leq 1/8$. Randomized entangled blocks remove unknown bias and choose their noise scale from observations. An exact rational code converts the estimate into one reusable description. The detailed qubit boundary geometry and the supporting instrument, preparation, coherent, and readout description theorems are retained with full proofs.

1. INTRODUCTION

The repeated use of a measurement distinguishes perturbations that have the same one-use size. Near a projection, a change of eigenspace can be resolved on a scale of order N^{-1} , while an interior spectral perturbation has scale $N^{-1/2}$. These scales meet at singular effects and at repeated eigenvalues. We determine their joint geometry for the entire body of ordered binary measurements in every fixed finite input dimension, and derive the resulting description entropy.

Let

$$\mathfrak{E}_d = \{E = E^* : 0 \preceq E \preceq I_d\}, \quad \mathcal{M}_E(\rho) = \text{tr}(E\rho)|1\rangle\langle 1| + \text{tr}((I_d - E)\rho)|0\rangle\langle 0|.$$

The output labels are ordered and the device has no residual quantum output. For two such memoryless devices, d_N is the supremum of the unhalved trace distance between the final joint states of a common experiment using at most N calls. The experiment may retain an arbitrary reference, entangle inputs across calls, use feedback, and stop publicly. In particular $0 \leq d_N \leq 2$. The one-use identity $d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}$ is known [5, Lemmas III.7–III.8]. We write d_N^{na} for the restriction to nonadaptive experiments with the same reference access.

1.1. The matrix comparison. For $M = (E + F)/2$, $H = F - E$, and $V = M(I_d - M)$, define

$$Q_N(E, F)^2 = N\langle H, (L_V + R_V + N^{-1} \text{Id})^{-1} H \rangle_{\text{HS}},$$

where $L_V(X) = VX$ and $R_V(X) = XV$. The inverse acts on the real Hilbert space of Hermitian matrices. Our first main result is Theorem 2.1:

$$\frac{\min\{1, Q_N(E, F)\}}{8192d} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

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This comparison is uniform in N , in the effects, and in their spectral multiplicities. It gives a horizon-independent comparison between adaptive and nonadaptive discrimination at each fixed d ; it does not assert equality of their optimum values. The expression Q_N is an intrinsic comparison modulus, with no asserted triangle inequality.

The upper bound uses a horizontal dilation of a binary measurement. A regularized Sylvester equation splits an arbitrary matrix tangent into a horizontal part and a remainder controlled by the hybrid inequality. Operator concavity of $G(I_d - G)$ then integrates the tangent estimate into the displayed midpoint formula on the closed body. The lower bound uses one-dimensional Bernoulli tests and two-dimensional compressions, whose full biased geometry is proved in Section 5. This argument retains both spectral endpoints and does not choose coordinates on a singular eigenvector chart.

1.2. Entropy and nested support scales. Theorem 3.1 gives, for every fixed $d \geq 1$,

$$C_N(\mathfrak{E}_d, \delta) \asymp_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}, \quad N \geq 1, \quad 0 < \delta \leq \delta_d.$$

Here centres are arbitrary legal memoryless devices of the same ordered binary interface; rational centres attain the same order. The constant $\delta_d > 0$ and both comparison constants depend only on d . The corresponding fixed-length payload is

$$\frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The factor $[\log(N+2)]^{\lfloor d/2 \rfloor}$ comes from nested pairs of eigenvalues approaching zero and one. In the spectral integral, the cumulative exponent of each ordered endpoint sector is one half of the square of its signed prefix sum. Each balanced prefix contributes a logarithm. The argument first bounds the weighted volume of actual operational balls uniformly at every boundary point; the spectral integral is used only after that metric step has been established.

1.3. A common learner and exact descriptions. For an unknown effect in $\mathfrak{E}_2$, there is also a single learning experiment with the uniform guarantee

$$\sup_{E \in \mathfrak{E}_2} \Pr_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

Theorem 4.1 proves that the optimum worst-case training call count has order $N\delta^{-2} \log(1/\eta)$ for small δ and $0 < \eta \leq 1/8$. Here N is the future horizon in the loss. Every entangled training block has length at most N . Variance-sensitive product observations handle small contrast; randomized GHZ experiments remove unknown bias and select their block length from the data when the contrast is large. Fresh endpoint observations complete the estimate, including at zero eigenvalues. A scalar Bernoulli subfamily supplies the converse.

The rational code of Theorem 6.2 converts this estimate into one reusable word without additional device calls. Its payload charges bias, contrast, and direction together. The call bound, the payload length, and the classical cost of finding the word are distinct resources. The arbitrary-dimensional rational covering theorem asserts existence of optimal-order codebooks; the implemented exact joint encoder treats qubits.

1.4. Relation to prior work and organization. Horizontal Kraus gauges and channel-extension metrology have established antecedents [22, 7, 23, 24]; integrating channel distinguishability along a parameter path also predates this work [25, 26]. Our matrix theorem provides the explicit binary variance form, its uniform finite-horizon midpoint comparison, and a matching nonadaptive lower bound. The spectral volume calculation uses classical matrix integration [27, 28]; its balanced-prefix accumulation gives the logarithmic multiplicity above.

Finite-use measurement discrimination with entangled references is developed in [16, 20, 21]. The differing numbers of outcomes and ancillary interfaces are compared in Section 7. The common

learner uses multiscale phase ideas [31, 32] in a fully unknown biased measurement experiment. Its future-horizon loss is distinct from the one-use tomography guarantees of [5, 29].

Sections 2 and 3 prove the matrix comparison and entropy law. The common learning theorem follows before the detailed qubit geometry and exact code. The appendices retain the supporting instrument, preparation, coherent, readout, and unbiased-boundary developments with their complete proofs. The structural companion and the complete research edition retain the stochastic-realization theory and its arithmetic, profinite, and causal-process refinements.

2. MATRIX GEOMETRY IN ARBITRARY INPUT DIMENSION

The binary effect body has a finite-use description that does not require eigenvector coordinates. Its regularization is the same at every spectral rank, and two-dimensional compressions supply lower witnesses for the full matrix problem. Throughout this section $d, N \geq 1$ are integers and

$$(2.1) \quad \mathfrak{E}_d = \{E \in \text{End}(\mathbb{C}^d) : E = E^*, 0 \preceq E \preceq I_d\}.$$

The measurement $\mathcal{M}_E$ has the ordered effects $E, I_d - E$ and no residual quantum output. The distance d_N has the same reference-assisted adaptive meaning as in Theorem 5.1. Write d_N^{na} for its restriction to nonadaptive experiments, allowing entanglement across the calls and a retained reference. These experiments may depend on the known pair of hypotheses.

Put $\tau = N^{-1}$ and $V_E = E(I_d - E)$. On the real Hilbert space of Hermitian matrices with Hilbert–Schmidt inner product, define

$$(2.2) \quad \mathcal{A}_E(H) = V_E H + H V_E + \tau H,$$

$$(2.3) \quad g_{N,E}(H)^2 = N \langle H, \mathcal{A}_E^{-1}(H) \rangle_{\text{HS}}.$$

The operator $\mathcal{A}_E$ is strictly positive, including at a singular effect. For a pair E, F set

$$(2.4) \quad M = \frac{1}{2}(E + F), \quad H = F - E, \quad Q_N(E, F) = g_{N,M}(H).$$

In any orthonormal eigenbasis of M , with eigenvalues m_i and $v_i = m_i(1 - m_i)$, this is

$$(2.5) \quad Q_N(E, F)^2 = N \sum_{i,j=1}^d \frac{|H_{ij}|^2}{v_i + v_j + N^{-1}}.$$

Definition (2.4) makes the expression independent of all choices at repeated eigenvalues. We use it as a comparison modulus; a triangle inequality for Q_N is not required.

The inverse Sylvester operator is familiar from the Bures metric g^{B} on the positive definite cone [27]. Its normalization gives the algebraic identity

$$g_{N,E}(H)^2 = 2N g_{V_E + \tau I_d/2}^{\text{B}}(H, H).$$

Here H remains the effect tangent; the differential of the variance map $E \mapsto V_E$ is $H - EH - HE$.

Theorem 2.1 (Finite-use geometry of the matrix effect body). *For every $E, F \in \mathfrak{E}_d$,*

$$(2.6) \quad \frac{1}{8192d} \min\{1, Q_N(E, F)\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

The constants are uniform in the horizon and in both effects, without a spectral margin or a condition on eigenvalue multiplicities. The adaptive upper bound permits arbitrary retained references, common feedback, and bounded public stopping. Each lower witness uses inputs in one fixed subspace of dimension at most two. Moreover,

$$(2.7) \quad d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}.$$

The theorem includes every rank of projection and all of the intermediate singular strata. The variance $E(I_d - E)$, rather than the smallest eigenvalue of either outcome, determines the matrix weights. The factor N^{-1} makes the formula finite on the entire closed body. The proof first establishes a path inequality valid for every common adaptive tester, and then obtains the converse from one- and two-dimensional experiments.

2.1. A horizontal measurement dilation. Kraus-gauge bounds for channel estimation were developed in [22, 7, 23]. Their integration along channel paths appears in [25, Section III, equation (26)] and, with general adaptive bounds, in [26, Section IV.A]. We use an explicit horizontal lift for binary effects and control the finite-horizon regularization by a trace-norm remainder.

Lemma 2.2 (Horizontal derivative and regularized path bound). *Every continuously differentiable path $G : [0, 1] \rightarrow \mathfrak{E}_d$ satisfies*

$$(2.8) \quad d_N(\mathcal{M}_{G(0)}, \mathcal{M}_{G(1)}) \leq 4 \int_0^1 g_{N,G(t)}(G'(t)) dt.$$

Proof. We begin with an interior effect G , so that G and $I_d - G$ are invertible. Set $S = \sqrt{G(I_d - G)}$. For a Hermitian matrix K put $H_0 = SK + KS$. At the two factors $A_1 = \sqrt{G}$, $A_0 = \sqrt{I_d - G}$ prescribe derivatives

$$(2.9) \quad B_1 = \sqrt{I_d - G} K, \quad B_0 = -\sqrt{G} K.$$

They obey

$$(2.10) \quad \begin{aligned} A_1^* B_1 + B_1^* A_1 &= H_0, & A_0^* B_0 + B_0^* A_0 &= -H_0, \\ \sum_{y=0}^1 A_y^* B_y &= 0, & \sum_{y=0}^1 B_y^* B_y &= K^2. \end{aligned}$$

These tangent equations can be realized by genuine dilations. Let $G_1(s) = G + sH_0$, $G_0(s) = I_d - G - sH_0$ for s in a sufficiently small open interval, and put $R_y(s) = \sqrt{G_y(s)}$. The first line of (2.10) implies that

$$T_y = (B_y - R'_y(0))R_y(0)^{-1}$$

is anti-Hermitian. Indeed, multiplying $T_y + T_y^*$ on the left and right by $R_y(0)$ gives zero by differentiating $R_y(s)^2 = G_y(s)$. Thus $\tilde{A}_y(s) = \exp(sT_y)R_y(s)$ has derivative B_y at zero and satisfies $\tilde{A}_y(s)^* \tilde{A}_y(s) = G_y(s)$ exactly. A measurement isometry is

$$W(s)\psi = \sum_{y=0}^1 |y\rangle_C |y\rangle_L \tilde{A}_y(s)\psi.$$

Here C is the observed outcome, and L and the factor space of $\tilde{A}_y$ are environments. Tracing the environments implements $\mathcal{M}_{G(s)}$, also on inputs entangled with a reference. At zero, (2.10) gives

$$(2.11) \quad W^* \dot{W} = 0, \quad \dot{W}^* \dot{W} = K^2.$$

Fix a common tester with N slots. Purify its initial state and all intervening operations, and retain the measurement environments only for this comparison. The derivative of its final purification in the direction H_0 is the sum of N terms, each having one differentiated occurrence of W . The terms are mutually orthogonal. For two different differentiated slots, cancel the common isometries after the later slot; the remaining overlap contains $W^* \dot{W} \otimes I = 0$. Each term has norm at most $\|\dot{W}\|_{\text{op}} = \|K\|_{\text{op}}$, independently of the reference and memory dimensions. Hence the

derivative of the purification has norm at most $\sqrt{N}\|K\|_{\text{op}}$. The pure-state trace-norm derivative bound, followed by contraction under the partial trace, bounds the observed output derivative by

$$(2.12) \quad 2\sqrt{N}\|K\|_{\text{op}}.$$

A publicly stopped tester is padded to N slots by calls on fixed dummy inputs, whose outputs are discarded. The same purification argument then applies to its original output. Classical feedback is included by purifying the common controlled operations.

Now let H be an arbitrary Hermitian tangent at G , and set $\kappa = N^{-1/2}$. There is a unique Hermitian solution of

$$(2.13) \quad SK + KS + \kappa K = H.$$

Write $H = H_0 + R$, with $H_0 = SK + KS$ and $R = \kappa K$. Both are legitimate tangents at the interior effect G . The derivative of a fixed tester's output is linear in the inserted channel tangent. The part arising from H_0 is bounded by (2.12). The part arising from R is bounded by $2N\|R\|_{\text{op}}$. To verify the latter bound, the one-call tangent has two opposite reference blocks

$$B_\rho = \text{tr}_A[(R \otimes I)\rho], \quad -B_\rho.$$

Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|R\|_{\text{op}}$ for every input state. Insert this tangent into each of the N slots in turn; all subsequent common channels contract the Hermitian trace norm. This proves the bound for its sum. Thus the actual tangent H has output derivative bounded by

$$(2.14) \quad 2\sqrt{N}\|K\|_{\text{op}} + 2N\|R\|_{\text{op}} = 4\sqrt{N}\|K\|_{\text{op}}.$$

The splitting is used only for the differential at an interior effect; it does not postulate a physical mixture realizing the entire path.

In an eigenbasis of G , with $v_i = g_i(1 - g_i)$, equation (2.13) reads

$$K_{ij} = \frac{H_{ij}}{\sqrt{v_i} + \sqrt{v_j} + \kappa}.$$

Since $(\sqrt{v_i} + \sqrt{v_j} + \kappa)^2 \geq v_i + v_j + \tau$,

$$(2.15) \quad \sqrt{N}\|K\|_{\text{op}} \leq \sqrt{N}\|K\|_{\text{HS}} \leq g_{N,G}(H).$$

For an interior path, integrate the resulting output derivative bound for each fixed tester, and then take the supremum. This proves (2.8) without exchanging a supremum with a derivative.

For a path in the closed body, apply the result to $G_\epsilon(t) = (1 - 2\epsilon)G(t) + \epsilon I_d$, where $0 < \epsilon < 1/2$. At fixed N , the form $g_{N,G}(H)$ is continuous on the whole body and bounded above by $N\|H\|_{\text{HS}}$. The integrals therefore converge as $\epsilon \downarrow 0$. The hybrid inequality $d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 2N\|E - F\|_{\text{op}}$ gives continuity of the distances at both endpoints. Passing to the limit proves the asserted closed-body statement. $\square$

2.2. The midpoint comparison.

Proof of Theorem 2.1. For the upper bound use $G(t) = (1 - t)E + tF$. The map $G \mapsto V_G = G - G^2$ is operator concave. For $0 < t \leq 1/2$, write $G(t) = (1 - 2t)E + 2tM$ and discard the positive multiple of V_E in the concavity inequality. Applying the same argument from the other endpoint gives

$$(2.16) \quad V_{G(t)} \succeq a(t)V_M, \quad a(t) = 2\min\{t, 1 - t\}.$$

Left and right multiplication preserve this order as quadratic forms on Hermitian matrices. Since $a(t) \leq 1$, $\mathcal{A}_{G(t)} \succeq a(t)\mathcal{A}_M$. Inverting positive operators reverses their order, whence

$$g_{N,G(t)}(H) \leq a(t)^{-1/2}Q_N(E, F).$$

The scalar factor has integral 2 on $[0, 1]$. Lemma 2.2 yields $d_N \leq 8Q_N$; the trace-norm bound $d_N \leq 2$ gives the stated upper estimate.

For the lower bound choose any orthonormal eigenbasis e_i of M and put

$$(2.17) \quad q_{ij} = |H_{ij}| \sqrt{\frac{N}{v_i + v_j + \tau}}, \quad Q_N^2 = \sum_{i,j} q_{ij}^2.$$

Repeated input e_i gives Bernoulli probabilities $s = m_i - H_{ii}/2$, $t = m_i + H_{ii}/2$. Their variances satisfy

$$\max\{s(1-s), t(1-t)\} \leq s(1-s) + t(1-t) \leq 2m_i(1-m_i).$$

Lemma 5.2 therefore gives

$$(2.18) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{64} \min\{1, q_{ii}\}.$$

For $i \neq j$, restrict each input to $\mathcal{S} = \text{span}\{e_i, e_j\}$. The resulting channels have the compressed effects $E_{\mathcal{S}}, F_{\mathcal{S}}$, including on inputs entangled with a retained reference. Their midpoint is $M_{\mathcal{S}} = \text{diag}(m_i, m_j)$. Write their spectral data as (p, q, u) and (p', q', v) in the notation of Section 5, with gaps r, r' and angle h when both gaps are positive. Put

$$V_E^{\mathcal{S}} = p(1-p) + q(1-q), \quad V_F^{\mathcal{S}} = p'(1-p') + q'(1-q').$$

For the compressed matrices the exact identity

$$V_{M_{\mathcal{S}}} = \frac{1}{2}V_{E_{\mathcal{S}}} + \frac{1}{2}V_{F_{\mathcal{S}}} + \frac{1}{4}(F_{\mathcal{S}} - E_{\mathcal{S}})^2$$

shows, on taking traces, that

$$(2.19) \quad v_i + v_j + \tau \geq \frac{1}{2}(V_E^{\mathcal{S}} + \tau), \quad v_i + v_j + \tau \geq \frac{1}{2}(V_F^{\mathcal{S}} + \tau).$$

The scalar part of $F_{\mathcal{S}} - E_{\mathcal{S}}$ has zero off-diagonal entry. Its traceless part has operator norm $|ru - r'v|/2$. Decomposing this vector along the smaller radius gives

$$(2.20) \quad \begin{aligned} |H_{ij}| &\leq \frac{1}{2}|r - r'| + \min\{r, r'\} \sin(h/2) \\ &\leq \frac{1}{2}(|p - p'| + |q - q'| + \min\{r, r'\}h). \end{aligned}$$

When one radius is zero, the angular term is zero and the same inequality holds without choosing its direction. Each endpoint variance is bounded by the appropriate $V_E^{\mathcal{S}}$ or $V_F^{\mathcal{S}}$. Equations (2.19)–(2.20) therefore imply

$$(2.21) \quad q_{ij} \leq \frac{1}{\sqrt{2}} (T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\}h).$$

The nonadaptive assertion of Theorem 5.1 applied to this compressed pair yields

$$(2.22) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{8192} \min\{1, q_{ij}\}.$$

Every such witness is a legitimate experiment for the original effects by a fixed isometric embedding of $\mathcal{S}$. Since $\max_{i,j} q_{ij} \geq Q_N/d$, taking the strongest of (2.18) and (2.22) proves the lower bound. For $d = 1$ only the diagonal argument is needed.

Finally, a one-use reference-assisted difference has output blocks B_{ρ} and $-B_{\rho}$, with $B_{\rho} = \text{tr}_A[(E - F) \otimes I]\rho$. Trace-norm duality against Hermitian contractions gives $\|B_{\rho}\|_1 \leq \|E - F\|_{\text{op}}$. An eigenstate of $E - F$ at an extremal eigenvalue attains equality. This proves (2.7). $\square$

Corollary 2.3 (Uniform comparison with nonadaptive experiments). *For all $E, F \in \mathfrak{E}_d$ and all $N \geq 1$,*

$$(2.23) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 65536d d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

Proof. If $Q_N \leq 1$, divide the upper estimate $8Q_N$ by the lower estimate $Q_N/(8192d)$. If $Q_N \geq 1$, the upper estimate is 2 and the lower estimate is $1/(8192d)$, which gives a smaller constant. The identical pair needs no division. $\square$

2.3. All spectral ranks and local forms.

Lemma 2.4 (Spectral noncancellation in every rank). *Let $\lambda_1(E) \geq \dots \geq \lambda_d(E)$ and $\lambda_1(F) \geq \dots \geq \lambda_d(F)$ be the ordered spectra. For every $1 \leq k \leq d$,*

$$(2.24) \quad B_N(\lambda_k(E), \lambda_k(F)) \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

For effects diagonal in the same ordered orthonormal basis,

$$(2.25) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \sum_{k=1}^d B_N(\lambda_k(E), \lambda_k(F)).$$

Proof. Suppose $\lambda_k(E) \geq \lambda_k(F)$. The span of the first k eigenvectors of E intersects the orthogonal complement of the first $k-1$ eigenvectors of F in a nonzero subspace. A unit vector in that intersection has success probabilities $s \geq \lambda_k(E) \geq \lambda_k(F) \geq t$ under the two hypotheses. The monotonicity in Lemma 5.2 implies $B_N(s, t) \geq B_N(\lambda_k(E), \lambda_k(F))$. Repeated use of this vector proves the first assertion; reverse the hypotheses for the opposite ordering. Multiplicities cause no change in the dimension argument.

For the upper bound, first measure the common eigenbasis and then produce a bit whose success probability is its corresponding eigenvalue. Supply N independent bits in each of the d rows before the experiment. Every common adaptive tester is a common channel of these row arrays, including its reference blocks and all publicly stopped histories. The tensor-product triangle inequality gives (2.25); unused row bits are discarded. $\square$

The following local comparison will permit volume estimates at every boundary stratum. It concerns matrices themselves and does not require a coordinate chart for eigenvectors.

Lemma 2.5 (Uniform local comparison of the variance forms). *Set $W_G = G(I_d - G) + \tau I_d/2$. For a Hermitian H , put $q = g_{N,G}(H)$. Then*

$$(2.26) \quad \|W_G^{-1/2} H W_G^{-1/2}\|_{\text{HS}} \leq 2q.$$

For either sign for which $G \pm H/2 \in \mathfrak{E}_d$, one has

$$(2.27) \quad \|W_G^{-1/2} (W_{G \pm H/2} - W_G) W_G^{-1/2}\|_{\text{op}} \leq q + \frac{3}{4}q^2.$$

In particular, for $q \leq 1/4$,

$$(2.28) \quad \frac{1}{2}W_G \preceq W_{G \pm H/2} \preceq 2W_G.$$

Thus, if either $Q_N(E, F) \leq 1/4$ or $g_{N,E}(F - E) \leq 1/4$, the midpoint and the fixed-endpoint quadratic forms are comparable within factors 2.

Proof. In an eigenbasis of W_G , with eigenvalues $w_i \geq \tau/2$,

$$\frac{w_i + w_j}{N w_i w_j} \leq 4.$$

Multiplying this inequality by $N|H_{ij}|^2/(w_i + w_j)$ and summing proves (2.26). Put $Z = W_G^{-1/2} H W_G^{-1/2}$. The exact expansion

$$W_{G \pm H/2} - W_G = \pm \frac{1}{2}(H - GH - HG) - \frac{1}{4}H^2$$

has normalized linear part $\pm((I_d - 2G)Z + Z(I_d - 2G))/4$, since G commutes with W_G . Its operator norm is at most $\|Z\|_{\text{op}}/2 \leq q$. The normalized quadratic part is $-ZW_G Z/4$. Since $\tau \leq 1$, one has $\|W_G\|_{\text{op}} \leq 3/4$, and its norm is at most $3q^2/4$. This proves (2.27) and hence (2.28). The same order holds for left plus right multiplication, and inversion gives the quadratic-form comparison. Apply it first at $G = M$ or at $G = E$, respectively, to obtain the final assertion. $\square$

3. DESCRIPTION ENTROPY IN ARBITRARY MATRIX DIMENSION

The intrinsic comparison of Theorem 2.1 permits the full effect body to be treated without choosing eigenvector coordinates near repeated spectra. Throughout this section the input dimension $d \geq 1$ is fixed. Constants may depend on d , but not on the integer horizon $N \geq 1$ or on the accuracy parameter.

Write $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ for the covering number under d_N , with centres in the same legal memoryless ordered binary interface, and write $\mathcal{C}_N^{\text{rat}}$ when the real and imaginary parts of every matrix entry of each centre are rational.

Theorem 3.1 (Entropy of the full effect body). *For every integer $d \geq 1$ there are constants $c_d, C_d, \delta_d > 0$ such that, for all integers $N \geq 1$ and all $0 < \delta \leq \delta_d$,*

$$\begin{aligned} c_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2} &\leq \mathcal{C}_N(\mathfrak{E}_d, \delta) \\ &\leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}_d, \delta) \\ &\leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}. \end{aligned} \quad (3.1)$$

Consequently the optimum fixed-length reusable description has length

$$(3.2) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The estimates include all support and spectral multiplicities. The rational-centre assertion is an existence statement in every dimension; Theorem 6.2 gives the separate exact encoder in dimension two.

For $d = 1$, the result is the Bernoulli covering law $N^{1/2} \delta^{-1}$. For $d = 2$, it recovers Theorem 6.1. The next two cases have orders $N^{9/2} \log(N+2) \delta^{-9}$ and $N^8 [\log(N+2)]^2 \delta^{-16}$. The logarithmic power counts independent nested approaches of paired eigenvalues to the two support endpoints. We first establish a uniform measure estimate for actual operational balls, and then evaluate the total measure.

3.1. Operational balls and a regularized volume form. Set $n = d^2$, $\tau = N^{-1}$ and

$$W_E = E(I - E) + \frac{\tau}{2} I.$$

As in Section 2, let

$$(3.3) \quad g_{N,E}(H)^2 = N \langle H, (L_{W_E} + R_{W_E})^{-1} H \rangle_{\text{HS}}$$

on the real Hilbert space of Hermitian matrices. Let $G_{N,E}$ denote this positive quadratic form in Hilbert–Schmidt orthonormal coordinates, and define

$$(3.4) \quad \rho_N(E) = \sqrt{\det G_{N,E}}, \quad d\mu_N(E) = \rho_N(E) dE,$$

where dE is Lebesgue measure in those coordinates. The cutoff makes ρ_N positive and continuous on the whole closed body for each N .

Lemma 3.2 (Uniform volume of operational balls). *For fixed d there are $a_d, A_d, r_d > 0$ such that*

$$(3.5) \quad a_d r^{d^2} \leq \mu_N(B_{d_N}(E, r)) \leq A_d r^{d^2}$$

for every $N \geq 1$, every $E \in \mathfrak{E}_d$, and every $0 < r \leq r_d$. Balls are taken inside $\mathfrak{E}_d$.

Proof. We make explicit the two local consequences of Lemma 2.5 that will be used. If H is Hermitian and $q = g_{N,E}(H)$, diagonalizing W_E gives

$$(3.6) \quad \|W_E^{-1/2} H W_E^{-1/2}\|_{\text{HS}}^2 \leq 4q^2,$$

since its eigenvalues w_i are at least $\tau/2$ and

$$\frac{w_i + w_j}{Nw_iw_j} = \tau(w_i^{-1} + w_j^{-1}) \leq 4.$$

For any legal $E + tH$, direct expansion yields

$$W_{E+tH} - W_E = \frac{t}{2}((I - 2E)H + H(I - 2E)) - t^2H^2.$$

The matrices E and W_E commute, $\|I - 2E\|_{\text{op}} \leq 1$, and $\|W_E\|_{\text{op}} \leq 3/4$. Thus the norm of the normalized difference is at most $2|t|q + 3t^2q^2$. For $|t| \leq 1$ and q below an absolute constant, it follows that

$$(3.7) \quad \frac{1}{2}W_E \preceq W_{E+tH} \preceq 2W_E.$$

Loewner comparison passes to $L_W + R_W$ and, by inversion, to the quadratic forms. It also gives

$$(3.8) \quad 2^{-n/2}\rho_N(E) \leq \rho_N(E + tH) \leq 2^{n/2}\rho_N(E).$$

Apply this comparison from an endpoint to the midpoint and from the midpoint to either endpoint. Theorem 2.1 then provides constants $A, B, r_0 > 0$, depending only on d , such that

$$(3.9) \quad d_N(E, F) \leq r \implies g_{N,E}(F - E) \leq Ar,$$

$$(3.10) \quad g_{N,E}(F - E) \leq Br \implies d_N(E, F) \leq r,$$

for legal E, F and $0 < r \leq r_0$. Decrease r_0 so that Ar_0 lies in the small-form range of (3.7).

Let ω_n be the Euclidean volume of the unit ball in $\mathbb{R}^n$. The ellipsoid $g_{N,E}(H) \leq u$ has Lebesgue volume $\omega_n u^n / \rho_N(E)$. Equations (3.9) and (3.8) therefore imply

$$\mu_N(B_{d_N}(E, r)) \leq 2^{n/2}\omega_n(Ar)^n.$$

For the lower bound we construct a full legal ellipsoid, including when E lies on a support boundary. Choose a small constant $a > 0$ depending only on d , set $s = ar\tau$, and move towards the scalar interior point:

$$(3.11) \quad E_s = (1 - 2s)E + sI.$$

Take r_0 sufficiently small that $s \leq 1/4$. Then $sI \preceq E_s \preceq (1 - s)I$. Since the displacement commutes with E , its form is bounded by

$$g_{N,E}(E_s - E)^2 = N \sum_i \frac{s^2(1 - 2\lambda_i)^2}{2\lambda_i(1 - \lambda_i) + \tau} \leq d \frac{s^2}{\tau^2} = da^2r^2.$$

Choose $a\sqrt{d} \leq B/4$. By (3.10), $d_N(E, E_s) \leq r/4$.

Let $X = X^*$ satisfy $g_{N,E_s}(X) \leq br$, with $b > 0$ to be chosen. Write ℓ_i for the eigenvalues of E_s and $w_i = \ell_i(1 - \ell_i) + \tau/2$. Since $\ell_i \geq s$ and $w_i \leq \ell_i + \tau/2$, we have

$$\begin{aligned} \|E_s^{-1/2}XE_s^{-1/2}\|_{\text{HS}}^2 &\leq g_{N,E_s}(X)^2 \max_{i,j} \frac{w_i + w_j}{N\ell_i\ell_j} \\ &\leq b^2r^2 \left(\frac{2\tau}{s} + \frac{\tau^2}{s^2} \right) = b^2 \left(\frac{2r}{a} + \frac{1}{a^2} \right). \end{aligned}$$

The same bound holds with $I - E_s$ in place of E_s , because $1 - \ell_i \geq s$ and $w_i \leq 1 - \ell_i + \tau/2$. Taking $b \leq a/4$ and $ar_0 \leq 1$ makes both operator norms at most $1/2$. Consequently every point of the entire ellipsoid

$$\mathcal{A}_s = \{E_s + X : g_{N,E_s}(X) \leq br\}$$

is a legal effect. Choose also $b \leq B/4$. Equation (3.10) and the triangle inequality give $\mathcal{A}_s \subset B_{d_N}(E, r)$. The density comparison about E_s and the exact ellipsoid volume now imply

$$\mu_N(B_{d_N}(E, r)) \geq 2^{-n/2} \omega_n (br)^n.$$

All choices depend only on d . This proves the assertion on every boundary stratum as well as in the interior. $\square$

3.2. The total regularized volume.

Proposition 3.3 (Nested endpoint accumulation). *For fixed $d \geq 1$, uniformly over $N \geq 1$,*

$$(3.12) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} [\log(N+2)]^{[d/2]}.$$

Proof. In a spectral basis, the quadratic form has weight $N/(2w_i)$ on the i th diagonal direction and weight $N/(w_i + w_j)$ on each of the two Hilbert–Schmidt orthonormal real directions belonging to $i < j$. Hence

$$(3.13) \quad \rho_N(E) = 2^{-d/2} N^{d^2/2} \prod_i w_i^{-1/2} \prod_{i < j} (w_i + w_j)^{-1}.$$

We use the classical spectral change of variables [28]. For a simple spectrum, differentiation of $E = U \operatorname{diag}(\lambda) U^*$ gives

$$U^*(dE)U = d \operatorname{diag}(\lambda) + [U^* dU, \operatorname{diag}(\lambda)].$$

Each complex off-diagonal pair therefore contributes $(\lambda_i - \lambda_j)^2$ to the real Jacobian. Integration over the compact unitary orbit and over permutations supplies a positive constant depending only on d . The repeated-spectrum set is Lebesgue null. This spectral change of variables, together with (3.13), yields

$$(3.14) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} I_d(\tau),$$

where

$$(3.15) \quad I_d(\tau) = \int_{[0,1]^d} \prod_i (v_i + \tau)^{-1/2} \prod_{i < j} \frac{(\lambda_i - \lambda_j)^2}{v_i + v_j + \tau} d\lambda, \quad v_i = \lambda_i(1 - \lambda_i).$$

Replacing $v_i + \tau/2$ by $v_i + \tau$ in the single-coordinate factors changes the integral by a factor between fixed positive constants depending only on d .

We prove

$$(3.16) \quad I_d(\tau) \asymp_d [\log(2 + \tau^{-1})]^{[d/2]}, \quad 0 < \tau \leq 1.$$

For the upper bound, partition each coordinate into $[0, 1/4]$, $[1/4, 3/4]$, and $[3/4, 1]$. Factors involving an eigenvalue in the middle interval are bounded above by constants, so these coordinates can be integrated out. For each of the remaining $l \leq d$ coordinates put $x_i = \lambda_i$ near zero and $x_i = 1 - \lambda_i$ near one. On a sector $0 \leq x_1 \leq \dots \leq x_l \leq 1/4$, let $\epsilon_i \in \{1, -1\}$ record the endpoint and put

$$S_j = \sum_{i=1}^j \epsilon_i, \quad S_0 = 0, \quad y_j = x_j + \tau.$$

The individual factor is at most $C y_j^{-1/2}$. For $i < j$, a same-endpoint pair factor is at most $C y_j$, whereas an opposite-endpoint factor is at most C/y_j . The sector integrand is consequently bounded by

$$(3.17) \quad C_d \prod_{j=1}^l y_j^{a_j-1}, \quad a_j = \frac{1}{2} + \epsilon_j S_{j-1} = \frac{S_j^2 - S_{j-1}^2}{2}.$$

Its cumulative exponents satisfy

$$(3.18) \quad A_j = \sum_{i=1}^j a_i = \frac{S_j^2}{2} \geq 0.$$

Set $T = 1/4 + \tau$ and $L = \log(T/\tau)$. Substitute $y_j = T \exp(-z_j)$, where $z_1 \geq \dots \geq z_l \geq 0$, and then $u_j = z_j - z_{j+1}$ for $j < l$, $u_l = z_l$. The sector integral is bounded by

$$(3.19) \quad C_d \int_{u_j \geq 0, \sum_j u_j \leq L} \exp \left(- \sum_j A_j u_j \right) du.$$

The omitted factor T^{A_l} is bounded above uniformly in $0 < \tau \leq 1$. Each $A_j > 0$ is at least $1/2$, so its variable has an integrable exponential tail. Each vanishing A_j contributes at most one factor $1 + L$. A vanishing prefix sum requires an even j , and there are at most $\lfloor l/2 \rfloor$ such indices. Summing the finitely many sectors proves the upper estimate in (3.16). The case $l = 0$ contributes only a bounded constant.

For the lower bound let $m = \lfloor d/2 \rfloor$. Choose scales

$$(3.20) \quad \tau \leq t_1, \quad 8t_j \leq t_{j+1} \ (j < m), \quad t_m \leq 1/32,$$

and restrict $2m$ eigenvalues in pairs by

$$(3.21) \quad \lambda_j^- = a_j \in [t_j, 2t_j], \quad \lambda_j^+ = 1 - b_j, \quad b_j \in [t_j, 2t_j], \quad 1 \leq j \leq m.$$

If d is odd, restrict the remaining eigenvalue to $[3/8, 5/8]$; all its factors are bounded above and below by positive constants. For one pair the two individual spectral factors and its opposite-endpoint factor have product comparable to t_j^{-2} . For two pairs at scales $t_i \leq t_j/8$, the two same-endpoint factors are comparable to t_j and the two opposite-endpoint factors to t_j^{-1} . Their product is therefore bounded above and below by fixed constants. Thus the integrand on the box is comparable to $\prod_j t_j^{-2}$ and its volume to $\prod_j t_j^2$. Each box contributes at least $c_d > 0$.

Take the scales from $\tau, 8\tau, 8^2\tau, \dots$ up to $1/32$. The boxes for distinct increasing m -tuples are disjoint. Their number is a binomial coefficient bounded below by $c_d [\log(\tau^{-1})]^m$ once $\log(\tau^{-1})$ is sufficiently large depending on d . In the remaining bounded range, a fixed box of distinct eigenvalues strictly inside $(0, 1)$ has a positive uniform integral and supplies the lower estimate. For $m = 0$ this interior argument suffices on the full range. This proves (3.16) and, by (3.14), the proposition. $\square$

The balanced-prefix identity (3.18) accounts for all nested endpoint scales. A simultaneous approach of all eigenvalues to the endpoints detects only one possible critical scaling; the separated pairs in (3.21) show why the logarithmic multiplicity can be larger.

3.3. Covering, rational centres, and description length.

Proof of Theorem 3.1. Choose δ_d smaller than the radius in Lemma 3.2. A cover by L balls of radius δ satisfies

$$\mu_N(\mathfrak{E}_d) \leq LA_d \delta^{d^2},$$

which gives the required lower bound. Conversely, take a maximal δ -separated subset of $\mathfrak{E}_d$. Its balls of radius $\delta/3$ are pairwise disjoint, and the lower bound in (3.5) bounds its cardinality by

$$C_d \mu_N(\mathfrak{E}_d) \delta^{-d^2}.$$

It is a radius- δ cover by maximality. Such finite maximal sets exist: the effect body is compact, d_N is continuous by the hybrid bound, and $d_1(E, F) = 2\|E - F\|_{\text{op}}$ separates distinct effects. Proposition 3.3 now gives both estimates for legal centres.

Rational legal effects are dense. Indeed, first move towards $I/2$ by (3.11), and then approximate all independent real matrix coordinates by rational numbers while remaining strictly between 0 and I . For fixed N , continuity in the operational distance follows again from the hybrid bound. Replace every centre of a radius- $\delta/2$ cover by a rational legal effect at distance less than $\delta/2$. The resulting radius- δ cover has the same cardinality. This proves the rational-centre upper bound.

A fixed-length payload indexes the centres of a cover, and conversely each legal reusable program with a fixed payload supplies a centre. Taking base-two logarithms of (3.1) and rounding the index length gives (3.2). $\square$

4. LEARNING THE COMPLETE BINARY QUBIT EFFECT BODY

We now pass from supplied descriptions to an unknown device. Throughout this section the input dimension is two. Write

$$E = \frac{1}{2}((1+b)I + ru \cdot \sigma), \quad 0 \leq r \leq 1, \quad |b| + r \leq 1,$$

with the direction absent when $r = 0$. The bias, contrast, and direction are all unknown. The outputs are ordered and classical, and each call consumes its input without a residual quantum output. A learning procedure may prepare entangled input blocks, retain the unused input qubits, and use classical feedback between blocks. It has no access to a measurement environment. Its choices depend only on the public parameters and the observed record.

The integer N below is the horizon of a future comparison, measured by d_N . The number M of training calls is a separate resource. The output is a classical description of a legal effect. For a unit vector w set $P_w = (I + w \cdot \sigma)/2$ and write $\mathcal{M}_{p,q,w}$ for the measurement associated with $pP_w + q(I - P_w)$, where $p = (1 + b + r)/2$ and $q = (1 + b - r)/2$. We use the spectral quantities

$$V = p(1 - p) + q(1 - q), \quad K_N(p, q) = r\sqrt{N/(V + N^{-1})}.$$

Theorem 4.1 (Common learning at the finite-use scale). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every $N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common learning procedure returns an effect $\hat{E} \in \mathfrak{E}_2$ with*

$$(4.1) \quad \sup_{E \in \mathfrak{E}_2} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record it uses at most

$$(4.2) \quad M \leq CN\delta^{-2} \log(1/\eta)$$

calls, and every entangled input block has length at most N . Conversely, every common procedure satisfying (4.1), with at most M calls on every record, obeys

$$(4.3) \quad M \geq cN\delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. The constants are uniform over the closed effect body, including scalar, deterministic, and projective effects.

The proof constructs one decision tree for the entire parameter body. Its experiments are not selected from a known pair of hypotheses. The angular part uses multiscale phase estimation, a method with antecedents in [31, 32]. We give the phase and confidence bounds explicitly. A data-dependent amplitude threshold selects the block length without prior calibration of the spectrum.

4.1. Bernoulli errors and estimated axes. For $s, t \in [0, 1]$ put

$$(4.4) \quad \mathfrak{h}(s, t)^2 = (\sqrt{s} - \sqrt{t})^2 + (\sqrt{1-s} - \sqrt{1-t})^2.$$

Write $B_N(s, t)$ for the unhalved distance between the two N -fold Bernoulli product laws. We use the following elementary estimates at both support endpoints.

Lemma 4.2 (Endpoint estimation). *For every $s, t \in [0, 1]$,*

$$(4.5) \quad B_N(s, t) \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For each fixed $A > 0$ there is C_A such that, if $0 < \lambda \leq 1$ and

$$(4.6) \quad |s - t| \leq A(\sqrt{t(1-t)\lambda} + \lambda),$$

then $\mathfrak{h}(s, t) \leq C_A \sqrt{\lambda}$. If two estimates of an ordered pair (p, q) each have $\mathfrak{h}$ error at most ε , sorting the estimates preserves this bound up to an absolute constant.

Proof. The root fidelity is $1 - \mathfrak{h}^2/2$. Tensorization and the trace-fidelity inequality give

$$B_N(s, t) \leq 2\sqrt{1 - (1 - \mathfrak{h}(s, t)^2/2)^{2N}} \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For the first square root in (4.4), if $t \geq \lambda$ divide the difference by $\sqrt{t}$ in (4.6). If $t < \lambda$, both t and s are at most $C_A \lambda$. Apply the same argument to their complements. For sorting, set $\phi(t) = \arcsin \sqrt{t}$. Then

$$\mathfrak{h}(s, t) = 2 \sin(|\phi(s) - \phi(t)|/2).$$

The two distances are uniformly comparable on $[0, \pi/2]$. Sorting is nonexpansive for the sup norm against an already ordered pair of real numbers. Apply this fact in the ϕ coordinate. $\square$

If $\hat{t}$ is the empirical frequency in n Bernoulli trials, Bernstein's inequality gives, with failure at most $2e^{-L}$,

$$(4.7) \quad |\hat{t} - t| \leq \sqrt{2t(1-t)L/n} + \frac{2L}{3n}.$$

Thus Lemma 4.2 applies with $\lambda = L/n$, including at $t = 0, 1$.

For a unit estimate w of u , write

$$(4.8) \quad h = \angle(u, w), \quad \gamma = \frac{r}{2}(1 - u \cdot w).$$

The input states on the axes $+w$ and $-w$ have respective success probabilities

$$(4.9) \quad s_+ = p - \gamma, \quad s_- = q + \gamma, \quad p = \frac{1}{2}(1 + b + r), \quad q = \frac{1}{2}(1 + b - r).$$

If $\hat{x}$ estimates $x = ru$ with Euclidean error at most e , normalizing $\hat{x}$ gives

$$(4.10) \quad rh \leq \pi \min(r, e), \quad \gamma \leq \min(r, e^2/r) \quad (r > 0).$$

Indeed $\|u - w\| \leq 2e/r$, $h \leq (\pi/2)\|u - w\|$, and $\gamma = r\|u - w\|^2/4$. If $\hat{x} = 0$, choose a fixed direction; then $e \geq r$ gives the same bounds. At $r = 0$ both quantities vanish.

Lemma 4.3 (Product learning away from unit contrast). *Suppose $r \leq 7/8$. There is a product-input procedure, not requiring b, r, u as input, which uses eight groups of n calls and returns a legal effect F such that, for $1 \leq L \leq n$,*

$$(4.11) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq C(\sqrt{NL/n} + NL/n)$$

outside an event of probability at most Ce^{-L} .

Proof. First use the positive and negative eigenstates of each of the three Pauli axes, with n calls per state. Encode an observed outcome as $Z \in \{-1, 1\}$. The two means on axis j are $b + x_j$ and $b - x_j$. Estimate x_j by half their empirical difference. Put

$$w_0 = 1 - b^2, \quad \lambda = L/n.$$

The two variances on axis j sum to $2(w_0 - x_j^2)$. Bernstein's inequality and a union bound therefore give

$$(4.12) \quad e = \|\hat{x} - x\| \leq 3(\sqrt{w_0\lambda} + \lambda).$$

More explicitly, the coordinate error is at most $\sqrt{2w_0\lambda} + 4\lambda/3$; the sign observation has range length two. Taking the Euclidean norm gives (4.12), since $\sqrt{6} < 3$ and $4\sqrt{3}/3 < 3$.

Normalize to a direction w . Feasibility and $r \leq 7/8$ imply

$$(4.13) \quad w_0 \geq r(2 - r), \quad V = \frac{1}{2}(w_0 - r^2) \geq w_0/9.$$

Use (4.10) and Theorem 5.4. The square-root term in (4.12) contributes at most $C\sqrt{N\lambda}$ to $K_N(p, q)h$, and its additive term contributes at most $CN\lambda$. Consequently,

$$(4.14) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w}) \leq C(\sqrt{N\lambda} + N\lambda).$$

This calculation does not require a lower bound on r or w_0 .

It remains to control both endpoint probabilities in (4.9). For either $t = p$ or $t = q$ put $v_t = t(1 - t)$. Since

$$|p(1 - p) - q(1 - q)| = r|b| \leq r, \quad w_0 = 2V + r^2 \leq 4v_t + 3r,$$

we claim

$$(4.15) \quad \gamma \leq 9\sqrt{v_t\lambda} + 72\lambda.$$

Equation (4.12) gives $e^2 \leq 18(w_0\lambda + \lambda^2)$. If $r \leq \lambda$, use $\gamma \leq r$. If $r > \lambda$, use

$$\gamma \leq 72v_t\lambda/r + 72\lambda, \quad \gamma \leq r,$$

and $\min(r, 72v_t\lambda/r) \leq \sqrt{72v_t\lambda}$. This proves (4.15).

Make two fresh groups of n calls, along $+w$ and $-w$. Conditional on the previous record, these are Bernoulli trials with the parameters in (4.9). Their variances differ from v_t by at most γ . Equations (4.7) and (4.15) imply

$$|\hat{t}_{\text{raw}} - t| \leq C(\sqrt{v_t\lambda} + \lambda).$$

For the additional term use $\sqrt{\gamma\lambda} \leq C(\sqrt{v_t\lambda} + \lambda)$, which follows from $\sqrt{AB} \leq (A + B)/2$ with $A = \sqrt{v_t\lambda}$ and $B = \lambda$. Lemma 4.2 gives root-distance error $C\sqrt{\lambda}$ for both endpoints. Sort the two empirical parameters to obtain $\hat{p} \geq \hat{q}$ and return $F = \hat{p}P_w + \hat{q}(I - P_w)$. The aligned programme bound (5.9) and (4.5) give a spectral contribution at most $C\sqrt{N\lambda}$. Combine with (4.14). All concentration bounds for the fresh groups hold conditionally, so their failure probabilities may be added to those of the first six groups. $\square$

4.2. A common entangled experiment with unknown bias.

Lemma 4.4 (Two-plane phase observation). *Fix an orthonormal frame (e_1, e_2, w) known to the experimenter. For each $i = 1, 2$ and integer $m \geq 1$, there is an m -call experiment with an observation $Y \in \{-1, 1\}$ whose mean, for a chosen phase φ , is*

$$(4.16) \quad \mathbb{E}_E Y = \Re(e^{i\varphi} Z_i), \quad Z_i = [x \cdot w + ix \cdot e_i]^m, \quad x = ru.$$

Two phases estimate the complex number Z_i . For $0 < \zeta \leq 1$ and $0 < \epsilon \leq 1/2$, four settings, using the two planes, estimate both complex numbers with error at most ζ and failure at most ϵ , with at most $C\zeta^{-2} \log(C/\epsilon)$ independent blocks per setting.

Fix $a = 1/64$. If $r > 0$ and $h = \angle(u, w) \leq 2a/m$, then

$$(4.17) \quad .99r^m \leq |Z_i| \leq r^m, \quad \text{Arg } Z_i = m \arctan \frac{u \cdot e_i}{u \cdot w},$$

where the principal argument is used. If the amplitudes are bounded below by a fixed positive constant and ζ is sufficiently small, these observations produce a direction estimate w' with

$$(4.18) \quad \angle(u, w') \leq C\zeta/m.$$

Proof. Choose a computational basis along the axis perpendicular to the plane spanned by w, e_i , with phase convention such that the off-diagonal matrix element of $O = 2E - I = bI + ru \cdot \sigma$ is $r(u \cdot w + iu \cdot e_i)$. Prepare

$$|G_{\varphi,s}\rangle = \frac{|0\rangle^{\otimes m} + se^{i\varphi}|1\rangle^{\otimes m}}{\sqrt{2}},$$

where s is a fresh fair sign. Pass its qubits through the measurement and record $Y = s \prod_{j=1}^m Z_j$, where each output is encoded as $Z_j \in \{-1, 1\}$. Memorylessness makes the joint outcome law the product POVM on the entangled input; the qubits may be supplied sequentially while the unused ones are retained.

The expansion of $\langle G_{\varphi,s} | O^{\otimes m} | G_{\varphi,s} \rangle$ has diagonal term

$$\frac{1}{2}((b + ru \cdot f_i)^m + (b - ru \cdot f_i)^m),$$

where f_i is the chosen perpendicular axis. It is independent of s and disappears after multiplication by s and averaging. The off-diagonal terms give (4.16). At $x = 0$ the mean is zero and no direction is needed. In particular the experiment removes the unknown bias exactly. The phases $0, \pi/2$ give the real part and the negative imaginary part. Hoeffding's inequality for four bounded sample means proves the stated block count.

For the phase conclusions assume $r > 0$ and put $\theta_i = \arctan((u \cdot e_i)/(u \cdot w))$. Then $|m\theta_i| \leq 2a$ and

$$|Z_i| = r^m(1 - (u \cdot f_i)^2)^{m/2} \geq r^m \cos(h)^m \geq r^m(1 - 2a^2/m).$$

This proves (4.17), with no phase ambiguity. For the estimates set

$$\hat{\theta}_i = \text{Arg}(\hat{Z}_i)/m, \quad w' = \frac{w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2}{\|w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2\|}.$$

The exact formula with θ_i recovers u . For $|\hat{Z}_i - Z_i| \leq |Z_i|/2$,

$$|\text{Arg}(\hat{Z}_i) - \text{Arg}(Z_i)| \leq 2|\hat{Z}_i - Z_i|/|Z_i|.$$

On $|\theta_i| \leq 1/4$ the displayed map from two angles to a unit vector is Lipschitz; a bound of 16 for angular distance against the largest coordinate error follows from differentiation of $\tan$ and normalization. This proves (4.18). Our fixed cone lies strictly inside the principal-argument interval, and the argument uses no unwrapping threshold from a separate phase-estimation theorem. $\square$

Lemma 4.5 (Selecting a block length from the observations). *Suppose $r \geq 5/8$ and an initial known direction w satisfies $\angle(u, w) \leq a = 1/64$. A common adaptive procedure selects a length m_* and a direction w_* , for $N \geq 1$ and $0 < \eta \leq 1/8$, using at most $CN \log(C/\eta)$ calls on every record. With failure at most $\eta/8$,*

$$(4.19) \quad \frac{1}{8} \min\{N, (1-r)^{-1}\} \leq m_* \leq N,$$

$$(4.20) \quad r^{m_*} \geq 1/5, \quad \angle(u, w_*) \leq a/m_*.$$

The reciprocal at $r = 1$ is interpreted as infinity. New observations at m_*, w_* have both complex amplitudes at least $1/6$.

Proof. Fix $\kappa = a/1024$, and let H be the largest power of two not exceeding N . Try lengths $m = 1, 2, 4, \dots, H$. At length m , estimate both complex numbers in Lemma 4.4 to error at most κ , assigning conditional failure allowance

$$(4.21) \quad \epsilon_m = \frac{\eta m}{32N}.$$

Accept the length if both empirical amplitudes are at least $1/4$. On acceptance update the direction by the formula in the proof of Lemma 4.4, and double the length unless the cap has been reached. On rejection retain the preceding accepted length and direction. If none was accepted, use a prescribed fallback with $m_* = 1$. All choices are functions of the record.

We prove the assertions on the event that the successive complex estimates obey their bounds. Before a length m is tried the angular error is at most $2a/m$: this holds initially, and is preserved by each accepted update. On acceptance the true amplitudes are at least $1/4 - \kappa > 1/5$. The argument bound and the Lipschitz constant in Lemma 4.4 give an updated angular error at most

$$\frac{32\kappa}{(1/4 - \kappa)m} \leq \frac{a}{4m}.$$

Furthermore $r^m \geq 1/4 - \kappa$, whence $m(1 - r) \leq 2$.

On rejection, (4.17) gives $r^m < 1/3$. Since $r \geq 5/8$,

$$-\log r \leq (1 - r)/r \leq 2(1 - r), \quad m(1 - r) \geq (\log 3)/2.$$

The preceding accepted length $m/2$ is therefore at least $(\log 3)/(4(1 - r))$. Length one is always accepted on the good event, since its true amplitudes exceed $.99(5/8)$. If the cap is reached, the selected length is $H \geq N/2$. These facts prove (4.19) and (4.20). At the updated direction the amplitude is at least $r^{m_*} \cos(a/m_*)^{m_*} > 1/6$.

The concentration bounds are conditional on each preceding good record. Their failure probabilities sum to at most $\eta/8$, because the sum of the tested powers of two is at most $2N$. By Lemma 4.4, the number of calls is bounded by an absolute constant times

$$(4.22) \quad \sum_{m=1,2,4,\dots,H} m \log \frac{CN}{\eta m} \leq C' N \log(C'/\eta).$$

To see this, write $m = H/2^k$ and use the convergent sums of 2^{-k} and $k2^{-k}$, together with $1 \leq N/H < 2$. The public cap and these budgets bound the cost on every record, including records on which an estimate fails. Rejection can only shorten the sum. $\square$

4.3. The learning bound and its converse.

Proof of Theorem 4.1. We choose constants in the following order. Fix the local cone $a = 1/64$ and $\kappa = a/1024$ as above. Choose a sufficiently small absolute constant $c_* > 0$ in the final phase tolerance below. Then choose the absolute multipliers in the Bernoulli sample counts sufficiently large. Finally reduce δ_0 , if needed, so that $\delta_0 \leq 1/4096$. No optimized value of these constants is used.

First make the six Pauli-axis product experiments, with a sufficiently large absolute multiple of $\log(C/\eta)$ calls per input. With failure at most $\eta/8$, the half-difference estimate satisfies

$$\|\hat{x} - x\| \leq \varepsilon_0, \quad \varepsilon_0 \leq a/100.$$

If $\|\hat{x}\| < 3/4$, choose the product procedure. On the good event $r \leq 7/8$, so Lemma 4.3, with

$$L = \log(C/\eta), \quad n = \left\lceil CN\delta^{-2}L \right\rceil,$$

gives error at most $\delta/2$ with the asserted call count, after increasing C .

Otherwise $r \geq 5/8$, and the normalized initial estimate has angular error at most a . Apply Lemma 4.5. At the selected length and updated direction, make fresh two-plane observations with complex error tolerance

$$(4.23) \quad \zeta_* = c_* \delta \sqrt{m_*/N},$$

and failure allowance $\eta/8$. The resulting direction w_f satisfies

$$(4.24) \quad h_f = \angle(u, w_f) \leq C c_* \delta / \sqrt{N m_*}.$$

The calls used in this step are at most

$$(4.25) \quad C m_* \zeta_*^{-2} \log(C/\eta) \leq C' N \delta^{-2} \log(C/\eta).$$

Put $\nu = 1 - r$. Since $|b| \leq \nu$ and $r \geq 5/8$,

$$r\nu \leq V = \frac{1}{2}(1 - r^2 - b^2) \leq \nu.$$

Equation (4.19) implies

$$(4.26) \quad m_*(V + N^{-1}) \geq c_0 > 0$$

for an absolute c_0 . Thus (4.24) and Theorem 5.4 give

$$(4.27) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w_f}) \leq C c_* \delta.$$

The endpoint misalignment is also bounded:

$$(4.28) \quad \gamma = \frac{r}{2}(1 - u \cdot w_f) \leq r h_f^2 / 4 \leq C c_*^2 \delta^2 / N.$$

Make fresh groups of $n = \lceil C N \delta^{-2} \log(C/\eta) \rceil$ calls along $+w_f$ and $-w_f$. Conditional Bernstein bounds, Lemma 4.2, and $\mathfrak{h}(s, t)^2 \leq 2|s - t|$ give

$$(4.29) \quad \begin{aligned} \mathfrak{h}(\hat{p}_{\text{raw}}, p), \mathfrak{h}(\hat{q}_{\text{raw}}, q) &\leq C \sqrt{\log(C/\eta)/n} + C \sqrt{\gamma} \\ &\leq c' \delta / \sqrt{N}. \end{aligned}$$

The constant c' can be made as small as required by the preceding choices of c_* and the sample multiplier. Sort the estimates and return $\hat{E} = \hat{p}P_{w_f} + \hat{q}(I - P_{w_f})$. Equations (5.9) and (4.5) make the spectral loss at most $\delta/4$; choose c_* so that (4.27) contributes at most $\delta/4$. The total failure probability is less than η .

On exceptional records fix arbitrary legal conventions for a zero complex estimate, an ambiguous principal argument, an estimated angle outside $[-1/4, 1/4]$, or an absent accepted length. Each output remains a legal effect and each block length remains at most N . The public budgets in (4.22), (4.25), and the product experiments give (4.2) on every record. This proves the upper bound, with error slack.

For the converse consider the scalar effects

$$(4.30) \quad E_0 = \frac{1}{2}I, \quad E_1 = (\frac{1}{2} + \Delta)I, \quad \Delta = 256\delta/\sqrt{N}.$$

Since $\delta \leq 1/4096$, both parameters lie in $[1/2, 9/16]$. The finite Bernoulli lower bound in Lemma 5.2 yields

$$(4.31) \quad \begin{aligned} d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_1}) &= B_N(1/2, 1/2 + \Delta) \\ &\geq \frac{1}{64} \min\{1, \Delta \sqrt{N/(5/4)}\} > 2\delta. \end{aligned}$$

For these devices every call gives a fresh Bernoulli bit independent of the input. On an entangled input, each conditional reference block is the same reduced state multiplied by the scalar outcome probability. Thus any adaptive learning record, including its final classical estimate, is a common

stochastic processing of M independent Bernoulli bits and parameter-independent resources. Publicly stopped records may be padded. The relative entropy before processing is at most

$$(4.32) \quad MD(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) = -\frac{M}{2} \log(1 - 4\Delta^2) \leq 3M\Delta^2.$$

The two radius- δ success balls are disjoint by (4.31). Testing whether the estimated effect belongs to the first ball therefore gives two errors at most η . For classical laws P, Q , Jensen's inequality gives root fidelity at least $e^{-D(P\parallel Q)/2}$. The trace-fidelity inequality then implies that the sum of testing errors is at least $\frac{1}{2}e^{-D(P\parallel Q)}$. Consequently

$$2\eta \geq \frac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2 \delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this proves (4.3). $\square$

Corollary 4.6 (A learned joint description). *For rational δ in the range of Theorem 4.1, the learner may return one word of the rational joint code, with the same risk and call-count bounds. Its payload length is*

$$(4.33) \quad 2\log_2 N + \log_2 \log(N + 2) + 4\log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The decoding centre is a legal rational effect. This is a statement about device calls and payload length; classical arithmetic, workspace, and state-preparation costs are separate resources.

Proof. Use the learner with loss budget $\delta/2$. Its output is a finite computable expression determined by a finite record. Choose a positive rational $\varepsilon \leq \delta/(100N)$ and rational approximations (b_0, x_0) to its bias and Bloch vector with

$$|b_0 - \hat{b}| + \|x_0 - \hat{x}\| \leq \varepsilon.$$

Set $(b_1, x_1) = (b_0, x_0)/(1 + 2\varepsilon)$. Then

$$|b_1| + \|x_1\| \leq \frac{1 + \varepsilon}{1 + 2\varepsilon} \leq 1,$$

and the exact one-use formula (5.6) bounds the distance from the unrounded hypothesis by 3ε . The N -use hybrid inequality adds at most $3N\varepsilon$. Apply Theorem 6.2 to this rational estimated effect with public accuracy $\delta/2$. Its certified radius is $\delta/4$, so the total loss is less than δ . The capacity estimate in that theorem, at a constant rescaling of accuracy, gives (4.33). This postprocessing uses no additional device calls and receives no exact description of the unknown target. $\square$

The loss in Theorem 4.1 is a future adaptive distance. One-use measurement tomography is studied in [5, 29]; combining the stated one-use accuracy bounds only with the hybrid inequality gives a quadratic dependence on the future horizon. The argument here uses finite-use geometry and data-dependent entangled blocks to obtain the linear dependence in (4.2), with a matching scalar converse. Learning a stored programme from which a measurement is later retrieved, as in [30], has a different output resource. The present theorem concerns a classical estimate over the entire ordered binary qubit body; it does not assert this learning bound for the arbitrary-dimensional matrix body of Theorem 2.1.

5. THE FULL BINARY EFFECT BODY

The spectral boundary of a biased measurement has two distinct endpoints. A vanishing eigenvalue of one effect need not give a coherent angular scale: the other outcome can still destroy coherence. We therefore retain both endpoint coordinates and determine their joint finite-use geometry.

Let

$$(5.1) \quad \mathcal{E} = \left\{ (a, x) \in \mathbb{R} \times \mathbb{R}^3 : 0 \leq a - |x| \leq a + |x| \leq 2 \right\}, \quad E_{a,x} = \frac{1}{2}(aI + x \cdot \sigma).$$

We identify a point (a, x) of $\mathcal{E}$ with its effect $E_{a,x}$. Write $\mathcal{M}_{a,x}$ for the measurement with ordered effects $E_{a,x}, I - E_{a,x}$ and no residual quantum output. Its maximum and minimum eigenvalues, spectral gap, and bias are

$$p = \frac{1}{2}(a + |x|), \quad q = \frac{1}{2}(a - |x|), \quad r = p - q = |x|, \quad b = p + q - 1.$$

Thus $0 \leq q \leq p \leq 1$ and $|b| + r \leq 1$. When $r > 0$ put $u = x/r$, so that $E_{a,x} = pP_u + q(I - P_u)$, where $P_u = (I + u \cdot \sigma)/2$. Directions are not identified under outcome exchange.

For an integer $N \geq 1$ set $\tau = N^{-1}$ and define

$$(5.2) \quad T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + \tau}}, \quad 0 \leq s, t \leq 1,$$

$$V(p, q) = p(1-p) + q(1-q) = \frac{1}{2}(1 - b^2 - r^2),$$

$$(5.3) \quad K_N(p, q) = r \sqrt{\frac{N}{V(p, q) + \tau}}.$$

For two effects with data (p, q, u) and (p', q', v) let $h \in [0, \pi]$ be the angle between u, v when both spectral gaps are positive. If either gap is zero, set the following angular term equal to zero, without choosing a direction at the repeated eigenvalue:

$$(5.4) \quad H_N(E, F) = T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\} h.$$

Theorem 5.1 (Finite-use geometry of binary qubit effects). *For every $E, F \in \mathcal{E}$ and $N \geq 1$,*

$$(5.5) \quad \frac{1}{8192} \min\{1, H_N(E, F)\} \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 3H_N(E, F)\}.$$

The distance permits all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping, and uses the unhalved trace norm. The constants are independent of both spectra, both directions, and the horizon. In particular,

$$(5.6) \quad d_1(\mathcal{M}_{a,x}, \mathcal{M}_{a',x'}) = |a - a'| + |x - x'|.$$

The lower comparison is witnessed by pair-dependent nonadaptive tests. It does not require a common estimator of all four target coordinates.

The denominator in (5.3) involves the two outcome variances. Replacing it by the distance to the nearest spectral face would give an incorrect angular scale on a face such as $q = 0$, $0 < p < 1$. The proof below supplies both a scalar-overlap dilation upper bound and explicit finite-distance lower witnesses.

5.1. The spectral endpoints. Put

$$B_N(s, t) = \|\text{Bern}(s)^{\otimes N} - \text{Bern}(t)^{\otimes N}\|_1.$$

Lemma 5.2 (A Bernoulli estimate at both support endpoints). *For $s, t \in [0, 1]$,*

$$(5.7) \quad \frac{1}{64} \min\{1, T_N(s, t)\} \leq B_N(s, t) \leq \min\{2, 3T_N(s, t)\}.$$

For a fixed one of the two parameters, B_N increases as the other parameter moves away from it on either side.

Proof. Assume $s < t$, put $\Delta = t - s$, and write $v = \max\{s(1-s), t(1-t)\}$. The identical pair is immediate. A coupling of individual bits gives $B_N \leq 2N\Delta$. The root fidelity $F =$

$\sqrt{st} + \sqrt{(1-s)(1-t)}$ satisfies

$$\begin{aligned} 1 - F &= \frac{\Delta^2}{2} \left(\frac{1}{(\sqrt{s} + \sqrt{t})^2} + \frac{1}{(\sqrt{1-s} + \sqrt{1-t})^2} \right) \\ &\leq \frac{\Delta^2(1+\Delta)}{2t(1-s)} \leq \frac{\Delta^2}{2v}, \end{aligned}$$

whenever $v > 0$. For the last inequality, both endpoint variances are at most $t(1-s)/(1+\Delta)$: after cancellation the required inequalities are $s(1+\Delta) \leq t$ and $(1-t)(1+\Delta) \leq 1-s$. They follow respectively from $\Delta(1-s) \geq 0$ and $t\Delta \geq 0$. The product-fidelity inequality gives $B_N \leq 2\sqrt{N}\Delta/\sqrt{v}$. For $z = Nv$, the inequality $\min\{2, 2/\sqrt{z}\}\sqrt{z+1} \leq 2\sqrt{2} < 3$ combines the two upper bounds. At $v = 0$ use the coupling bound.

For the lower bound, complement both bits if necessary to have $s + t \leq 1$. If $t > 3/4$, then $\Delta \geq 2t - 1 > 1/2$ and $B_N \geq B_1 = 2\Delta > 1$, which suffices. Otherwise $0 < t \leq 3/4$. Set

$$k = \min \left\{ N, \max \left\{ 1, \left\lfloor \frac{1}{2t} \right\rfloor \right\} \right\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, t^{-1}\}$, $m \geq N/(2k)$, and $(k-1)t \leq 1/2$. The event of no success in a block of k trials has probabilities $e_s = (1-s)^k$ and $e_t = (1-t)^k$, with

$$e_s - e_t \geq k\Delta(1-t)^{k-1} \geq k\Delta/2.$$

There is a common stochastic processing of this binary event whose success probability is $d + ce$, where

$$c = \frac{1}{\max\{e_s + e_t, 2 - e_s - e_t\}} \geq \frac{1}{2}, \quad d = \frac{1}{2}(1 - c(e_s + e_t)).$$

Indeed $d \geq 0$, $d + c \leq 1$, and the processed pair has parameters $(1 \pm c(e_s - e_t))/2$. Apply Lemma M.3 to m independent blocks and discard the unused calls. Contraction under this processing gives

$$B_N(s, t) \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, t^{-1})}\}.$$

Since $v \geq t(1-t) \geq t/4$, $T_N(s, t) \leq 2\Delta\sqrt{N \min(N, t^{-1})}$. This proves the stated lower constant.

For monotonicity, the likelihood ratio of the two binomial count laws is monotone in the count. For a fixed larger parameter their total variation is the maximum, over upper count intervals, of the larger-parameter probability minus the smaller-parameter probability. The latter probabilities decrease when the smaller parameter decreases. Fixing the smaller parameter gives the analogous statement with lower count intervals. Endpoint cases follow by continuity of the finite sums. $\square$

Lemma 5.3 (Spectral projection without cancellation). *For arbitrary directions and spectra,*

$$(5.8) \quad B_N(p, p') \leq d_N(\mathcal{M}_E, \mathcal{M}_F), \quad B_N(q, q') \leq d_N(\mathcal{M}_E, \mathcal{M}_F).$$

If the two directions agree, then

$$(5.9) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p',q',u}) \leq B_N(p, p') + B_N(q, q').$$

Consequently, with $d = d_N(\mathcal{M}_E, \mathcal{M}_F)$,

$$(5.10) \quad \begin{aligned} d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) &\leq 3d, \\ d_N(\mathcal{M}_{p',q',u}, \mathcal{M}_{p',q',v}) &\leq 3d. \end{aligned}$$

Proof. Suppose first $p \geq p'$. Repeatedly input a maximum-eigenvalue eigenstate of E . Its success probabilities under the two hypotheses are p and some $t \leq p'$. The monotonicity in Lemma 5.2 gives $B_N(p, p') \leq B_N(p, t) \leq d$. If $p < p'$, interchange the hypotheses. For $q \leq q'$, use a minimum-eigenvalue eigenstate of E ; the other success probability is at least q' . Reverse the hypotheses

if $q > q'$. These are separate admissible witnesses; no single input is asserted to witness both endpoint changes.

For aligned directions, first make the common projective measurement $\{P_u, I - P_u\}$ and, according to its outcome, generate a bit with success probability p or q . This implements exactly the measurement channel, including on inputs entangled with a reference: the reference blocks are the corresponding partial matrix elements weighted by the displayed success probabilities. Supply in advance N independent p -bits and N independent q -bits; at a call consume the next unused bit from the selected array. Every fixed adaptive tester is a common channel of these two arrays. The triangle inequality for tensor products therefore bounds its output distance by $B_N(p, p') + B_N(q, q')$. This proves (5.9) with arbitrary references, feedback, and stopping; all unqueried row bits are discarded. Combining it with (5.8) bounds every aligned spectral change by $2d$. The triangle inequality, aligning at u or at v , then proves (5.10). $\square$

5.2. Angular geometry with both eigenvalues retained.

Theorem 5.4 (Angular transition for a biased measurement). *For fixed $0 \leq q \leq p \leq 1$, $N \geq 1$, and directions at angle $h \in [0, \pi]$,*

$$(5.11) \quad \frac{1}{512} \min\{1, K_N(p, q)h\} \leq d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) \leq \min\{2, \sqrt{2}K_N(p, q)h\}.$$

More precisely, put

$$A = \sqrt{p(1-p)} + \sqrt{q(1-q)}, \quad z = (p-q) \sin(h/2).$$

When $A > 0$,

$$(5.12) \quad d_N \leq \min \left\{ 2, 2Nz, 2\sqrt{1 - \left(\frac{A^2}{A^2 + z^2} \right)^N} \right\}.$$

When $A = 0$, the bound $\min\{2, 2Nz\}$ remains valid.

Proof. The one-use distance is $2z$: the effect difference is traceless with operator norm z , and the same reference trace-norm duality as in Lemma M.2 applies. Replacing the calls one at a time gives the adaptive bound $2Nz$.

We next construct a dilation with scalar cross overlap. Common unitary conjugation puts the two directions in one real meridian. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_t = \exp(tJ), \quad t = h/2, \quad E = \text{diag}(p, q), \quad F = R_t E R_t^\top.$$

A measurement dilation is

$$W_E \psi = |+\rangle_C |+\rangle_L \sqrt{E} \psi + |-\rangle_C |-\rangle_L \sqrt{I-E} \psi.$$

The square-root vectors belong to a further two-dimensional environment. Tracing both environments leaves the ordered classical output and implements the measurement even with a reference. Put

$$f = \sqrt{pq} + \sqrt{(1-p)(1-q)}.$$

For any ϕ , apply the common rotation $R_{\phi-t}$ to the two-dimensional environment of the dilation W_F . The cross overlap is

$$\begin{aligned} W_E^\dagger (I \otimes R_{\phi-t}) W_F &= (\sqrt{E} R_\phi \sqrt{E} + \sqrt{I-E} R_\phi \sqrt{I-E}) R_{-t} \\ &= (\cos \phi I + f \sin \phi J) R_{-t}. \end{aligned}$$

Here the rotation acts as the identity on the outcome registers. When $f > 0$, choose

$$D = \sqrt{f^2 \cos^2 t + \sin^2 t}, \quad \cos \phi = \frac{f \cos t}{D}, \quad \sin \phi = \frac{\sin t}{D}.$$

Then the displayed overlap is cI , where $c = f/D \in [0, 1]$. This construction also covers $t = 0$ and $t = \pi/2$. To express it in the spectral coordinates, put $T = \sqrt{p(1-q)} + \sqrt{q(1-p)}$. Direct multiplication gives

$$fT = A, \quad (1 - f^2)T^2 = (p - q)^2.$$

If $A > 0$ then $f, T > 0$, and therefore

$$(5.13) \quad c^2 = \frac{f^2}{f^2 \cos^2 t + \sin^2 t} = \frac{A^2}{A^2 + (p - q)^2 \sin^2 t}.$$

Fix any common adaptive tester. Purify its initial state and all intermediate operations, and retain the measurement environments in this proof. The environments are never accessed by the tester. At each call the scalar identity for the cross overlap multiplies the inner product of the two purifications by c , independently of the current input and reference. The common intervening isometries preserve the inner product. A tester that stops early is padded to N calls on dummy inputs, with their outcomes ignored in its final output. Thus two purifications of that final output have inner product c^N . Trace-norm contraction and the pure-state formula give $d_N \leq 2\sqrt{1 - c^{2N}}$, proving (5.12). This is a pairwise dilation comparison; it does not require a processor for the entire effect body or an unknown-input environment-seizing procedure.

Since $V \leq A^2 \leq 2V$ and $1 - (1 + y)^{-N} \leq Ny$ for $y \geq 0$, the two nontrivial upper bounds imply

$$d_N \leq \min\{Nrh, \sqrt{N}rh/\sqrt{V}\} \leq \sqrt{2}K_N(p, q)h$$

when $V > 0$. The final inequality follows by putting $z = NV$ and using $\min\{1, z^{-1/2}\}\sqrt{z+1} \leq \sqrt{2}$. If $V = 0$, the only possibilities are scalar deterministic effects or a projective measurement, and the hybrid bound suffices.

For the lower bound, two explicit witnesses suffice. First suppose $r > 0$ and $h > 0$. Input a pure state in the direction $u - v$. Its success probabilities are

$$\frac{1 + b + r \sin(h/2)}{2}, \quad \frac{1 + b - r \sin(h/2)}{2}.$$

Their difference is $r \sin(h/2)$. The larger of their Bernoulli variances is at most $1 - b^2$: after complementing both output laws we may assume $a = 1 + b \leq 1$; both success probabilities lie in $[0, a]$, so their variances are at most $a \leq a(2 - a) = 1 - b^2$. Lemma 5.2, together with $\sin(h/2) \geq h/\pi$, now gives

$$(5.14) \quad d_N \geq \frac{1}{64\pi} \min \left\{ 1, rh \sqrt{\frac{N}{1 - b^2 + \tau}} \right\}.$$

The repeated pure input is a nonadaptive test.

For the second witness, choose a half-turn of the input qubit about an axis normal to the plane of u, v . It sends both Bloch directions to their negatives. With probability 1/2 use the original measurement, and with probability 1/2 apply this half-turn before the call and swap the two recorded outcomes afterwards. Discard the random choice. The resulting measurement has effects $(I \pm ru \cdot \sigma)/2$ or $(I \pm rv \cdot \sigma)/2$ under the respective hypotheses: the bias has been removed and the visibility is r . The same random preprocessing and postprocessing are used under both hypotheses and on every call. Any test for this unbiased pair is consequently an admissible test

for the original pair. Theorem M.1 gives

$$(5.15) \quad d_N \geq \frac{1}{64} \min \left\{ 1, rh \sqrt{\frac{N}{1-r^2+\tau}} \right\}.$$

The witness is the finite entangled-block test of Lemma M.4, with this common processing. If the directions coincide or are antipodal, the required half-turn is chosen about any perpendicular axis.

Finally, feasibility $|b| + r \leq 1$ implies

$$\min\{1 - b^2, 1 - r^2\} \leq 2(1 - b^2 - r^2) = 4V.$$

For example, write $x = \max\{|b|, r\}$ and $y = \min\{|b|, r\}$; then $x \leq 1 - y$, $0 \leq y \leq 1/2$, and $x^2 + 2y^2 \leq (1 - y)^2 + 2y^2 \leq 1$, which is the asserted inequality. Hence the larger of the two coefficients in (5.14) and (5.15) is at least $K_N(p, q)/2$. Their stronger lower bound is at least $(128\pi)^{-1} \min\{1, K_N(p, q)h\}$, which implies (5.11). If $r = 0$ or $h = 0$, the two channels coincide and the assertion is immediate. $\square$

Proof of Theorem 5.1. For positive gaps use a path whose angular leg has the spectrum with the smaller angular coefficient. Lemma 5.3, followed by Lemma 5.2 and Theorem 5.4, gives

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 3T_N(p, p') + 3T_N(q, q') + \sqrt{2} \min\{K_N(p, q), K_N(p', q')\}h \leq 3H_N(E, F).$$

If one gap vanishes, that scalar effect may be aligned with the other direction at no cost, so the same estimate holds with angular term zero.

For the lower bound, put $A = T_N(p, p')$, $B = T_N(q, q')$, and $C = \min\{K_N(p, q), K_N(p', q')\}h$. The spectral projection lemma gives $d_N \geq 64^{-1} \min\{1, A\}$ and $d_N \geq 64^{-1} \min\{1, B\}$. Its angular projections and Theorem 5.4 give $d_N \geq 1536^{-1} \min\{1, C\}$. The largest of these three bounds is at least $4608^{-1} \min\{1, A + B + C\}$, which proves the weaker displayed constant $1/8192$. At a scalar effect only the two spectral bounds are needed. To check the assertion about lower witnesses, let d_N^{na} be the supremum over nonadaptive tests. The two spectral lower bounds hold with d_N^{na} , and the aligned spectral upper bound is consequently at most $2d_N^{\text{na}}$. Apply the triangle inequality to the output of a nonadaptive angular witness on the three-channel path. Its actual-pair leg is at most d_N^{na} , and its aligned spectral leg is at most $2d_N^{\text{na}}$. The same lower comparison thus holds with d_N^{na} in place of d_N .

It remains to verify the one-use identity. For a reference-assisted input state ρ , the difference has output blocks B_ρ and $-B_\rho$, where $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An extremal eigenstate of $E - F$ attains equality. Thus $d_1 = 2\|E - F\|_{\text{op}}$; the two eigenvalues of $E - F$ are $((a - a') \pm |x - x'|)/2$. This proves (5.6). $\square$

Remark 5.5 (Boundary faces and repeated eigenvalues). On $p = q = t$, the measurement is the input-independent coin $\text{Bern}(t)$ and all directions represent the same effect. Its exact finite-use distance along this scalar interval is $B_N(t, t')$. On the face $q = 0$, $0 < p < 1$, the angular coefficient is

$$K_N(p, 0) = p \sqrt{\frac{N}{p(1-p) + N^{-1}}}.$$

For a fixed p in this interval it has square-root-horizon order, despite the zero eigenvalue. At the projective edge $(p, q) = (1, 0)$ it equals N . At the two deterministic apices $p = q = 0$ and $p = q = 1$, the angular coefficient vanishes while the spectral moduli retain their finite-use support cutoff. These cases are part of the same theorem, without a spectral margin imposed on the target.

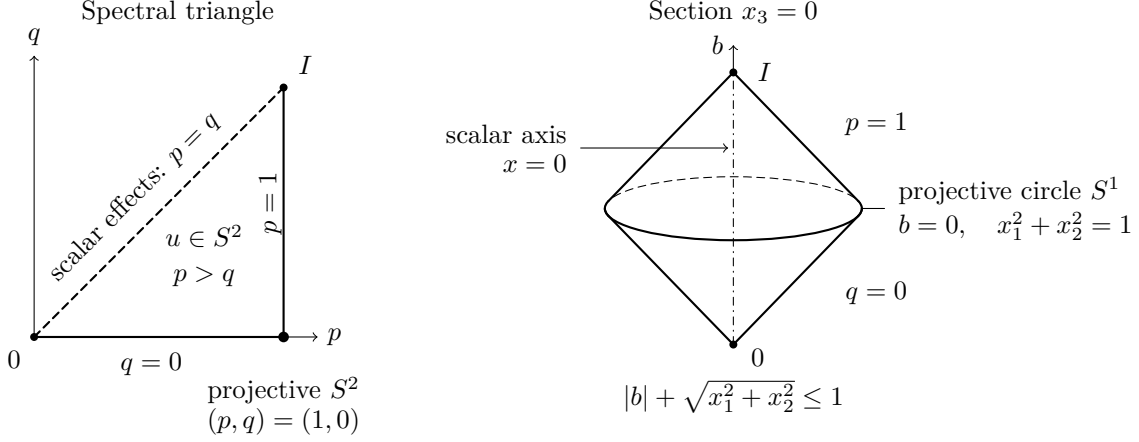


FIGURE 1. The four-dimensional effect body $|b| + |x| \leq 1$. Each point of the spectral triangle with $p > q$ carries a direction $u \in S^2$; on the scalar diagonal all directions represent the same effect. The boundary pieces $q = 0$ and $p = 1$ meet at the projective stratum $(p, q) = (1, 0)$, which carries the full sphere S^2 . The right panel is a projection of the three-dimensional section $x_3 = 0$; its equatorial circle is the intersection of that sphere with the section.

6. THE DESCRIPTION ENTROPY OF ALL BINARY QUBIT EFFECTS

Let

$$\mathfrak{E} = \{E = E^* \in \mathbb{C}^{2 \times 2} : 0 \leq E \leq I\}.$$

An effect E specifies the ordered measurement with effects $E, I - E$. Write $\mathcal{C}_N(\mathfrak{E}, \delta)$ for its covering number under d_N , allowing arbitrary legal memoryless centres of this interface, and write $\mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta)$ when the centres have rational Cartesian coordinates. Both eigenvalues and the eigenspaces are charged. The spectral representation $E = pP_u + q(I - P_u)$, $p \geq q$, represents a scalar effect only once, independently of u .

Theorem 6.1 (The full effect body). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every integer $N \geq 1$ and $0 < \delta \leq \delta_0$,*

$$(6.1) \quad cN^2 \log(N+2) \delta^{-4} \leq \mathcal{C}_N(\mathfrak{E}, \delta) \leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta) \leq CN^2 \log(N+2) \delta^{-4}.$$

Consequently the optimum fixed-length reusable payload is

$$(6.2) \quad 2 \log_2 N + \log_2 \log(N+2) + 4 \log_2(1/\delta) + O(1).$$

The constants are uniform over the whole closed effect body, including scalar, deterministic and projective effects. For rational accuracy input, rational target data admit an exact rational encoder attaining this order.

The logarithm records accumulation near the projective stratum, where both outcome probabilities can approach their support endpoints. We prove the upper bound by a finite rational construction and the lower bound by separated families of projective-corner boxes.

6.1. A rational spectral and angular code. For $N \geq 1$ and rational $0 < \delta \leq 1/4$, put

$$(6.3) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad s_i = \begin{cases} \frac{i^2}{B^2 + i^2}, & 0 \leq i \leq B, \\ 1 - \frac{(2B-i)^2}{B^2 + (2B-i)^2}, & B < i \leq 2B. \end{cases}$$

Thus $0 = s_0 < s_1 < \dots < s_{2B} = 1$. For $0 \leq j < i \leq 2B$ define

$$(6.4) \quad K_{ij}^2 = \frac{N(s_i - s_j)^2}{s_i(1 - s_i) + s_j(1 - s_j) + N^{-1}}, \quad A_{ij} = \max \left\{ 1, \left\lceil \sqrt{256K_{ij}^2/\delta^2} \right\rceil \right\}.$$

Use the six signed stereographic charts

$$(6.5) \quad u_a = \epsilon \frac{1 - |z|^2}{1 + |z|^2}, \quad u_b = \frac{2z_b}{1 + |z|^2} \quad (b \neq a), \quad a \in \{1, 2, 3\}, \quad \epsilon \in \{-1, 1\},$$

with $z \in \{-A_{ij}, \dots, A_{ij}\}^2/A_{ij}$ for the spectral pair (s_i, s_j) . Every chart word decodes to $s_i P_u + s_j(I - P_u)$. A diagonal pair $i = j$ has the single word $s_i I$. The exact number of words, including redundant chart words, is

$$(6.6) \quad L_{N,\delta} = (2B + 1) + \sum_{0 \leq j < i \leq 2B} 6(2A_{ij} + 1)^2.$$

Theorem 6.2 (Exact rational realization). *The words in (6.6) form a radius- $\delta/2$ cover of $\mathfrak{E}$ by legal rational effects, and*

$$(6.7) \quad L_{N,\delta} \leq CN^2 \log(N + 2) \delta^{-4}$$

for an absolute C . If

$$E = \frac{1}{2}((1 + b)I + x \cdot \sigma), \quad b \in \mathbb{Q}, \quad x \in \mathbb{Q}^3, \quad |b| + |x| \leq 1,$$

then a covering word is computable by rational arithmetic and integer comparisons. Its payload is one fixed-length index in the set of (6.6).

Proof. Write $f(t) = t^2/(1 + t^2)$, $0 \leq t \leq 1$. For a probability in $[0, 1/2]$, round its inverse coordinate t to the nearest multiple of $1/B$; for a probability in $[1/2, 1]$, use the reflected chart $1 - f(t)$. Choose the multiple of smaller magnitude at ties. This defines a nondecreasing quantizer on $[0, 1]$, so separately quantizing $p \geq q$ preserves their order.

The square-root vector associated with $f(t)$ is $(t, 1)/\sqrt{1 + t^2}$; its derivative has norm $1/(1 + t^2) \leq 1$. The reflected chart has the same bound. Hence the Euclidean square-root distance between each probability and its quantization is at most $1/(2B)$. If H is this distance, the product-fidelity inequality gives unhalved distance at most $2\sqrt{N}H$ between the corresponding N -fold Bernoulli laws. The common two-coin programme in Lemma 5.3 therefore gives a spectral error at most

$$2\sqrt{N}/B \leq \delta/4.$$

This argument includes adaptive testers and entangled references.

If the quantized eigenvalues coincide, this completes the encoding. Otherwise choose a signed chart whose distinguished coordinate has maximal absolute value in the target direction. Its inverse coordinate belongs to $[-1, 1]^2$. Rounding its two coordinates to the grid of denominator A_{ij} gives angular error at most $2\sqrt{2}/A_{ij}$, since geodesic distance is bounded by the length of the mapped straight segment and the differential of stereographic projection has norm at most two. The angular upper bound in Theorem 5.4 gives error at most

$$\sqrt{2}K_{ij} \frac{2\sqrt{2}}{A_{ij}} \leq \delta/4.$$

The triangle inequality proves the covering assertion. Formula (6.5) preserves the unit sphere exactly, so every decoded effect is legal and rational.

We verify exact computability even when the input eigenvalues are irrational. Put $Q = |x|^2 \in \mathbb{Q}$. The eigenvalues are $(1 + b \pm \sqrt{Q})/2$. Each decision in the spectral rounding compares one of these numbers with a rational threshold $f((k + 1/2)/B)$ or its reflection. Such a comparison is the sign of a rational number plus or minus $\sqrt{Q}$; checking signs before squaring makes it an exact rational

comparison. If $Q = 0$, the two quantized eigenvalues coincide. Otherwise choose the smallest index a maximizing $|x_a|$ and put $\epsilon = \text{sgn}(x_a)$. The inverse chart is

$$z_b = \frac{x_b}{\sqrt{Q} + |x_a|} \quad (b \neq a).$$

Its comparison with any rational rounding threshold again reduces to the sign of a rational number plus a rational multiple of $\sqrt{Q}$. The integers B, A_{ij} are computed by integer squaring and rational comparison. Thus no irrational normalization is required. The decoder uses only the displayed rational formulae.

It remains to count the words. Set $m_i = \min(i, 2B - i)$ and $U = B^2/N$. Directly from (6.3),

$$\frac{m_i^2}{4B^2} \leq s_i(1 - s_i) \leq \frac{m_i^2}{B^2}, \quad K_{ij}^2 \leq \frac{4NB^2}{m_i^2 + m_j^2 + U}.$$

Each value of m_i occurs at most twice. Since $(2A_{ij} + 1)^2 \leq C(1 + K_{ij}^2/\delta^2)$, we obtain

$$(6.8) \quad L_{N,\delta} \leq CB^2 + \frac{CNB^2}{\delta^2} \sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U}.$$

Here $U \geq 1$. The part with $\max(a, b) \leq \sqrt{U}$ is bounded by an absolute constant. Each successive dyadic square ring contributes at most another absolute constant. There are at most $C \log(N + 2)$ such rings, because $B/\sqrt{U} = \sqrt{N}$. Consequently

$$\sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U} \leq C \log(N + 2).$$

Finally $B^2 \leq CN/\delta^2$, so (6.8) proves (6.7). The support cutoff in the denominator is retained throughout the count, including the endpoint grid words. $\square$

6.2. Packing the projective corner.

Proof of Theorem 6.1. The upper bound follows from Theorem 6.2. For a real δ , choose a rational δ' with $\delta \leq \delta' \leq 2\delta$ and use its radius- $\delta'/2$ code; decrease δ_0 to at most $1/8$. For the lower bound first let $N \geq 16$. For every

$$t = 8^j/N \leq 1/16, \quad j \geq 0,$$

consider the effects with

$$(6.9) \quad p = 1 - \alpha, \quad q = \beta, \quad t \leq \alpha, \beta \leq 2t, \quad u = u(z), \quad z \in [-1/4, 1/4]^2,$$

where $u(z)$ is the chart in (6.5) with $a = 3, \epsilon = 1$. On this square, spherical distance is bounded above and below by absolute positive multiples of Euclidean distance in z . Throughout (6.9), the eigenvalue gap is at least $3/4$ and

$$p(1 - p) \asymp q(1 - q) \asymp t, \quad p(1 - p) + q(1 - q) + N^{-1} \asymp t.$$

The finite-distance comparison of Theorem 5.1 therefore supplies an absolute $c_1 > 0$ such that any two points in this box satisfy

$$(6.10) \quad d_N \geq c_1 \min \left\{ 1, \sqrt{N/t} |(\alpha, \beta, z) - (\alpha', \beta', z')|_\infty \right\}.$$

Choose an absolute sufficiently large A and then an absolute sufficiently small δ_0 . In each of the four coordinates take a grid of spacing $A\delta\sqrt{t/N}$. Since $Nt \geq 1$, this spacing is at most half the

width of either spectral interval when $\delta \leq \delta_0$. The resulting points have pairwise distance at least 4δ by (6.10), and their number is at least

$$c \left(\frac{t}{\delta \sqrt{t/N}} \right)^2 \left(\frac{1}{\delta \sqrt{t/N}} \right)^2 = cN^2 \delta^{-4}.$$

Points from different boxes are also uniformly separated. Indeed, if $t' \geq 8t$, then $\alpha' \geq t'$ and $\alpha \leq 2t \leq t'/4$. The spectral modulus of the corresponding upper eigenvalues is bounded below by an absolute positive multiple of $\sqrt{Nt'} \geq 1$. The spectral noncancellation bound in Theorem 5.1 gives $d_N \geq c_2 > 0$. Shrinking δ_0 ensures that this is at least 4δ . There are at least $c \log(N+2)$ permitted boxes. Their union is therefore a 4δ -packing of cardinality $cN^2 \log(N+2) \delta^{-4}$.

For $1 \leq N < 16$, use instead a fixed spectral rectangle $5/8 \leq p \leq 3/4$, $1/4 \leq q \leq 3/8$ and the same angular chart. Theorem 5.1 gives a uniform local Euclidean lower bound, and a grid of spacing $C\delta$ has at least $c\delta^{-4}$ points. The factor $N^2 \log(N+2)$ is bounded on this finite range. This proves the same lower estimate for all $N \geq 1$.

A radius- δ ball about any legal centre contains at most one point of a 4δ -packing, by the triangle inequality. Hence the packing proves the lower bound with the stated centre convention. Taking binary logarithms and rounding the index length proves (6.2). $\square$

6.3. The source of the logarithm. The spectral and angular weights also give the following integral form of the same accumulation. Put $v_p = p(1-p)$, $v_q = q(1-q)$ and $\tau = N^{-1}$. Then

$$(6.11) \quad \int_{0 \leq q \leq p \leq 1} \frac{N^2 (p-q)^2 dp dq}{(v_p + v_q + \tau) \sqrt{(v_p + \tau)(v_q + \tau)}} \asymp N^2 \log(N+2).$$

To verify this, near $p=1, q=0$ put $\alpha = 1-p, \beta = q$. After removing the factor N^2 , the integrand is comparable to

$$\frac{1}{(\alpha + \beta + \tau) \sqrt{(\alpha + \tau)(\beta + \tau)}}.$$

For $s = \alpha + \beta$, integration over $0 \leq \alpha \leq s$ is bounded above by an absolute constant times $1/(s + \tau)$. For $s \geq 2\tau$, restriction to $s/4 \leq \alpha \leq 3s/4$ gives the matching lower bound. Integration in s proves logarithmic growth. Near the scalar endpoint $p=q=0$, the factor $(p-q)^2$ is at most $(p+q)^2$, while $v_p + v_q \asymp p+q$; the remaining integral is uniformly bounded. Complementation gives the same conclusion near $p=q=1$. Away from these endpoint neighbourhoods, $v_p + v_q$ is bounded below and the factors $v_p^{-1/2}, v_q^{-1/2}$ are integrable. Bounded N is absorbed into the constants. Thus the extra logarithm is contributed by the projective corner. The finite covering construction and packing above also include the collapsed scalar coordinates, where the spectral representation ceases to be a chart.

7. MEASUREMENT DISCRIMINATION, GEOMETRY, AND LEARNING

The present binary interface retains the ordered classical outcome and an arbitrary external reference, but no postmeasurement quantum system from the device. This fixes the comparison with the measurement-discrimination literature. The projective theorem of Puchała and collaborators [16, Corollary 1 and Theorem 2] gives an exact multiple-use formula with ancillary access. It is used in the retained observable-readout theorem. The restricted ancillary interface in Fiurášek–Mičuda [19] leads to a different comparison of adaptive and parallel probes.

Krawiec–Pawela–Puchała [20, Section 4] construct a pair of nine-outcome qutrit rank-one POVMs that admit perfect adaptive discrimination with two calls and no perfect discrimination by any finite parallel experiment. Datta–Biswas–Saha–Augusiak [21, Theorems 3–5] study entanglement-assisted single-use discrimination, including perfect constructions and a qubit entanglement advantage. Their qubit construction has three outcomes. These are direct antecedents for

general POVM discrimination with a retained reference. They do not supply a full binary-effect family covering law. Our comparison (2.23) permits an adaptive advantage and controls its size uniformly in N for fixed input dimension.

The differential argument in Lemma 2.2 belongs to the established horizontal-gauge and channel-extension method [22, 7, 23, 24]. Finite adaptive discrimination bounds obtained by integrating a local channel bound appear in [25, 26]. The binary Sylvester choice here gives the specific regularized variance $M(I - M)$, which can be integrated explicitly and matched by compressed nonadaptive tests. The inverse-Sylvester expressions of Bures geometry [27] and its spectral integration [28] are related methods. Our form acts on the effect displacement H with variance $V = M(I - M)$; it is not the ordinary Bures pullback under $M \mapsto V$, whose differential would be $H - MH - HM$. The operational-ball lemma and the nested endpoint integral are both required for Theorem 3.1.

Mele-Bittel [5, Lemmas III.7–III.8 and Corollary III.9] give the binary one-use identity and an operator-norm learner. Zambrano-Ramos-Calderer-Kueng [29] study one-use outcome variation under their single-copy probe model. Combining a one-use ϵ guarantee with the hybrid bound gives a future- N guarantee after setting ϵ of order δ/N . Theorem 4.1 instead proves and matches the order $N\delta^{-2}\log(1/\eta)$ for the future adaptive loss on all of $\mathfrak{E}_2$. The learning network of [30] stores quantum resources for later measurement retrieval and has a different output interface.

Multiscale phase estimation and robust phase recovery are established techniques [31, 32]. Our proof supplies its own small-cone argument and conditional confidence allocation; it does not invoke the numerical unwrapping threshold corrected by the erratum to [32]. Randomized signs remove the unknown bias exactly, and empirical amplitudes choose the entangled block length without a supplied noise parameter. The learning theorem is a common experiment for all unknown qubit effects. The pair-dependent tests used in the matrix metric comparison and the supplied-target exact encoder retain their separate roles.

All matrix entropy constants are for fixed dimension and small error. The proof includes every rank and spectral multiplicity of the binary effect body. Multi-outcome POVMs, residual quantum outputs, growing-dimension constants, and full submaximal-error matrix coverings require additional arguments. The inherited full-error preparation and fixed-readout results retain their own stated ranges.

APPENDIX A. RATIONAL INSTRUMENTS STABLE UNDER EXTERNAL REFERENCES

A rounding operation on density matrices need not be affine. In particular, a nonlinear state repair cannot be tensored with the identity on an unknown reference system. We therefore round a *description of the instrument*, once and independently of its input, rather than round the state on which it acts. The resulting maps are completely positive and jointly trace preserving. All constants below are explicit in the input and output dimensions and the number of outcomes.

A.1. An integral repair of Choi data. Write M_d for the complex matrices of order d . An instrument from M_d to M_n with m outcomes is a family of completely positive maps Φ_y whose sum is trace preserving. We use the input-first, *unnormalized* Choi convention

$$(A.1) \quad J_y = \sum_{i,j=1}^d |i\rangle\langle j| \otimes \Phi_y(|i\rangle\langle j|), \quad J_y \geq 0, \quad \sum_{y=1}^m \text{Tr}_n J_y = I_d.$$

Here Tr_n traces the output factor; in particular $\sum_y \text{tr} J_y = d$, not one. These are the classical Choi criteria [1, Theorems 2.22 and 2.26]. The channel with the outcome retained is $\mathcal{F}(X) = \bigoplus_y \Phi_y(X)$. We write $\|\cdot\|_\diamond$ for the completely bounded trace norm.

Lemma A.1 (Choi trace norm and reference stability). *For any linear map $\Lambda : M_d \rightarrow M_n$ with unnormalized Choi matrix $J(\Lambda)$,*

$$d^{-1} \|J(\Lambda)\|_1 \leq \|\Lambda\|_\diamond \leq \|J(\Lambda)\|_1.$$

Consequently the diamond distance of two instruments, with the outcome retained, is at most the sum of the trace norms of their Choi-block differences.

Proof. Applying $\text{id}_d \otimes \Lambda$ to the normalized maximally entangled state gives the lower bound. For the upper bound, let u, v be unit vectors in $\mathbb{C}^r \otimes \mathbb{C}^d$. With $\Omega = \sum_i e_i \otimes e_i$, write $u = (A \otimes I_d)\Omega$ and $v = (B \otimes I_d)\Omega$. The matrices $A, B : \mathbb{C}^d \rightarrow \mathbb{C}^r$ have Hilbert–Schmidt norm one and therefore operator norm at most one. Thus

$$(\text{id}_r \otimes \Lambda)(|u\rangle\langle v|) = (A \otimes I_n)J(\Lambda)(B^* \otimes I_n)$$

has trace norm at most $\|J(\Lambda)\|_1$. The trace-norm unit ball is the convex hull of rank-one partial isometries (including zero as a convex combination of opposite such matrices). A singular-value decomposition of an arbitrary input matrix therefore proves the same induced trace-norm bound, independently of r . Taking the supremum over r proves the assertion. For an instrument the Choi matrix is, up to a fixed permutation of factors, the direct sum of its blocks. $\square$

Let B be a positive integer. The supplied approximations T_y are Hermitian matrices with entries in $B^{-1}\mathbb{Z}[i]$. Assume that every independent real diagonal coordinate, and every independent real and imaginary off-diagonal coordinate, differs from that of J_y by at most B^{-1} . The T_y need not be positive or satisfy the marginal constraint. Put

$$(A.2) \quad R = I_d - \sum_y \text{Tr}_n T_y, \quad Q_y = T_y + \mathbf{1}_{\{y=1\}} R \otimes |e_1\rangle\langle e_1|,$$

$$(A.3) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad \tilde{J}_y = \frac{BQ_y + cI_{dn}}{D}.$$

The choice of the first outcome and first output coordinate is merely a fixed convention. Any fixed outcome and output basis coordinate give the same legality and uniform error bound. If specified zero outcomes must remain exactly zero, remove them before applying the theorem and restore them afterwards; m then counts only the repaired outcomes.

Theorem A.2 (Reference-stable rational instrument rounding). *For every $d, n, m, B \geq 1$, the matrices in (A.3) are Choi matrices of an instrument. Their numerators are positive semidefinite matrices with Gaussian-integer entries, their common denominator is D , and*

$$(A.4) \quad \|\mathcal{F} - \tilde{\mathcal{F}}\|_\diamond \leq \sum_y \|J_y - \tilde{J}_y\|_1 \leq \frac{3mndc}{B + mnc} \leq \frac{K(d, n, m)}{B}, \quad K(d, n, m) = 6mn^2d^2(1 + mn).$$

The construction uses integer additions, multiplication by specified integers, and the supplied grid entries. It uses no spectral factorization, positivity projection, or rational Kraus factorization. Its complete output description has

$$O(md^2n^2 \log(B + mnc + 1))$$

bits, with an absolute implicit constant for the fixed encoding.

Proof. Set $E_y = T_y - J_y$. Counting the real coordinates gives $\|E_y\|_{\text{op}} \leq \|E_y\|_F \leq 2nd/B$. For a Hermitian E , the inequalities $-\|E\|_{\text{op}}I \leq E \leq \|E\|_{\text{op}}I$ imply $\|\text{Tr}_n E\|_{\text{op}} \leq n\|E\|_{\text{op}}$. It follows that

$$\|R\|_{\text{op}} \leq 2mn^2d/B, \quad \|Q_y - J_y\|_{\text{op}} \leq c/B.$$

Since $J_y \geq 0$, the numerator $BQ_y + cI_{dn}$ is positive. All its entries are Gaussian integers. The partial trace constraint is exact, because

$$\sum_y \text{Tr}_n(BQ_y + cI_{dn}) = (B + mnc)I_d.$$

The Choi criteria therefore give a completely positive instrument, including when some of the original outcomes are zero.

Let $F_y = Q_y - J_y$. The identity

$$D(\tilde{J}_y - J_y) = BF_y + cI_{dn} - mncJ_y$$

and $\|F_y\|_1 \leq dnc/B$ give

$$D \sum_y \|\tilde{J}_y - J_y\|_1 \leq mndc + mndc + mnc \sum_y \text{tr } J_y = 3mndc.$$

Lemma A.1 proves the norm assertion. Finally each diagonal entry of a positive numerator is at most D , by the summed partial trace constraint. Positivity bounds every off-diagonal modulus by the geometric mean of two diagonal entries, hence also by D . Encoding $m(dn)^2$ entries and the common denominator proves the bit bound. All intermediate entries have polynomially bounded bit length in the stated parameters and the supplied data. $\square$

For exact rational input, directed truncation supplies T_y by integer division. For uncertain real input, the hypothesis is a certified error bound on the *final grid coordinates*; an uncertified floating estimate is not a premise of the theorem. Although B may be a power of two, $D = B + mnc$ generally is not. The output is rational, not in general dyadic. The construction may replace a zero outcome by a small nonzero one; its mass is charged in (A.4). Known zero outcomes can instead be removed before rounding and restored as zero blocks afterwards, with m counting the remaining outcomes.

Corollary A.3 (A finite-description instrument net). *For fixed d, n, m and $0 < \delta \leq 1$, the preceding construction yields a finite diamond- δ net of legal instruments. Each member has an integer description of length*

$$O(md^2n^2 \log(2 + K(d, n, m)/\delta + mnc)).$$

This is a deterministic description net, not a statistically learned net. A particular member is computable in polynomial bit time from rational input data and the grid precision; enumeration of the entire net is not required.

Proof. Every Choi diagonal is at most one by (A.1), and positivity bounds every entry modulus by one. There are finitely many possible directed grid truncations at $B = \lceil K/\delta \rceil$. Apply the theorem to those arising from valid instruments. Computation of one truncation and its repair uses a polynomial number of elementary integer operations on polynomial-length operands. $\square$

The CP/TP constraints and adding a scalar identity buffer are not new principles. Projected least-squares process tomography already repairs Choi estimates and uses identity mixing to finish a TP-preserving positivity correction [2, Section 4]. Here the particular integer formula supplies an explicit denominator and a deterministic diamond-error budget from coordinate errors. No nearest-projection, statistical sample-complexity, or rank-optimality claim is made.

A.2. Adaptive use and stopped observations. A tester may retain an arbitrary finite-dimensional quantum memory, start with a state entangled across that memory and the instrument input, and insert arbitrary channels and finite-outcome measurements between uses. The command port is classical: its value may be selected from the public history and measured tester memory.

Coherent access to a superposition of commands is not an interface assumed here. The *same* tester is used in the two compared experiments.

Theorem A.4 (Adaptive reference-stable precision budget). *Suppose that at use t , uniformly over permitted commands and public histories, $\|\mathcal{F}_{t,a} - \tilde{\mathcal{F}}_{t,a}\|_\diamond \leq \delta_t$. For every common tester with a classical command port (not coherent superpositions of commands), every initial state, and every public stopping rule $\tau \leq N$, the final states, with the stopped history and tester memory retained, satisfy*

$$(A.5) \quad \|\Omega_\tau - \tilde{\Omega}_\tau\|_1 \leq \sum_{t=1}^N \delta_t.$$

The bound is independent of the tester-memory dimension. In particular, rounding each command by Theorem A.2 with $B \geq NK(d, n, m)2^L$ gives stopped joint-state error at most 2^{-L} and transcript total-variation error at most 2^{-L-1} .

Proof. Replace the instruments one use at a time. Just before the replaced use, both members of a neighboring hybrid pair have the same joint state. Decomposing a classical command/history register into blocks, the diamond bound applies to each positive block and is multiplied by its trace. These traces sum to one. The error introduced at this use is therefore at most δ_t , including any quantum reference. All subsequent operations in the hybrid pair are the same channels; complete trace-norm contraction bounds their final difference by the same quantity. Summing the N hybrid differences proves (A.5). For a stopping rule retain an absorbing halt flag, use the identity on halted blocks, and finish all experiments at use N . A public history-dependent stopping rule is then part of the same tester. The same argument applies, without conditioning on a possibly rare stopping history. Taking traces and dividing by two gives the total-variation assertion. $\square$

The hybrid principle is classical; see [1, Section 3.3, equation (3.306)] and the operational strategy norms of [3]. The conclusion here is a uniform finite-description approximation by *genuine instruments*. It is not a classical device that accepts an unknown physical entangled state and prepares its output. A numerical simulation of a specified joint state must charge the joint matrix dimension. Nor does the nonlinear state repair in the preceding section become a channel by this theorem.

Proposition A.5 (Posterior control at a stopping time). *Let $A_h = p_h \rho_h$ and $\hat{A}_h = q_h \hat{\rho}_h$ be the positive blocks at a common bounded stopping time, with both total traces one. At zero simulated mass assign any density matrix to $\hat{\rho}_h$. If $\sum_h \|A_h - \hat{A}_h\|_1 \leq \delta$, then*

$$(A.6) \quad \sum_h p_h \|\rho_h - \hat{\rho}_h\|_1 \leq 2\delta, \quad \Pr_p\{\|\rho_h - \hat{\rho}_h\|_1 > \eta\} \leq \min\{1, 2\delta/\eta\}.$$

The tail estimate holds for every $\eta > 0$. For any event of true probability $\alpha > 0$, the two pooled conditional states have distance at most $\min\{2, 2\delta/\alpha\}$, with an arbitrary conditional state when the simulated event has probability zero.

Proof. Writing $e_h = \|A_h - \hat{A}_h\|_1$, trace contraction gives $|p_h - q_h| \leq e_h$. For positive p_h ,

$$p_h \|\rho_h - \hat{\rho}_h\|_1 \leq e_h + |p_h - q_h| \leq 2e_h.$$

The inequality remains valid if $q_h = 0$. Sum and apply Markov's inequality. Applying the same calculation to the sums over an event gives the final assertion; two density matrices have distance at most two. Zero true mass makes no contribution. $\square$

The proposition gives weighted, tail and pooled-event posterior control, not simultaneous accuracy of every normalized rare history.

APPENDIX B. EXACT VALIDATION OF RATIONAL DESCRIPTIONS

The rational codes below require an exact positivity test. We record the arithmetic fact in the form used by both encoders.

Lemma B.1 (Reduced Schur entries and bit length). *Let A be a Hermitian matrix of order q with Gaussian-integer entries, whose real and imaginary parts have absolute value at most 2^h . In exact Schur elimination, let I be the ordered indices of the positive pivots already eliminated. Every remaining entry is*

$$(B.1) \quad S_{ij} = \frac{\det A[I \parallel i, I \parallel j]}{\det A[I, I]},$$

where $I \parallel i$ appends i in that order, and the empty principal determinant is one. Reduced numerators and denominators have $O(q(h + \log(q + 1)))$ bits. Hermitian positive semidefiniteness is therefore decidable by this elimination in polynomial bit time and space in the explicit input length. No optimized polynomial exponent is asserted.

Proof. When the leading pivot t is positive, block congruence reduces

$$\begin{pmatrix} t & b^* \\ b & C \end{pmatrix} \geq 0 \iff C - bb^*/t \geq 0.$$

A negative pivot rejects. At a zero diagonal, each 2×2 principal minor forces the corresponding row and column to vanish; otherwise reject, and if they vanish delete that row and column. Positive pivots make $A[I, I]$ positive definite. The bordered determinant identity

$$\det \begin{pmatrix} A[I, I] & A[I, j] \\ A[i, I] & A_{ij} \end{pmatrix} = \det A[I, I] (A_{ij} - A[i, I]A[I, I]^{-1}A[I, j])$$

proves (B.1), including after zero rows are skipped. The specified ordering prevents a spurious permutation sign.

A minor of order $k \leq q$ has determinant modulus at most $k^{k/2}(\sqrt{2}2^h)^k$, by the product of its row norms. Its real and imaginary parts are integers with $O(k(h + \log(k + 1)))$ bits. The principal denominator is a positive integer. Cancelling common factors cannot enlarge these lengths. Each arithmetic operation on reduced Gaussian-rational entries thus has polynomial operand size. There are polynomially many operations and at most q^2 stored entries; ordinary multiplication, division and the Euclidean algorithm give the stated bit bounds. For rational rather than integral input, first clear a common positive denominator. Its bit length is at most the sum of the encoded denominator lengths, so this preliminary step also has polynomial cost. $\square$

APPENDIX C. INTRINSIC DESCRIPTION ENTROPY OF INSTRUMENTS

We now determine the high-resolution description exponent, rather than only an upper bound on the number of stored matrix entries. The objects being described remain genuine quantum instruments. A binary description of such an object is not a physical classical implementation of it.

Fix positive integers d, n, m , put $q = dn$, and use the input-first, unnormalized Choi convention of Section A. Let

$$\mathcal{C} = \mathcal{C}_{d,n,m} = \{(J_y)_{y=1}^m : J_y \in \text{Herm}(q), J_y \geq 0, \sum_y \text{Tr}_n J_y = I_d\}.$$

The corresponding outcome-retaining channel is denoted by $\mathcal{F}_J$. On tuples of Hermitian matrices write

$$\|X\|_2 = \left(\sum_y \|X_y\|_F^2 \right)^{1/2}, \quad \|X\|_\diamond = \|\mathcal{F}_X\|_\diamond.$$

Here $\mathcal{F}_X$ is the linear map defined by the Choi blocks, whether or not they are positive. Distances in this section are unhalved norms. The instrument diamond norm means exactly the diamond norm of this flagged direct-sum channel, not the sum of the individual outcome-map diamond norms. All m labeled outcome slots are part of the body. A legal η -description code is a finite set of codewords with one member of $\mathcal{C}$ assigned to each codeword, such that every target is within diamond distance η of a decoded member. Each codeword selects one fixed memoryless instrument, not a target-dependent random mixture with uncharged mixing probabilities. A rational code requires the decoded matrices to have rational real and imaginary parts. Dimensions, the outcome names and the requested tolerance are fixed public parameters in the code-length statements. Their encodings are additional data when they vary.

Lemma C.1 (Affine dimension and an interior ball). *The affine hull of $\mathcal{C}$ has real dimension*

$$(C.1) \quad s = mq^2 - d^2 = d^2(mn^2 - 1).$$

Let $V = \{X : \sum_y \text{Tr}_n X_y = 0\}$ and $J_y^\circ = I_q/(mn)$. For $X \in V$,

$$(C.2) \quad |X|_2 \leq \sum_y \|X_y\|_1 \leq d\|X\|_\diamond, \quad \|X\|_\diamond \leq \sqrt{mq} |X|_2.$$

The $\|\cdot\|_\diamond$ -ball in V of radius $r_0 = (dmn)^{-1}$, translated by J° , is contained in $\mathcal{C}$. Moreover $\mathcal{C} - J^\circ$ lies in the diamond ball of radius two.

Proof. The map from $\text{Herm}(q)^m$ to $\text{Herm}(d)$ given by summing output partial traces is surjective: an arbitrary $H \in \text{Herm}(d)$ is the image of the tuple with first block $H \otimes |e_1\rangle\langle e_1|$ and other blocks zero. Its kernel has codimension d^2 . All blocks of J° are positive definite, so small perturbations in that kernel remain legal. This proves (C.1) and that V is the translation space of the full affine hull, not merely of a containing affine space.

The first two inequalities follow from Lemma A.1 and the direct-sum Choi matrix. The last follows from that lemma, $\|X_y\|_1 \leq \sqrt{q}\|X_y\|_F$, and Cauchy–Schwarz over y . If $\|X\|_\diamond \leq r_0$, every block has operator norm at most $|X|_2 \leq 1/(mn)$, which proves positivity of $J^\circ + X$. The affine constraint is unchanged. Finally a channel has diamond norm one, so the distance between any two outcome-retaining channels is at most two. This also follows from complete trace-norm contractivity of channels, as in [1, Corollary 3.40]. $\square$

The lower Choi estimate from Section A thus has a specific converse use: it transfers a diamond ball into the positive Choi body. Neither this step nor the dimension count uses a spectral gap.

Theorem C.2 (Sharp high-resolution description exponent). *Suppose $s > 0$. Let $\mathcal{N}_\diamond(\eta)$ and $\mathcal{N}_\diamond^\mathbb{Q}(\eta)$ be the smallest cardinalities of legal and rational legal η -description codes for $\mathcal{C}$. For $0 < \eta \leq 1$,*

$$(C.3) \quad \max\{1, (r_0/\eta)^s\} \leq \mathcal{N}_\diamond(\eta) \leq \mathcal{N}_\diamond^\mathbb{Q}(\eta) \leq (1 + 8/\eta)^s.$$

In particular, for fixed d, n, m , both optimal fixed-length binary code lengths are

$$(C.4) \quad s \log_2(1/\eta) + O_{d,n,m}(1) \quad (\eta \downarrow 0).$$

When $s = 0$, $m = n = 1$, the unique instrument is the trace map and the code length is zero.

Proof. Equip V with Lebesgue measure and the diamond norm restricted to V . By Lemma C.1, the target set contains a ball of radius r_0 . A union of M balls of radius η covering this ball has volume at most $M\eta^s$ times the unit-ball volume. Hence $M \geq (r_0/\eta)^s$. Centres anywhere in the affine hull give the same lower bound.

For the upper bound, take a maximal set of points of $\mathcal{C}$ whose mutual distances are greater than $\eta/2$. Balls of radius $\eta/4$ about them are disjoint up to null boundaries and lie in the ball of radius $2 + \eta/4$ about J° . The number of points is therefore at most $(1 + 8/\eta)^s$, and maximality

makes them an $\eta/2$ -net. Rational legal instruments are dense, by the explicit construction in Theorem C.3 below. Replace each centre by a rational legal point within $\eta/2$. This yields a rational η -net of the same cardinality. This dependence is not circular: the construction below is proved directly from the Choi constraints.

A codebook with M words can be indexed by $\lceil \log_2 M \rceil$ bits, and a code of at most b fixed-length bits has at most 2^b words. Taking logarithms in (C.3) proves (C.4). The singleton case follows directly from the constraint $J_1 = I_d$. $\square$

This is an elementary volumetric theorem for the precise legal Choi body and norm. The volume comparison is not a new general covering principle. The coefficient s is sharp in the high-resolution limit with dimensions fixed; the estimate does not assert optimal constants in a simultaneous dimension-accuracy limit.

C.1. An intrinsic rational coordinate code. Fix an outcome y_* and an output basis index α_* . The independent real coordinates of a Hermitian matrix are its real diagonal entries and the real and imaginary parts of its strictly upper-triangular entries. Omit all d^2 such coordinates belonging to the submatrix

$$((J_{y_*})_{i\alpha_*,j\alpha_*})_{1 \leq i,j \leq d}.$$

There are exactly s remaining real coordinates, in a fixed order. Every coordinate of a legal target has absolute value at most one: positivity and the marginal identity bound each diagonal by one, and $|J_{uv}|^2 \leq J_{uu}J_{vv}$ bounds the off-diagonal entries.

For $B \in \mathbb{N}$, truncate the s retained coordinates toward zero on the grid $B^{-1}\mathbb{Z}$, producing digits $z_1, \dots, z_s \in [-B, B] \cap \mathbb{Z}$. Fill the omitted submatrix by the exact marginal identity:

$$(C.5) \quad (Q_{y_*})_{i\alpha_*,j\alpha_*} = \delta_{ij} - \sum_{(y,\alpha) \neq (y_*,\alpha_*)} (Q_y)_{i\alpha,j\alpha}.$$

All summands on the right were retained. Lower-triangular entries are Hermitian copies, not additional digits.

Theorem C.3 (Exact intrinsic codec). *Let*

$$(C.6) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad K = 3mndc.$$

The matrices

$$(C.7) \quad \hat{J}_y = (BQ_y + cI_q)/D$$

are positive semidefinite with Gaussian-integer numerators, satisfy $\sum_y \text{Tr}_n \hat{J}_y = I_d$, and obey

$$(C.8) \quad \|\mathcal{F}_J - \mathcal{F}_{\hat{J}}\|_{\diamond} \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/D \leq K/B.$$

They can be encoded by one integer in $[0, (2B + 1)^s)$, using at most $\lceil \log_2((2B + 1)^s) \rceil$ bits. Thus $B = \lceil K/\eta \rceil$ achieves the leading coefficient s in (C.4) by a direct coordinate code, without enumerating a net. For rational input, encoding, decoding and validity testing have polynomial bit complexity in the explicit input and output lengths. This is not a workspace bound for a trajectory simulator.

Proof. Non-omitted independent-coordinate errors are at most $1/B$. An omitted coordinate is reconstructed from at most $mn - 1$ errors, so its error is at most $(mn - 1)/B$. In particular every independent real error is at most mn/B . Counting the Hermitian copies gives the safe estimate

$$\|Q_y - J_y\|_{\text{op}} \leq \|Q_y - J_y\|_F \leq 2qmn/B \leq c/B.$$

The affine constraint is exact by (C.5). Therefore $BQ_y + cI_q \geq 0$ and its summed output partial trace is $(B + mnc)I_d$. With $E_y = Q_y - J_y$,

$$D(\hat{J}_y - J_y) = BE_y + cI_q - mncJ_y.$$

The sum of the trace norms of each of the three terms is bounded by $mndc$, using $\sum_y \text{tr } J_y = d$ for the last term. Lemma A.1 proves (C.8).

Set $M = 2B + 1$. The integer with base- M digits $z_j + B$ determines all retained coordinates, and (C.5) then determines the omitted coordinates. A fixed number of digits is used, including leading zeros. Only integers whose reconstructed, buffered blocks are positive are valid codewords. They can be recognized by the exact Schur test in Lemma B.1. Every code produced from a legal target is valid by the preceding proof. If a total decoder is required, assign each invalid word the canonical rational instrument J° ; unused binary strings can be treated in the same way. The valid codebook already covers all targets. In the singleton case the construction decodes the trace map from the empty string.

For rational data the coordinate digits are obtained by exact signed integer division. Matrix filling, buffering and base conversion are finite integer operations with polynomial operand lengths. Exact PSD validation has polynomial cost by Lemma B.1. No eigenvector computation, rational Kraus factorization, random draw, statistical sample or floating-point decision is involved. For an arbitrary real target, the coordinate construction proves existence and density; it is not an algorithm for recovering unknown real input from finitely many unverified measurements. $\square$

Remark C.4 (Known zero outcomes and public headers). If a specified set of outcomes is known to be zero, remove it before coding and restore it afterwards. With $m' \geq 1$ active outcomes the intrinsic dimension is $d^2(m'n^2 - 1)$, and m' replaces m in the constants. If the active set is not a fixed public parameter, its mask must also be encoded. The generic construction does not preserve zeros without this operation. The choice of (y_*, α_*) affects neither the dimension nor the bounds. Headers for dimensions, B , outcomes and the chosen coordinate convention are not part of a fixed-parameter codeword; a self-contained file carries and charges them separately.

APPENDIX D. DESCRIPTION ENTROPY UNDER REPEATED ADAPTIVE USE

The number of calls changes the precision needed in a reusable instrument description. On a strictly positive Choi body the relevant scale is $N^{-1/2}$, not the N^{-1} scale supplied by a general one-slot hybrid. We prove a finite, multiparameter coding statement against arbitrary common adaptive quantum testers.

For two instruments $J, L \in \mathcal{C}_{d,n,m}$, define

$$(D.1) \quad d_N(J, L) = \sup_{\mathsf{T}} \|\Omega_J^{\mathsf{T}} - \Omega_L^{\mathsf{T}}\|_1.$$

Here T is the same causal tester in both experiments, uses the same memoryless instrument at most N times, and may have arbitrary finite-dimensional quantum memory, initially entangled references, intermediate quantum channels and measurements, and classical feedback. Fresh ancillary systems of arbitrary finite dimension are allowed: each tester has finite-dimensional registers at every slot, but no common dimension bound is imposed in the supremum. Outcomes and final memory are retained. A public stopping time bounded by N is allowed. A changed input dimension can be handled by a tester channel between calls. There is no coherent superposition of command ports, no tester informed of the hidden choice J versus L , and no stopping based on inaccessible simulator state. Taking suprema of trace distances gives the triangle inequality. In particular $d_N \leq 2$. Public stopping is included by an absorbing flag, as in Theorem A.4.

For a fixed a with $0 < a < 1/(mn)$ let

$$\mathcal{C}_a = \{J \in \mathcal{C} : J_y \geq aI_{dn} \text{ for every } y\}.$$

This is a Choi eigenvalue margin, not a spectral gap for an action or a mixing hypothesis on trajectories. Its affine dimension is the same $s = d^2(mn^2 - 1)$ as before.

Lemma D.1 (Two-point classical programme bound). *Suppose $J_y, L_y \geq \lambda I_{dn}$ for every outcome and $r = \|J - L\|_2 \leq \lambda$, where $\lambda > 0$. Then, for every $N \geq 1$,*

$$(D.2) \quad d_N(J, L) \leq \min\{2, 2\sqrt{N}r/\lambda\}.$$

The same bound covers arbitrary initially entangled references and bounded public stopping in (D.1).

Proof. There is nothing to prove for $r = 0$. Put $K = (J + L)/2$ and $U = (J - L)/r$. Since $U \in V$ and each $\|U_y\|_{\text{op}} \leq 1$,

$$K^+ = K + \lambda U, \quad K^- = K - \lambda U$$

are legal instruments. With

$$p_{\pm} = \frac{1}{2} \pm \frac{r}{4\lambda}, \quad q_{\pm} = \frac{1}{2} \mp \frac{r}{4\lambda},$$

we have $J = p_+K^+ + p_-K^-$ and $L = q_+K^+ + q_-K^-$. All four probabilities are at least $1/4$. The elementary inequality $\log x \leq x - 1$ implies

$$D(p\|q) \leq \sum_{b \in \{+, -\}} \frac{(p_b - q_b)^2}{q_b} \leq \frac{2r^2}{\lambda^2}.$$

Choose the programme signs independently in advance for all N slots. Conditional on a sign string, a fixed tester acts on exactly the same sequence of genuine quantum instruments in the two experiments. It therefore defines one channel from the classical sign string to the final quantum-classical state. This remains true with adaptive feedback: conditioning fixes the instruments, not the tester's outcomes. If the tester stops early, the unused signs are ignored. Contractivity and the classical product identity give

$$\|\Omega_J^T - \Omega_L^T\|_1 \leq 2 \text{TV}(p^{\otimes N}, q^{\otimes N}) \leq \sqrt{2ND(p\|q)} \leq 2\sqrt{N}r/\lambda.$$

The middle bound is the usual Pinsker inequality. To recall an elementary proof, coarse-grain two distributions to the set where the first is larger. The log-sum inequality reduces relative entropy to binary relative entropy. Its second derivative in the first argument is $1/[x(1-x)] \geq 4$, so integration about its minimum proves $D \geq 2 \text{TV}^2$. Taking the supremum over testers proves the assertion. $\square$

The classical-programme idea is an established method in quantum metrology; Demkowicz-Dobrzański, Kołodyński and Guţă [7, Equations (4)–(6), Methods] use parameter-dependent mixtures of fixed channels to bound distinguishability. The lemma above gives the finite two-point trace-norm estimate needed here, including common adaptive testers and public stopping. It is not a claim that the mixture method or the square-root scale is a new metrological principle. The operations $K^{\pm}$ remain quantum operations, not classical physical substitutes for unknown input.

Lemma D.2 (Repeated Choi tests). *For any two instruments $J, L \in \mathcal{C}$, with $t = \|J - L\|_2$,*

$$(D.3) \quad d_N(J, L) \geq 2 - 4 \exp\left(-\frac{Nt^2}{8d^2}\right).$$

A negative right-hand side is simply a vacuous lower bound.

Proof. At each call supply half of a normalized maximally entangled state and retain its other half and the outcome. The resulting density matrices are $A = d^{-1} \oplus_y J_y$ and $B = d^{-1} \oplus_y L_y$. They

have trace one. The projector on the positive part of $A - B$ gives a binary measurement whose probability gap is

$$g = \frac{1}{2} \|A - B\|_1 \geq t/(2d).$$

The same binary measurement is made on each of the N independent outputs. With a threshold halfway between the two means, each wrong-side probability is at most $\exp(-Ng^2/2)$. Indeed for a Bernoulli variable Z , the second derivative of $\log \mathbb{E} e^{u(Z - \mathbb{E}Z)}$ is the variance of a tilted Bernoulli variable and is at most $1/4$. The value and first derivative at zero vanish; integration and Chernoff's inequality give $\Pr\{N^{-1} \sum Z_i - \mathbb{E}Z \geq v\} \leq e^{-2Nv^2}$, and likewise for the other tail. Set $v = g/2$. The total variation distance between the two product experiments is at least $1 - 2e^{-Ng^2/2}$; the unhalved trace norm of their classical outputs is twice this. This particular nonadaptive tester is allowed in (D.1), and the result follows. The tester may depend on the pair J, L , as is permitted when lower-bounding a metric; it is the same tester for the two members of that pair. $\square$

Theorem D.3 (Sharp horizon exponent for interior descriptions). *Fix d, n, m with $s > 0$, a margin $0 < a < 1/(mn)$ and a tolerance $0 < \delta < 1$. Let $\mathcal{M}_N(\delta; a)$ be the minimum number of legal decoded instruments whose d_N -balls of radius δ cover $\mathcal{C}_a$. A decoded instrument need not itself have margin a . Write $\mathcal{M}_N^{\mathbb{Q}}$ when rational decoded instruments are required. Set*

$$R = \frac{1}{mn} - a, \quad c_\delta = \sqrt{8d^2 \log \frac{4}{1-\delta}}, \quad B_N = \left\lceil \frac{K}{a} \left(2 + \frac{4\lceil \sqrt{N} \rceil}{\delta} \right) \right\rceil,$$

where K is from (C.6). Then

$$(D.4) \quad \max\{1, (R\sqrt{N}/c_\delta)^s\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s.$$

Consequently the minimum fixed-length description satisfies

$$(D.5) \quad \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) = \frac{s}{2} \log_2 N + O_{d,n,m,a,\delta}(1),$$

and the same formula holds without the rationality restriction. The upper bound is achieved by the exact intrinsic codec applied once at grid B_N ; its decoded instrument is reused at every call.

Proof. The tuple-Frobenius ball in V of radius R about J° is contained in $\mathcal{C}_a$. Choose a maximal set of points in that ball with pairwise distances greater than $c_\delta/\sqrt{N}$. Balls of that radius about the selected points cover the original ball. Euclidean volume comparison therefore gives at least $(R\sqrt{N}/c_\delta)^s$ points. By Lemma D.2, their pairwise d_N distances are greater than

$$2 - 4e^{-c_\delta^2/(8d^2)} = 1 + \delta > 2\delta.$$

One d_N -ball of radius δ cannot cover two of these points, regardless of the location of its legal centre. This proves the lower bound.

For the upper bound use Theorem C.3. It gives

$$r = |J - \hat{J}|_2 \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/B_N \leq \min\{a/2, a\delta/(4\sqrt{N})\}.$$

Every block of $\hat{J}$ is consequently at least $(a/2)I$, and both J and $\hat{J}$ satisfy Lemma D.1 with $\lambda = a/2$ and $r \leq \lambda$. Its conclusion gives $d_N(J, \hat{J}) \leq 4\sqrt{N}r/a \leq \delta$. There are at most $(2B_N + 1)^s$ possible digit strings. All codewords arising from targets in $\mathcal{C}_a$ decode legally. The remaining codewords may be rejected or assigned J° ; this does not affect the cover. Since $B_N = O_{d,n,m,a,\delta}(\sqrt{N})$, taking logarithms in (D.4) proves (D.5). $\square$

Remark D.4 (What is charged). The lower bound concerns the number of possible reusable mathematical instrument descriptions, not the mutable memory of a simulator that receives and stores the entire instrument as input. It applies to all full-dimensional strictly positive instrument

families above, without an expanding unitary subsystem. It therefore complements, rather than replaces, the hypothesis-sensitive streaming-space lower bounds. For variable a, δ, d, n, m , the displayed finite inequalities remain valid; (D.5) holds with those parameters fixed. The estimates of this section alone assert neither an optimal uniform dimension constant nor a complete joint N - δ asymptotic. The latter is supplied, for fixed dimensions and margin, by Theorem E.3. The rational implementation takes rational a, δ and an asserted margin, which it can verify exactly. No samples of an unknown channel are supplied or required.

Proposition D.5 (The boundary is quantitatively different). *There is no constant C such that $d_N(J, L) \leq C\sqrt{N}|J - L|_2$ holds for every pair of qubit channels and every N . In particular the strict margin in Theorem D.3 cannot be removed from its proof by setting $a = 0$.*

Proof. Let $U_\theta = \text{diag}(1, e^{i\theta})$ and compare its unitary channel to the identity channel. For their unnormalized Choi matrices,

$$|J(U_\theta) - J(I)|_2 = 2\sqrt{2}|\sin(\theta/2)| \leq \sqrt{2}|\theta|.$$

At $\theta = \pi/N$, sequentially applying the unknown channel N times to $|+\rangle$ gives respectively $|-\rangle$ and $|+\rangle$. The final trace distance is two. The proposed upper bound would tend to zero. The general additive diamond hybrid remains valid at this boundary; the proposition asserts no global covering exponent for boundary or rank-constrained instrument classes. $\square$

D.1. Description, learning and physical simulation. A reusable rational description is different from a protocol that replaces quantum communication by classical messages. Naik, Zartab, Gisin and Banik [4, Theorems 2 and 5] show a finite-classical-message obstruction for the perfect qubit channel in a reference-sensitive joint-measurement model, and finite-resource possibility for noisy depolarizing channels. Their receiver retains a quantum system; the task is physical simulation. Here the decoded objects and the vertices in Lemma D.1 are genuine quantum instruments. Our numerical algorithms instead start from specified matrices. Neither construction supplies a classical physical device accepting unknown entangled input.

Unknown-channel learning charges channel queries used to infer a classical description. Mele and Bittel [5, Theorem I.1] prove dimension-optimal query complexity at fixed diamond accuracy; Oufkir and Girardi [6] give further accuracy-dependent query lower bounds. The input here is already specified, and the lower bound in (D.4) counts descriptions, not the queries needed to select one from unknown data. The elementary packing and testing ingredients are shared methods, not a priority claim that those learning theorems are consequences of this code.

APPENDIX E. JOINT HORIZON AND ACCURACY IN DESCRIPTION LENGTH

The exponential test in Lemma D.2 resolves pairs at fixed error, but its lower bound is vacuous when the two experiments are very close. A bounded statistic gives the local estimate needed to retain the accuracy parameter. Throughout this section $N \geq 1$ and the adaptive distance d_N is the unhalved final-state trace distance maximized over one common tester, as in (D.1). The unknown device is the same memoryless instrument at every call; commands are classical, and stopping is public and bounded by N .

Lemma E.1 (A local binary test). *For $0 \leq p, q \leq 1$,*

$$(E.1) \quad \text{TV}(\text{Bern}(p)^{\otimes N}, \text{Bern}(q)^{\otimes N}) \geq \frac{3}{32} \min\{\sqrt{N}|p - q|, 1\}.$$

The estimate is uniform up to the endpoints of the probability interval.

Proof. Assume $p < q$ and first suppose $q - p \leq N^{-1/2}$. Put $b = N(p + q)/2$ and let $f : \mathbb{R} \rightarrow [0, 1]$ be the nondecreasing function which is affine with slope $(8\sqrt{N})^{-1}$ on $[b - 4\sqrt{N}, b + 4\sqrt{N}]$, zero

to its left, and one to its right. For $S \sim \text{Bin}(N, t)$, differentiating the finite binomial polynomial gives the exact identity

$$\frac{d}{dt} \mathbb{E}_t f(S) = N \mathbb{E}_t \{f(T+1) - f(T)\}, \quad T \sim \text{Bin}(N-1, t).$$

It holds on $(0, 1)$ and extends continuously to both endpoints. For $t \in [p, q]$, the distance from $\mathbb{E}_t T$ to b is at most $\sqrt{N}/2 + 1$. Also $\text{Var}_t T \leq (N-1)/4$. Chebyshev's inequality gives

$$\Pr_t\{|T - \mathbb{E}_t T| \leq \sqrt{N}\} \geq 3/4.$$

On this event, both T and $T+1$ belong to the affine interval: their distance to b is at most $3\sqrt{N}/2 + 2 \leq 4\sqrt{N}$. Since f is nondecreasing everywhere,

$$\frac{d}{dt} \mathbb{E}_t f(S) \geq \frac{3}{32} \sqrt{N}.$$

Integration from p to q , and the characterization of total variation by $[0, 1]$ -valued statistics, prove the assertion in this case. The statistic depends only on the count, which is a function of the full binary string.

If $q - p > N^{-1/2}$, apply the same construction to p and $q_0 = p + N^{-1/2} < q$. Couple Bernoulli samples by common uniform random variables. Monotonicity of f then gives $\mathbb{E}_q f(S) \geq \mathbb{E}_{q_0} f(S)$, and the preceding lower bound is $3/32$. The cases $p = q$ and $q < p$ follow by equality and symmetry. $\square$

The proof uses only binomial differentiation and a bounded statistic; it requires neither a normal approximation nor a lower variance bound. In particular its constant does not deteriorate for a rare outcome.

Corollary E.2 (Local repeated Choi separation). *For the instrument body $\mathcal{C}_{d,n,m}$, let $t = |J - L|_2$. Then*

$$(E.2) \quad d_N(J, L) \geq \frac{3}{16} \min \left\{ \frac{\sqrt{N}}{2d} t, 1 \right\}.$$

Proof. Use the same normalized Choi inputs as in Lemma D.2. The positive spectral projection of $d^{-1} \oplus_y (J_y - L_y)$ defines a binary test with probability gap at least $t/(2d)$. Repeating this test independently and applying Lemma E.1 gives (E.2); classical trace distance is twice total variation. This is one admissible nonadaptive tester. For each pair it is the same tester in both experiments, and it is allowed in the supremum defining d_N . $\square$

Theorem E.3 (The joint interior coding law). *Fix d, n, m with $s = d^2(mn^2 - 1) > 0$ and $0 < a < 1/(mn)$. For all $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(E.3) \quad \max \left\{ 1, \left(\frac{R\sqrt{N}}{32d\delta} \right)^s \right\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s,$$

where $R = 1/(mn) - a$ and B_N is the explicit grid in Theorem D.3. Consequently the optimal legal fixed-length description, with or without rational decoded matrices, has length

$$(E.4) \quad \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O_{d,n,m,a}(1),$$

uniformly on this whole range of N and δ . The decoded instrument is reused unchanged, and no spectral-gap hypothesis occurs.

Proof. The tuple-Frobenius ball of radius R about J° in V is contained in $\mathcal{C}_a$. Take a maximal set in this ball whose pairwise Euclidean distances are greater than $h = 32d\delta/\sqrt{N}$. Maximality and s -dimensional volume give at least $(R/h)^s$ points. Corollary E.2 separates each pair by more than

2δ in d_N : before the minimum saturates its value is greater than 3δ , and after saturation it is $3/16 > 2\delta$. Thus no legal centre of a d_N -ball of radius δ can cover two packing points, even if that centre has no positive eigenvalue margin. This proves the lower bound.

The intrinsic rational codec and its classical-programme estimate in Theorem D.3 give the upper bound with the same B_N . Since $N \geq 1$ and $\delta \leq 1/32$, there is a constant $C = C(d, n, m, a)$ such that $2B_N + 1 \leq C\sqrt{N}/\delta$. The lower bound and this inequality give (E.4), including rounding a logarithm up to a fixed-length binary code. When the volumetric lower expression is less than one, its logarithm is negative; the nonnegative code length still gives the same uniform additive estimate. Neither bound charges the queries of a learning procedure. $\square$

The earlier fixed-error theorem is retained. The new assertion is the uniform additive remainder in (E.4), which is independent of both the horizon and the accuracy. It does not assert sharp dimension-dependent constants, nor is it a lower bound for the mutable workspace of the variable-description trajectory algorithm.

APPENDIX F. BOUNDARY-UNIFORM CODING OF PREPARATION INSTRUMENTS

Strict positivity is essential to the two-point programme argument, as Proposition D.5 shows. Nevertheless it is not necessary for a complete coding law in an irreversible family. We consider all outcome-retaining instruments that erase their input and prepare a specified quantum–classical state. This family contains zero outcomes and arbitrary rank-deficient output blocks.

For $m, n \geq 1$ put

$$\mathcal{S}_{m,n} = \{(\sigma_y)_{y=1}^m : \sigma_y \in \text{Herm}(n), \sigma_y \geq 0, \sum_y \text{tr} \sigma_y = 1\}.$$

Write $\sigma = \bigoplus_y \sigma_y$. For any input dimension $d \geq 1$, the associated instrument is

$$(F.1) \quad \mathcal{R}_{\sigma,y}(X) = \text{tr}(X)\sigma_y, \quad J_y(\mathcal{R}_\sigma) = I_d \otimes \sigma_y.$$

The output label y is retained. Its conditional quantum state is $\sigma_y/\text{tr} \sigma_y$ when that trace is positive. This is a specified mathematical quantum instrument; a description of it is not a numerical algorithm acting on an unknown physical input.

Lemma F.1 (Reduction to a product-state experiment). *For arbitrary $\sigma, \tau \in \mathcal{S}_{m,n}$,*

$$(F.2) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_\tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

Here the tester may have arbitrary quantum memory, initial reference entanglement, feedback and a public stopping time at most N .

Proof. Supply N independent program states σ to one fixed processor. At the next device call the processor discards the incoming device system and returns the next program state, including its classical outcome. Between calls it performs the prescribed tester operations. This reproduces the experiment exactly, even when the incoming system is entangled with the tester memory: the incoming system is traced out in (F.1). Unused program states are ignored after a public stop. The processor is the same channel for σ and τ , so trace-norm contraction gives the upper bound. Conversely, a tester can make exactly N calls and retain every outcome and quantum output. Its final state is the displayed product, which proves equality. The processor is a quantum channel, not a claimed finite-classical-message physical simulator. $\square$

F.1. Triangular factors and a rational code. Every $\sigma \in \mathcal{S}_{m,n}$ has triangular factors L_y with $\sigma_y = L_y L_y^*$ and nonnegative real diagonal entries. The usual no-pivot Cholesky procedure also covers singular matrices: a zero diagonal pivot forces the entire residual row and column to vanish and contributes a zero factor column. A block of rank r therefore has exactly r positive pivots. All factors together satisfy $\sum_y \|L_y\|_F^2 = 1$.

Fix public bounds $0 \leq r_y \leq n$, not all zero, and write

$$(F.3) \quad D = \sum_{y=1}^m r_y(2n - r_y), \quad v = D - 1, \quad \mathcal{S}_r = \{\sigma \in \mathcal{S}_{m,n} : \text{rank } \sigma_y \leq r_y\}.$$

If the positive pivot positions of L_y are $T_y \subseteq \{1, \dots, n\}$, record only the real diagonal and the real and imaginary coordinates strictly below each such pivot. Their total number is

$$h = \sum_y \sum_{j \in T_y} \{1 + 2(n - j)\} \leq D.$$

The resulting real coordinate vector $x \in \mathbb{R}^h$ has norm one. The inequality follows by placing each block's pivots as early as possible. It holds also when the actual ranks are smaller than the public bounds.

Choose a coordinate j of maximal absolute value, put $a = |x_j|$, and set $w = x/a$. For an integer $B \geq 1$, take

$$(F.4) \quad z_j = B \operatorname{sgn}(x_j), \quad z_i = \operatorname{sgn}(x_i) \lfloor B|w_i| \rfloor \quad (i \neq j).$$

Insert these integers in the recorded factor positions to obtain Gaussian-integer triangular matrices Z_y ; all unrecorded columns are zero. Define

$$(F.5) \quad T = \sum_y \|Z_y\|_F^2, \quad \hat{\sigma}_y = Z_y Z_y^* / T.$$

Since $|z_j| = B$, the denominator satisfies $B^2 \leq T \leq hB^2$.

Theorem F.2 (A boundary-uniform rational factor code). *The output (F.5) lies in $\mathcal{S}_r$, preserves every zero target outcome, and does not increase the rank of any individual block. For all $N \geq 1$,*

$$(F.6) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_{\hat{\sigma}}) \leq \min \left\{ 2, \frac{4\sqrt{N(h-1)}}{B} \right\} \leq \min \left\{ 2, \frac{4\sqrt{Nv}}{B} \right\}.$$

At fixed m, n, r, B a fixed-length code needs at most

$$(F.7) \quad mn + \lceil \log_2(2D(2B+1)^v) \rceil$$

bits. For rational target blocks, its encoder and decoder use exact rational and integer arithmetic, including integer square roots; no algebraic-number oracle, positive eigenvalue margin, or Kraus factorization is required.

Proof. The product $Z_y Z_y^*$ is positive semidefinite, and normalization makes the sum of its traces exactly one. Its number of possibly nonzero columns is no larger than that of L_y , so its rank does not exceed the target rank. When an outcome is zero it has no recorded columns and remains exactly zero.

Put $z' = z/B$. Its j th coordinate agrees with that of w ; every other coordinate differs by less than $1/B$. Both $\|w\|$ and $\|z'\|$ are at least one. The elementary normalization estimate gives

$$(F.8) \quad \left\| x - \frac{z}{\|z\|} \right\| \leq \frac{2\|w - z'\|}{\|w\|} \leq \frac{2\sqrt{h-1}}{B}.$$

Use the normalized vectors

$$|L\rangle = \sum_{y,i,j} (L_y)_{ij} |y, i\rangle |y, j\rangle, \quad |\hat{L}\rangle = T^{-1/2} \sum_{y,i,j} (Z_y)_{ij} |y, i\rangle |y, j\rangle.$$

Their reduced states on the first factor are σ and $\hat{\sigma}$. Their vector distance is the left side of (F.8). For any unit vectors u, v , diagonalization on their two-dimensional span gives

$$\| |u\rangle\langle u|^{\otimes N} - |v\rangle\langle v|^{\otimes N} \|_1 = 2\sqrt{1 - |\langle u, v \rangle|^{2N}} \leq 2\sqrt{N} \|u - v\|.$$

Indeed $1 - t^N \leq N(1 - t)$ on $[0, 1]$ and $1 - |\langle u, v \rangle|^2 \leq 2(1 - \Re \langle u, v \rangle)$. Taking partial traces and using Lemma F.1 proves (F.6). These are standard purification and product-fidelity estimates; see [1, Section 3.2]. Their short proof specifies the normalization used here.

The pivot sets use mn bits. Given them, there are at most $2h$ anchor/sign choices and $(2B+1)^{h-1}$ remaining digit strings. Pad their index to the maximum length in (F.7). An invalid header may be rejected, or sent by a total mathematical decoder to a fixed legal state. Valid factor words always decode positively, even if they did not arise from this encoder.

For rational input, maintain the exact rational Schur residual A . At a positive pivot $u = A_{jj}$, the factor diagonal is $\sqrt{u}$ and a lower entry is $A_{ij}/\sqrt{u}$. Thus the square of each of its real coordinates is rational, respectively u , or $(\Re A_{ij})^2/u$, or $(\Im A_{ij})^2/u$; its sign is known. Updating the residual uses rational arithmetic only. Choose the largest squared coordinate $b_j > 0$ by exact comparisons. If another coordinate has squared value b_i , then

$$\lfloor B|w_i| \rfloor = \text{isqrt}\left(\left\lfloor B^2 b_i/b_j \right\rfloor\right).$$

Consequently no square root of a rational number must be represented. The ratios-of-minors bounds in Lemma B.1 give polynomial operand lengths during this preprocessing. Integer square root, products and mixed-radix coding have polynomial bit complexity in the explicit input and output lengths. Encoding workspace and the expanded decoded matrix are not identified with the compressed payload. $\square$

F.2. Sharp description laws on the boundary. Let $\mathcal{P}_N(\delta; \mathbf{r})$ be the minimum number of legal preparation instruments whose d_N -balls of radius δ cover $\mathcal{S}_{\mathbf{r}}$; the decoded centre may have any output rank. Add a superscript $\mathbb{Q}$ when rational centres are required. For an upper construction our centres will obey the same rank bounds.

Theorem F.3 (Rank-stratified joint coding law). *Fix $m, n, \mathbf{r}$ as in (F.3). If $v > 0$, there exist constants $c, C > 0$, depending only on these fixed data, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(F.9) \quad \max\{1, (c\sqrt{N}/\delta)^v\} \leq \mathcal{P}_N(\delta; \mathbf{r}) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C(\sqrt{N}/\delta)^v.$$

The optimal legal and rational fixed-length descriptions therefore have length

$$(F.10) \quad \frac{v}{2} \log_2 N + v \log_2(1/\delta) + O_{m,n,r}(1)$$

uniformly in N and δ . The statement has no spectral-gap or positive-eigenvalue-margin hypothesis. It includes every rank-deficient boundary allowed by the public bounds. If $v = 0$, the family is a singleton and its optimal description length is zero.

Proof. For the upper bound take $B = \lceil 4\sqrt{Nv}/\delta \rceil$ in Theorem F.2; for rational implementation it suffices to replace $\sqrt{Nv}$ by its integer ceiling. Its error is at most δ , and its codebook cardinality is at most $2^{mn} 2D(2B+1)^v$. This has the asserted upper order uniformly on the stated parameter range.

For completeness we construct a full-dimensional local patch for the lower bound. For every block with $r_y > 0$, take an n by r_y lower-trapezoidal factor: entries above the first r_y diagonal positions vanish, and these diagonal entries are positive real numbers. These factors have D independent real coordinates. Impose $\sum_y \|L_y\|_F^2 = 1$ and work near the point with all these diagonal entries equal to $(\sum_y r_y)^{-1/2}$ and all other entries zero. An open patch of this unit sphere is a smooth v -dimensional manifold. The map

$$(L_y)_y \mapsto (L_y L_y^*)_y$$

is one-to-one there and has a smooth inverse on its image. To see the inverse explicitly, the leading r_y by r_y principal block is positive definite and determines its unique positive-diagonal Cholesky

factor A_y . The bottom-left block is $B_y A_y^*$, so it determines B_y by multiplication by $(A_y^*)^{-1}$. These operations are smooth near the chosen point. When $r_y = 0$, the entire block is identically zero and contributes no coordinate.

Choose a closed coordinate ball of sufficiently small positive radius inside this patch. The chart and its inverse have bounded derivatives on a slightly larger neighborhood; hence distances between parameters in the ball are bounded above and below by constant multiples of the tuple-Frobenius distance between their states. More explicitly one may bound the derivative of the inverse on a convex neighborhood of the chosen positive principal blocks, where the same Cholesky formulas extend smoothly. A maximal Euclidean packing of this parameter ball at scale $A\delta/\sqrt{N}$ therefore has at least $(c\sqrt{N}/\delta)^v$ points whose state distances exceed $32\delta/\sqrt{N}$, after increasing the fixed A .

The binary argument of Corollary E.2, applied directly to the trace-one block states (input dimension one), separates each pair by more than 2δ after N copies. By Lemma F.1 the same separation holds in d_N . One radius- δ ball, regardless of the rank of its legal centre, cannot contain two packing points. This proves the lower bound and then (F.10). Finally $D = 1$ requires $n = 1$ and exactly one bound $r_y = 1$; all other bounds vanish. The sole permitted state has unit mass on that outcome, as asserted. $\square$

Corollary F.4 (All ranks, with an explicit lower constant). *For the full family $\mathcal{S}_{m,n}$ put $v = mn^2 - 1 > 0$. Then*

$$(F.11) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{64mn\delta} \right)^v \right\} \leq \mathcal{P}_N(\delta) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta) \leq 2^{mn} 2mn^2 (2B + 1)^v, \quad B = \left\lceil \frac{4\sqrt{N}v}{\delta} \right\rceil.$$

In particular a general n -dimensional prepared state has coefficient $n^2 - 1$, while the class of pure prepared states has coefficient $2n - 2$ in (F.10). A classical m -outcome law has coefficient $m - 1$, including probabilities equal to zero.

Proof. The upper bound is Theorem F.2 with every $r_y = n$. For the lower bound, the affine space of block states has dimension $mn^2 - 1$. Its tuple-Frobenius ball of radius $1/(2mn)$ about the blocks $I_n/(mn)$ lies in $\mathcal{S}_{m,n}$. Repeat the packing proof of Theorem E.3 with input dimension one. For pure states take $m = 1, r_1 = 1$ in Theorem F.3; for classical laws take $n = 1$. $\square$

Remark F.5 (Operational and priority distinctions). Input erasure is decisive in (F.2). No analogous product-state reduction has been proved here for an arbitrary disturbing instrument; coherent accumulation for unitary channels remains governed by Proposition D.5. The present theorems classify reusable description lengths in the stated preparation families, not general simulator workspace. Constants remain dimension dependent.

Quantum population compression starts with physical copies of an unknown state and compresses their quantum information. Yang, Bai, Chiribella and Hayashi [8] obtain a half-parameter-count logarithmic memory law in that different task and show a limitation on purely classical compression of noncommuting families. Here the encoder is given a classical matrix description, and each codeword specifies a state or preparation instrument. It need not infer that word from unknown copies. The common half-dimension scale, purification inequalities and finite-dimensional covering methods are classical antecedents, not claimed inventions. The contribution established here is a uniform two-parameter legal-description law with a rational, zero-outcome-preserving, rank-nonincreasing encoder and matching lower bounds throughout the declared boundary families.

APPENDIX G. RETRACTION OF COVERING CENTRES FOR PREPARATION TARGETS

The centres in a covering problem need not belong to the family being covered. For preparation targets this distinction can be removed exactly, without a small-error or rank assumption. Throughout this section a centre is one memoryless instrument, reused at every call; it is not a device with an additional persistent environment.

Theorem G.1 (Preparation-centre retraction). *Let $\mathcal{C} = (\mathcal{C}_y)_{y=1}^m$ be any instrument from M_d to M_n , and set*

$$(G.1) \quad \tau_y = \mathcal{C}_y(I_d/d) = d^{-1} \text{Tr}_{\text{in}} J_y(\mathcal{C}), \quad \mathfrak{R}\mathcal{C} = \mathcal{R}_\tau.$$

Then $\mathfrak{R}$ maps legal instruments to preparation instruments, fixes every preparation instrument, and preserves rational Choi data. For every $N \geq 1$ and every $\sigma \in \mathcal{S}_{m,n}$,

$$(G.2) \quad d_N(\mathcal{R}_\sigma, \mathfrak{R}\mathcal{C}) \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Here d_N is the unhalved trace distance optimized over the common adaptive testers of Lemma F.1. Consequently, for any subset of preparation targets and any radius $\delta > 0$, the minimum covering cardinality is unchanged when arbitrary legal memoryless centres are restricted to preparation centres. The assertion also holds when centres are required to have rational Choi entries.

Proof. Complete positivity gives $\tau_y \geq 0$, and trace preservation gives $\sum_y \text{tr} \tau_y = 1$. The formula is rational and is the identity on the preparation family, so $\mathfrak{R}^2 = \mathfrak{R}$. Feed the state I_d/d into each of N calls to $\mathcal{C}$, use a fresh input at the next call, and retain all outputs and outcome labels. The resulting state is $\tau^{\otimes N}$, since the centre is memoryless. The same tester applied to $\mathcal{R}_\sigma$ returns $\sigma^{\otimes N}$. Therefore

$$\|\sigma^{\otimes N} - \tau^{\otimes N}\|_1 \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Lemma F.1 identifies the left side with the left side of (G.2). Retracting every centre in any cover preserves coverage and cannot increase cardinality. The reverse inequality follows because preparation centres are among the legal centres. The same argument preserves rationality. $\square$

The retraction does not preserve a target rank bound: for example the identity channel retracts to the maximally mixed preparation. Thus the centre restriction in Theorem G.1 is to the full preparation family, not necessarily to the target rank stratum. The rank-constrained rational upper centres of Theorem F.2 remain a separate, stronger construction. The theorem concerns the geometry of mathematical descriptions. Its maximally mixed test is a quantum experiment, not a finite-classical-message simulation of unknown inputs.

The reduction mechanism for input-erasing channels has a direct antecedent in Cooney–Mosonyi–Wilde [10]. Their discussion of two replacer channels reduces channel divergences to state divergences; their principal results concern asymptotic discrimination and strong-converse exponents for a channel versus a replacer. Lemma F.1 uses the same input-erasure mechanism in an exact finite-use trace-norm identity, whereas Theorem G.1 concerns the allowed centres of an entire covering problem. No exact nonadaptive optimality for an arbitrary non-replacer channel against a replacer is inferred from that identity. The processor in the product identity is quantum, and input erasure, rather than entanglement breaking alone, is its essential hypothesis.

APPENDIX H. COHERENT AND PREPARATION DIRECTIONS AT A SINGULAR BOUNDARY

The preparation theorem has a square-root horizon scale. Reversible quantum information can instead accumulate a coherent discrepancy at a linear scale. We now exhibit both scales in a single non-input-erasing family, and determine their combined description cost.

Fix $d \geq 2$, $n, m \geq 1$ and public rank bounds $0 \leq r_y \leq n$, not all zero. For $U \in U(d)$ and $\sigma \in \mathcal{S}_r$ define an instrument from M_d to M_{dn} by

$$(H.1) \quad \mathcal{F}_{U,\sigma,y}(X) = UXU^* \otimes \sigma_y.$$

Its outcome register is retained. Global phase is immaterial, so the unitary parameter lies in $PU(d)$. Put

$$u = d^2 - 1, \quad v = \sum_y r_y(2n - r_y) - 1.$$

The Choi matrix of each outcome is, up to a fixed tensor permutation, $|U\rangle\rangle\langle\langle U| \otimes \sigma_y$. Its rank is $\text{rank } \sigma_y$, not $d \text{rank } \sigma_y$ as in the input-erasing family. These instruments transmit the data system unitarily, including its entanglement with a reference. Their preparation supports may change and may lie on any of the permitted singular strata. Nevertheless, the outcome probabilities do not depend on the input. This is a specified boundary family, not the whole body of disturbing instruments.

Let $\mathcal{C}_N(\delta; \mathbf{r})$ denote its covering number in d_N , with centres allowed to be any legal memoryless instrument on the same input, output and outcome spaces. Let $\mathcal{C}_N^{\mathbb{Q}}$ impose the additional condition that the centres have Gaussian-rational Choi entries. A target is supplied as U and its preparation blocks; no unknown-device learning interface is used.

Theorem H.1 (Joint coding at a coherent-preparation boundary). *There are $0 < c \leq C < \infty$, depending only on the fixed dimensions and rank bounds, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(H.2) \quad c \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v \leq \mathcal{C}_N(\delta; \mathbf{r}) \leq \mathcal{C}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v.$$

The metric is the unhalved final-state trace norm, maximized over common adaptive quantum testers using the same memoryless instrument at most N times, with arbitrary finite references, quantum memory, feedback and bounded public stopping. In particular, the optimal fixed-length reusable description has length

$$(H.3) \quad (u + v/2) \log_2 N + (u + v) \log_2(1/\delta) + O_{d,n,m,\mathbf{r}}(1).$$

The upper centres can be chosen within the family (H.1), with rational unitary matrices and rational preparation blocks. They do not increase a target block rank and preserve its zero outcomes. A finite exact encoder exists for supplied rational unitaries and rational preparation blocks. Description length is not its workspace or running time, and decoding describes a quantum instrument, not a classical physical processor for unknown entangled inputs.

H.1. The two metric scales. For $U, V \in U(d)$ write

$$(H.4) \quad \chi(U, V) = \left(1 - \frac{|\text{tr}(U^*V)|^2}{d^2} \right)^{1/2}, \quad \Delta_N(\sigma, \tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

The first expression is the chordal distance between normalized Choi rays; it is a metric after quotienting global phase.

Lemma H.2 (Anisotropic adaptive comparison). *For the instruments (H.1),*

$$(H.5) \quad \max\{\min\{N\chi(U, V)/\sqrt{2}, 1\}, \Delta_N(\sigma, \tau)\} \leq d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}),$$

$$(H.6) \quad d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}) \leq \min\{2, 2dN\chi(U, V) + \Delta_N(\sigma, \tau)\}.$$

All lower testers are permitted to depend on the pair being compared.

Proof. Discarding the data-system outputs gives the product-state lower bound. For the unitary bound, discard the preparation systems and outcome labels, and apply the same known operation U^* to the data output after each call. The two resulting data operations are I and $W = U^*V$. If $e^{i\phi_j}$ are the eigenvalues of W , then

$$\chi(U, V)^2 = \frac{2}{d^2} \sum_{i,j} \sin^2((\phi_i - \phi_j)/2).$$

There is an eigenvalue pair whose principal half-angle $\theta \in [0, \pi/2]$ satisfies $\sin \theta \geq \chi(U, V)/\sqrt{2}$. Prepare the equal superposition of its orthogonal eigenvectors. After k calls its two possible pure

states have trace distance $2|\sin(k\theta)|$. If $N\theta \leq \pi/2$, take $k = N$ and use $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. The result is at least $N \sin \theta$. If $N\theta > \pi/2$, take $k = \lceil \pi/(4\theta) \rceil \leq N$. Then $\pi/4 \leq k\theta \leq 3\pi/4$, giving distance at least $\sqrt{2}$. The case $\theta = 0$ is immediate. This proves the first minimum in (H.5); the two lower witnesses need not coincide.

For the upper bound first change U to V , keeping σ fixed. The unnormalized Choi difference of the unitary channels has trace norm $2d\chi(U, V)$ and bounds their diamond distance. Appending a fixed preparation state does not increase that norm. A slot-by-slot hybrid and trace contraction for the common tester therefore cost at most $2dN\chi(U, V)$. Next hold V fixed and change σ to τ . All calls to V and all tester operations form one common quantum processor consuming N preparation states; unused copies are discarded at a public stop. This costs at most $\Delta_N(\sigma, \tau)$, not the sum of N one-copy trace distances. Triangle inequality gives (H.6). $\square$

Finite-use amplification of unitary differences is classical; see Acín [11] and Duan–Feng–Ying [12]. The elementary estimate above is included to specify its uniform constant, its pair-dependent controls and its compatibility with bounded stopping. Neither the existence of unitary discrimination nor the real dimension $d^2 - 1$ is claimed as a new ingredient.

H.2. A finite rational atlas for unitary channels. For $\ell \geq 1$, a signed stereographic chart on S^ℓ has the form

$$(H.7) \quad S_{j,\xi}(t)_j = \xi \frac{1 - \|t\|^2}{1 + \|t\|^2}, \quad S_{j,\xi}(t)_i = \xi \frac{2t_i}{1 + \|t\|^2} \quad (i \neq j), \quad \xi \in \{-1, 1\}.$$

For a target $q \in S^\ell$, choose j of largest absolute coordinate, $\xi = \text{sgn}(q_j)$ and $t_i = \xi q_i / (1 + |q_j|)$. Then $|t_i| \leq 1$ and $S_{j,\xi}(t) = q$. Direct expansion gives

$$(H.8) \quad \|S_{j,\xi}(t) - S_{j,\xi}(t')\| = \frac{2\|t - t'\|}{\sqrt{(1 + \|t\|^2)(1 + \|t'\|^2)}}.$$

Rounding each coordinate towards zero at denominator B thus gives a rational unit vector at distance at most $2\sqrt{\ell}/B$. Every decoded chart word is on the sphere, regardless of whether it is the canonical encoding of a target.

Lemma H.3 (Rational unitary code). *Put $L = d(d-1)/2$ and $K_d = 3L + 2(d-1)$. There is an explicitly enumerable set $\mathcal{U}_B \subset U(d, \mathbb{Q}(i))$ satisfying*

$$(H.9) \quad |\mathcal{U}_B| \leq 6^L 4^{d-1} (2B+1)^{d^2-1}, \quad \sup_U \min_{V \in \mathcal{U}_B} \chi(U, V) \leq K_d/B.$$

For rational U , membership in any prescribed rational squared-error ball $\chi(U, V)^2 \leq b$ is decidable by rational arithmetic alone.

Proof. The usual complex Givens elimination, in a fixed order of index pairs, expresses every unitary as a product of L embedded matrices

$$\begin{pmatrix} c & -\bar{s} \\ s & c \end{pmatrix}, \quad c \in \mathbb{R}, \quad c^2 + |s|^2 = 1,$$

and a diagonal unitary. For completeness, a column pair (a, b) is sent to one nonzero coordinate using $c = |a|/(|a|^2 + |b|^2)^{1/2}$ and $s = cb/a$ when $a \neq 0$; the cases $a = 0$ or $(a, b) = 0$ are obtained by $c = 0$ or by the identity. Applying the adjoint rotation eliminates the second coordinate. Successive eliminations give an upper triangular unitary, necessarily diagonal. Removing the phase of its first diagonal entry leaves $d-1$ free phases. All factors are taken in the corresponding fixed reversed order when reconstructing U .

Each Givens matrix has an S^2 parameter $(c, \Re s, \Im s)$; its operator-norm change equals the Euclidean change of that vector. Each remaining phase has an S^1 parameter. Apply (H.7) at grid B to these $L + d - 1$ factors. There are 6 chart headers for each S^2 and 4 for each S^1 , with

$2L + d - 1 = d^2 - 1$ grid digits altogether. Telescoping a product of unitaries gives, up to a global phase, operator error at most

$$(2\sqrt{2}L + 2(d - 1))/B \leq K_d/B.$$

For any fixed phase z of modulus one, $\chi(U, V)^2 \leq d^{-1}\|U - zV\|_F^2 \leq \|U - zV\|_{\text{op}}^2$. This proves (H.9). The formula (H.4) has rational square for rational U, V , proving the last assertion. Enumeration of these finite words followed by the squared-distance test is an exact encoder whenever the radius is at least K_d/B . This does not assert a fast nearest-centre algorithm. $\square$

H.3. The joint covering theorem.

Proof of Theorem H.1. For the upper bound use

$$B_U = \lceil 4dK_dN/\delta \rceil, \quad B_P = \max\{1, \lceil 8\sqrt{N}v/\delta \rceil\}.$$

Lemma H.3 makes the first term in (H.6) at most $\delta/2$. The factor code in Theorem F.2 makes the second term at most $\delta/2$. Its finite pivot headers and anchor code give at most $C_{n,m,r}(2B_P + 1)^v$ legal images. If $v = 0$ the preparation family is a singleton and no such digits are needed. Multiplication of the two cardinalities proves the upper bound and all legality, rank and zero-outcome assertions. For rational data, search the finite unitary list using the rational test $\chi(U, V)^2 \leq \delta^2/(16d^2N^2)$; the preceding estimate guarantees a result. The preparation encoder is already exact. Decoded Choi entries are rational products of the two rational outputs. The finite search and the expanded matrices are not charged to the reusable payload.

For the converse, a neighborhood of the identity in $PU(d)$ contains a quantitatively bi-Lipschitz u -dimensional Euclidean patch for the metric χ . Here is a direct verification. Use traceless Hermitian coordinates H near zero and the ray of e^{iH} . The differential of its normalized Choi projection at zero vanishes only when H is scalar, hence only when $H = 0$. On a sufficiently small convex coordinate ball its differential is uniformly closer than half its least singular value to that at zero. Integration along a line segment gives both Lipschitz inequalities. Since the projection Frobenius distance is $\sqrt{2}\chi$, the asserted metric comparison follows. Volumetric packing therefore gives at least $c_d(N/\delta)^u$ unitaries separated by more than $4\delta/N$, with a smaller fixed constant covering the coarse-scale cases as well.

The maximal-rank preparation patch used in Theorem F.3 has real dimension v and is bi-Lipschitz in tuple Frobenius norm. Pack it at separation greater than $32\delta/\sqrt{N}$, obtaining at least $c_{n,m,r}(\sqrt{N}/\delta)^v$ points. The local binary bound in Corollary E.2, applied to the preparation state (input dimension one), separates their N -copy states by more than 2δ . Indeed the bound at that scale is $(3/16)\min\{16\delta, 1\}$, which is 3δ before saturation and at least $3/16 > 2\delta$ afterwards. For $v = 0$ use its single point.

Take the Cartesian product of these packings. Two elements with different unitaries are separated by (H.5), since $\min\{2\sqrt{2}\delta, 1\} > 2\delta$. Two elements with the same unitary and different preparation blocks are separated by the other lower witness. Thus a ball of radius δ about any legal centre can contain at most one product point. This proves the lower bound even for centres outside the family. Taking base-two logarithms and rounding a fixed-length word length upward changes the answer by at most a fixed constant, proving (H.3). $\square$

The coefficient $u + v/2$ reflects two different reuse scales, rather than half of the total parameter dimension $u + v$. No spectral-gap assumption or positive eigenvalue margin is used. The constants are nevertheless fixed-dimensional. Nor is this a classification of every tangent direction of an arbitrary instrument: coupling an input-dependent measurement to its disturbance is not part of (H.1).

H.4. An explicit qubit codec. For $q = (q_0, q_1, q_2, q_3) \in S^3$ put

$$(H.10) \quad U(q) = \begin{pmatrix} q_0 + iq_3 & q_2 + iq_1 \\ -q_2 + iq_1 & q_0 - iq_3 \end{pmatrix}.$$

The points q and $-q$ describe the same channel. Choose a largest absolute coordinate, flip that common sign to make it positive, and use (H.7) with $\ell = 3$ and $\xi = 1$. Its header has four possibilities, and its body has exactly three signed digits.

Corollary H.4 (Exact rational coherent-preparation description). *For a supplied rational unit quaternion and rational preparation blocks, the preceding three-digit chart, at $B_U = \lceil 16N/\delta \rceil$, combined with the preparation codec at tolerance $\delta/2$, gives a legal rational instrument within d_N error δ . Its fixed-length description has*

$$(3 + v/2) \log_2 N + (3 + v) \log_2(1/\delta) + O_{n,m,r}(1)$$

bits, with no increase of a block rank and exact preservation of zero outcomes. It uses only integer and rational arithmetic, including the integer square roots in the preparation code. Its output is a mathematical instrument description, not physical classical simulation.

Proof. For quaternion matrices, $\|U(q) - U(q')\|_{\text{op}} = \|q - q'\|$. The unitary diamond error is at most twice this quantity. Thus the chart's coherent contribution is at most $4\sqrt{3}N/B_U < \delta/2$. Its squared upper certificate is $48N^2/B_U^2$. The preparation contribution is at most $\delta/2$. Triangle inequality proves the claim; one may certify its square by $2(e_U^2 + e_P^2)$, not by $e_U^2 + e_P^2$. The chart returns an exactly unit quaternion, and the preparation decoder returns positive blocks of total trace one. Formula (H.1) therefore gives exact complete positivity and trace preservation, including on singular inputs. The chart has at most $4(2B_U + 1)^3$ words, which gives the stated length after adding the inherited preparation payload. $\square$

The reference encoder replays both deterministic encoders when verifying a supplied target. Bare decoding checks only syntax, rational legality and the declared construction budget. It cannot certify proximity to an unspecified target. In particular, reference stability concerns the quantum instrument defined by the decoded Choi matrices, not access to a classical processor with equivalent action on an unknown reference-entangled state.

APPENDIX I. COVERINGS THROUGHOUT THE SUBMAXIMAL ERROR RANGE

A pairwise separation argument at radius δ requires distances larger than 2δ . Since our adaptive distance is an unhalved trace norm, this argument cannot by itself reach $\delta \geq 1$. We instead bound the number of targets that one centre can cover. The same measurement, rather than a different binary test for each pair, will be used in this part of the converse. Dimensions are fixed throughout. For a target family $\mathcal{F}$, write $\mathcal{M}_N(\delta; \mathcal{F})$ for its least covering cardinality in d_N with arbitrary legal memoryless centres, and add a superscript $\mathbb{Q}$ when the decoded Choi blocks must be rational.

Lemma I.1 (Multiplicity of a covering centre). *Let $J_1, \dots, J_K$ be targets and let one common tester with at most N calls have disjoint classical events $A_1, \dots, A_K$ such that $\Pr_{J_i}(A_i) \geq 1 - \eta$. If $0 \leq \eta < 1$ and $0 \leq \delta < 2(1 - \eta)$, every covering of these targets by M legal centres at adaptive radius δ satisfies*

$$(I.1) \quad M \geq (1 - \eta - \delta/2)K.$$

A centre need not belong to the target family. Indeed the conclusion still holds for a centre specified by a legal N -slot process with private memory, provided its distance to each assigned target controls this same tester.

Proof. Let Q be the tester's output distribution at a centre. For each target assigned to that centre, trace contraction under the terminal measurement gives $Q(A_i) \geq 1 - \eta - \delta/2$. The division by two is required because classical total variation is half the unhalved trace norm. Disjointness gives at most $(1 - \eta - \delta/2)^{-1}$ assigned targets. Sum this bound over centres. No product structure, unbiasedness or independence is required of Q . $\square$

Lemma I.2 (One common finite-dimensional estimator). *Put $b = m(dn)^2$. On $\mathcal{C}_{d,n,m}$ there is, for every $N \geq 1$, one nonadaptive tester and a Hermitian tuple estimator $\hat{J}$ satisfying*

$$(I.2) \quad \sup_{J \in \mathcal{C}_{d,n,m}} \mathbb{E}_J |\hat{J} - J|_2^2 \leq \frac{C_0}{N}, \quad C_0 = 2d^2b^2.$$

The estimator need not be positive or trace preserving. The tester and estimator are independent of the unknown target and of any later choice of a packing set.

Proof. One normalized maximally entangled input produces the flagged Choi state $\Omega_J = d^{-1} \bigoplus_y J_y$. Choose a Hilbert–Schmidt orthonormal Hermitian basis $H_1, \dots, H_b$ of the block-diagonal space. Every H_j has operator norm at most one. Thus $(I + H_j)/2, (I - H_j)/2$ is a binary POVM, and its signed outcome has mean $\text{tr}(H_j \Omega_J)$ and variance at most one. For $N \geq b$, use $k = \lfloor N/b \rfloor$ independent Choi inputs for each of the b measurements. There are at most N calls. Let $\hat{a}_j$ be the corresponding empirical signed mean and set $\hat{\Omega} = \sum_j \hat{a}_j H_j$. Orthonormality gives

$$\mathbb{E}_J \|\hat{\Omega} - \Omega_J\|_F^2 = \sum_j \text{Var}(\hat{a}_j) \leq b/k \leq 2b^2/N.$$

Set $\hat{J} = d\hat{\Omega}$, interpreted as a tuple. For $N < b$ use $\hat{J} = 0$ and make no calls. Then $|J|_2^2 = d^2 \text{tr} \Omega_J^2 \leq d^2 \leq C_0/N$. This proves every asserted range, without a rank or eigenvalue-margin condition. The binary POVMs are block diagonal in the public outcome, so coherent access to different classical flags is not needed. $\square$

This elementary informationally complete estimation argument is a standard finite-dimensional tomography construction; compare the process estimation setting of [2]. Here it is used only to build a common lower-bound witness. It does not constitute a new optimal-learning claim, and the description encoder below is still supplied its target.

Theorem I.3 (A full-range covering lower bound). *Let $\mathcal{F} \subseteq \mathcal{C}_{d,n,m}$ contain the image of a nonempty a -dimensional Euclidean ball under a bi-Lipschitz map for tuple Frobenius distance, with $a > 0$. For every fixed $0 < \delta_* < 2$ there is $c > 0$, depending on this patch, the fixed dimensions and δ_* , such that*

$$(I.3) \quad \mathcal{M}_N(\delta; \mathcal{F}) \geq cN^{a/2} \quad (N \geq 1, 0 < \delta \leq \delta_*).$$

Centres may be arbitrary legal memoryless instruments. The same lower bound holds under the relaxation to legal N -slot centres described in Lemma I.1.

Proof. Set $\eta = (2 - \delta_*)/4 > 0$ and $r_N = (C_0/(\eta N))^{1/2}$. The previous lemma and Markov's inequality give $\Pr_J\{|\hat{J} - J|_2 \leq r_N\} \geq 1 - \eta$ for every target. A volumetric packing of the fixed patch supplies at least $c_1 r_N^{-a}$ points at pairwise tuple distance greater than $2r_N$. The constant can be decreased to cover coarse scales: a singleton packing always exists. The estimator events consisting of the closed r_N -balls at these points are disjoint. Lemma I.1 applies because $1 - \eta - \delta/2 \geq 1 - \eta - \delta_*/2 = \eta$. It gives $\mathcal{M}_N \geq \eta c_1 r_N^{-a}$, as required. In particular one centre may cover several packing points when the error is large; the proof bounds that multiplicity rather than asserting it is one. $\square$

Corollary I.4 (Full-range interior and preparation laws). *Fix $0 < \delta_* < 2$. In Theorem E.3, with fixed d, n, m, a and $s = d^2(mn^2 - 1) > 0$, both real and rational legal-centre covering numbers satisfy*

$$(I.4) \quad \log_2 \mathcal{M}_N(\delta; a) = \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) + O(1) = \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O(1)$$

uniformly for $N \geq 1$ and $0 < \delta \leq \delta_$. The constants may depend on δ_* and the positive margin a . For the preparation family in Theorem F.3, the same extension holds with s replaced by $v = \sum_y r_y(2n - r_y) - 1$, with fixed dimensions and public output-rank bounds. Its constants require no positive eigenvalue margin. If $v = 0$ the family is a singleton and needs no payload.*

Proof. For $0 < \delta \leq 1/32$ these are the inherited joint laws. The interior ball and maximal-rank preparation chart used in their proofs are bi-Lipschitz patches of dimensions s and v , respectively. Theorem I.3 therefore supplies the lower term $(s/2) \log_2 N + O(1)$, or $(v/2) \log_2 N + O(1)$, on the remaining range $1/32 < \delta \leq \delta_*$. On that range $\log_2(1/\delta)$ is bounded above and below by constants depending only on δ_* . For the upper bound use the old code at accuracy $1/32$, which is also accurate to δ . This proves a uniform additive remainder for both kinds of centres. If $\delta_* \leq 1/32$ there is no remaining range. $\square$

The cap $\delta_* < 2$ is substantive: at unhalved error two every nonempty family is covered by a single legal centre. Nothing above is uniform as $\delta_* \uparrow 2$, or along a varying error sequence converging to two. The exact zero-error endpoint is also separate from the positive-error coding law. The older coherent/preparation theorem retains its own $\delta \leq 1/32$ range; the present common tomography argument gives the half-dimension scale, not its coherent N^{-1} scale.

APPENDIX J. INPUT-DEPENDENT BOUNDARY INSTRUMENTS

Fix an orthonormal basis $e_1, \dots, e_d$ of the input and the associated complete dephasing map $\mathcal{D}(X) = \sum_x \langle e_x, X e_x \rangle |e_x\rangle \langle e_x|$. For positive semidefinite $n \times n$ matrices σ_{xy} satisfying $\sum_y \text{tr } \sigma_{xy} = 1$ for every x , define

$$(J.1) \quad \Phi_{\sigma, y}(X) = \sum_{x=1}^d \langle e_x, X e_x \rangle \sigma_{xy}, \quad 1 \leq y \leq m.$$

These instruments measure a fixed input basis and then prepare an outcome-dependent state. Their public outcome probabilities $\sum_x \langle e_x, X e_x \rangle \text{tr } \sigma_{xy}$ may depend on the input. Measurement destroys input coherences; the family is not in general input erasing, and it is not the coherent tensor family of Theorem H.1.

The Choi blocks and their ranks are

$$(J.2) \quad J_y = \bigoplus_{x=1}^d \sigma_{xy}, \quad \text{rank } J_y = \sum_x \text{rank } \sigma_{xy}.$$

In particular the marginal constraint is exactly one unit-trace condition per input row. Fix public integers $0 \leq r_{xy} \leq n$, with $\sum_y r_{xy} > 0$ in every row, and let $\mathcal{F}_r$ be the family with $\text{rank } \sigma_{xy} \leq r_{xy}$. Put

$$(J.3) \quad v_x = \sum_y r_{xy}(2n - r_{xy}) - 1, \quad V = \sum_x v_x.$$

The ranks here are conditional output-state ranks, not independent rank bounds on arbitrary Choi blocks. The input basis, dimensions and rank bounds are fixed, not learned from an unknown device.

Theorem J.1 (Rank-boundary coding with input-dependent outcomes). *Fix the preceding data and $0 < \delta_* < 2$. If $V > 0$, the least covering cardinality by arbitrary legal memoryless centres and the least cardinality with rational decoded Choi blocks both obey*

$$(J.4) \quad \log_2 \mathcal{M}_N(\delta; \mathcal{F}_r) = \frac{V}{2} \log_2 N + V \log_2(1/\delta) + O(1)$$

uniformly for $N \geq 1$ and $0 < \delta \leq \delta_$. The error is the unhalved final-state trace distance over arbitrary common quantum testers with finite references, quantum memory, feedback and public stopping bounded by N . A rational upper code has decoded centres in (J.1); it never increases any conditional block rank and preserves every zero block exactly. Its encoder on rational target data uses integer and rational arithmetic only. If $V = 0$ the family is a known singleton.*

The remainder depends on the fixed dimensions, rank bounds and δ_ , but not on N , δ or a smallest positive output eigenvalue. Payload length is distinct from encoder workspace and expanded decoder storage. The decoded object is a mathematical quantum instrument, not a physical classical simulator on an unknown reference-entangled input.*

J.1. Centres and the adaptive metric. Write $\Omega_x = \bigoplus_y \sigma_{xy}$ for the flagged state of one input row. A useful general observation first identifies the allowed centres.

Proposition J.2 (Idempotent input retraction). *We use $\Phi\Pi = \Phi \circ \Pi$: the rightmost map acts on the input first. Let Π be a fixed completely positive trace-preserving idempotent input map and let every target satisfy $\Phi\Pi = \Phi$. Then*

$$(J.5) \quad d_N(\Phi, K\Pi) \leq d_N(\Phi, K)$$

for every legal memoryless centre K and every N . Thus covering numbers with all legal centres equal those with centres fixed by right composition with Π , at every radius. For $\Pi = \mathcal{D}$, the retracted centre has rows $\tau_{xy} = K_y(|e_x\rangle\langle e_x|)$ and is of the form (J.1). In the prescribed coordinate basis this operation preserves rationality. It need not preserve the target's rank bounds.

Proof. For any tester of Φ against $K\Pi$, insert Π on its input immediately before each use of the unknown device. This is a legal common tester of Φ against K : in the first hypothesis $\Phi\Pi = \Phi$, and in the second it implements $K\Pi$. References, feedback and stopping are unaffected. Taking the supremum proves (J.5). Idempotence says that $K\Pi$ is fixed by right composition with Π . Map every covering centre this way; cardinality cannot increase. The reverse covering inequality follows from inclusion of allowed centre classes. For dephasing, the displayed rows are positive and have unit total trace, since K is an instrument. They are diagonal input blocks of the original Choi matrices. $\square$

This is a data-processing retraction, not a rank-preserving projection. It generalizes the preparation-centre observation of Theorem G.1, whose fixed input map discards its input and supplies one chosen state. In the converse of Theorem J.1, no rank promise is imposed on centres.

Lemma J.3 (Row witnesses and a common programme). *For two instruments of (J.1),*

$$(J.6) \quad \max_x \|\Omega_x^{\otimes N} - \Lambda_x^{\otimes N}\|_1 \leq d_N(\Phi_\sigma, \Phi_\tau) \leq \left\| \bigotimes_x \Omega_x^{\otimes N} - \bigotimes_x \Lambda_x^{\otimes N} \right\|_1.$$

If unit purifications a_x, b_x of the corresponding row states are chosen and $\eta_x = \|a_x - b_x\|$, then

$$(J.7) \quad d_N(\Phi_\sigma, \Phi_\tau) \leq 2\sqrt{N \sum_x \eta_x^2}.$$

For one use there is the separate exact identity

$$(J.8) \quad d_1 = \max_x \|\Omega_x - \Lambda_x\|_1.$$

Proof. For the first inequality repeatedly feed $|e_x\rangle\langle e_x|$ and retain all flagged outputs. For the second, prepare, for each of the N slots, one copy of every row state. A common processor measures the actual input in the prescribed basis and uses only the programme state with that outcome x . The remaining row states in that slot are discarded. If the input is entangled with the tester's memory, its conditional memory operator is retained by this measurement; appending the selected row state gives exactly (J.1) on that joint input. The unobserved input outcome is internal to the processor, not new feedback granted to the tester. All later operations and the discard of unused slots at a stop form one common CPTP processor. Trace contraction proves the upper bound.

Purify all these programme states by the chosen vectors. For unit vectors, $1 - |\langle a_x, b_x \rangle|^2 \leq \eta_x^2$. The elementary product inequality $1 - \prod t_j \leq \sum(1 - t_j)$ for $t_j \in [0, 1]$ gives pure-programme trace distance squared at most $4N \sum_x \eta_x^2$. Partial trace and the common processor give (J.7). Finally, for an arbitrary one-use input with reference, the difference is $\sum_x \xi_R^x \otimes (\Omega_x - \Lambda_x)$, where $\xi_R^x \geq 0$ and $\sum_x \text{tr} \xi_R^x = 1$. The triangle inequality gives the one-use upper maximum, and a basis input attains it. $\square$

The product programme consumes up to dN row states in the proof, whereas the actual tester makes at most N instrument calls. It is only a comparison device, not an equality of these resource counts. Neither inequality in (J.6) is asserted to be an equality for general N . In particular the absence of an asymptotic adaptive advantage in a binary discrimination exponent does not imply a finite-use metric equality; compare [9, Sections II and V].

J.2. Exact rational decoding and the covering proof.

Lemma J.4 (A common-grid row code). *For rational target data, apply the singular factor encoder of Theorem F.2 separately in every input row at a common integer grid $B \geq 1$. If h_x is that row's actual real factor-coordinate count, the decoded instrument satisfies*

$$(J.9) \quad d_N(\Phi, \tilde{\Phi})^2 \leq \frac{16N \sum_x (h_x - 1)}{B^2} \leq \frac{16NV}{B^2}.$$

There are at most $C_{d,n,m,r}(2B+1)^V$ legal decoder images. The conditional block ranks do not increase; consequently no outcome Choi rank increases either. All zero conditional blocks remain zero. The expanded common denominator is at most $B^{2d} \prod_x h_x$ and thus has $O_{d,n,m}(1 + \log B)$ bits.

Proof. The inherited anchor construction gives a unit factor purification in row x within distance $2\sqrt{h_x - 1}/B$ of the target's chosen unit factor vector. Apply (J.7). Trace normalization is performed separately in each row, so (J.2) is exactly trace preserving. The inherited zero-pivot proof, rational squared coordinates and integer-square-root ratio calculation apply row by row. Their finite masks and anchor headers have fixed cardinality; the row body has at most v_x signed grid coordinates. Multiplying these counts gives the asserted exponent V . Only valid decoded words count. The block-rank and zero statements hold row by row and pass to the direct sum in (J.2). Row x has normalization denominator $T_x \leq h_x B^2$. A common denominator is the least common multiple of the T_x , bounded by their product. This proves the final assertion. $\square$

Proof of Theorem J.1. For the upper bound take $B = \max\{1, \lceil 4\sqrt{NV}/\delta \rceil\}$. The last lemma gives error at most δ and at most $C(2B+1)^V$ centres, hence the upper description law for $\delta \leq \delta_* < 2$. Real targets admit the same factor rounding construction; the decoded matrices are still rational. For supplied rational targets, no irrational factor is stored: the exact inherited square-comparison algorithm determines the digits. Thus the real-family rational-centre upper bound does not restrict the targets to a countable subclass.

For the lower bound choose, in every row with $v_x > 0$, the gauge-fixed maximal-rank factor patch used in Theorem F.3. Its inverse is the local leading-block Cholesky map. Rows with $v_x = 0$ are fixed. A Cartesian product of sufficiently small closed coordinate balls contains a V -dimensional ball. The map to the tuple (σ_{xy}) , and hence to J , is bi-Lipschitz; direct sum is an isometry for tuple Frobenius norm. This supplies one fixed V -dimensional patch with no claim of a global factor chart. For $\delta \leq 1/32$ pack it at separation greater than $32d\delta/\sqrt{N}$. Corollary E.2 gives distances larger than 2δ , so even an unrestricted legal centre covers at most one packing point. This gives $c(\sqrt{N}/\delta)^V$ centres. For $1/32 < \delta \leq \delta_*$ use Theorem I.3 on the same patch and the boundedness of $\log(1/\delta)$ on that interval. These lower bounds apply also to rational centres, which form a smaller class. Taking logarithms, with the usual ceiling for fixed-length words, completes the proof. If $V = 0$, each nonempty row consists of a specified one-dimensional unit block; its location is fixed by the public rank bounds. Thus the whole family is a known singleton. $\square$

For example, with scalar output $n = 1$ and all rank bounds equal to one, (J.4) has $V = d(m - 1)$: it is a boundary coding law for the rows of a classical stochastic matrix. With one quantum output and full row ranks, $V = d(n^2 - 1)$; with one pure output per row, $V = d(2n - 2)$. Mixed and pure conditional states need not commute. These dimensions are standard rank-stratum dimensions; the theorem identifies their joint coding scale under the declared adaptive metric. The measured input basis is fixed. No law for varying measurement bases, unknown instruments, or all irreversible quantum dynamics follows from this result.

APPENDIX K. VARYING PROJECTIVE READOUT AND CONDITIONAL PREPARATIONS

The fixed input basis can be allowed to vary when its outcome remains observable. This changes the coding scale: destruction of the measured quantum input does not make the readout coordinate a preparation coordinate. The distinguishing experiment may use entangled inputs.

Let $\mathfrak{F}_d$ be the space of ordered orthogonal rank-one projections $P = (P_1, \dots, P_d)$ on $\mathbb{C}^d$ with $\sum_x P_x = I_d$. It is the flag manifold $U(d)/\mathbb{T}^d$, of real dimension $b = d^2 - d$. The column phases are immaterial; permutation of the ordered outcomes is not a gauge. For each x let $\sigma_x = (\sigma_{xy})_{y=1}^m \in \mathcal{S}_{m,n}$ and set

$$(K.1) \quad \mathcal{F}_{P,\sigma,(x,y)}(X) = \text{tr}(P_x X) \sigma_{xy}, \quad J_{xy} = P_x^\top \otimes \sigma_{xy}.$$

Both labels (x, y) are retained. The transpose in this input-first Choi formula is essential, including for complex projections. Fix public bounds $0 \leq r_{xy} \leq n$, with at least one positive bound in each row, and put

$$v_x = \sum_y r_{xy}(2n - r_{xy}) - 1, \quad V = \sum_x v_x.$$

The family $\mathfrak{R}_r$ consists of (K.1) with these rank bounds. Let $\mathcal{R}_N(\delta)$ be its covering number for d_N , allowing arbitrary legal memoryless instrument centres of the same interface; $\mathcal{R}_N^{\mathbb{Q}}(\delta)$ requires rational centres. Public dimensions, outcome labels, ranks, horizon and tolerance are not payload.

Theorem K.1 (Mixed-scale coding with observable readout). *For fixed d, n, m, r there are constants $c, C > 0$ such that for $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(K.2) \quad c(N/\delta)^b (\sqrt{N}/\delta)^V \leq \mathcal{R}_N(\delta) \leq \mathcal{R}_N^{\mathbb{Q}}(\delta) \leq C(N/\delta)^b (\sqrt{N}/\delta)^V.$$

Consequently the optimal reusable description length is

$$(K.3) \quad (b + V/2) \log_2 N + (b + V) \log_2(1/\delta) + O(1),$$

with remainder uniform in N and δ . The upper centres belong to the displayed family, preserve its zero conditional blocks and do not increase any conditional rank. For supplied rational projections and rational row states there is a finite exact rational encoder, without choosing algebraic unit-vector

representatives or searching an exhaustive unitary net. Real targets admit the same rational-centre cover by the exact-real construction. If $b + V = 0$, the family is a known singleton and needs zero payload.

The error range here is not the full submaximal range of Theorem J.1. The readout part of the lower proof uses pairwise separation; it is not an application of the full-error half-dimensional estimator theorem.

K.1. The two distinguishing scales. Write $M_P(X) = (\text{tr}(P_x X))_x$ and $q(P, Q)^2 = \sum_x \|P_x - Q_x\|_F^2$.

Lemma K.2 (Readout modulus). *If $z = d_1(M_P, M_Q)$, then*

$$(K.4) \quad \sqrt{2/d} q(P, Q) \leq z \leq \sqrt{d} q(P, Q).$$

For $z < 2$ one has

$$(K.5) \quad d_N(M_P, M_Q) = 2 \sin(\min\{N \arcsin(z/2), \pi/2\}).$$

For $z = 2$ the distance is two at every $N \geq 1$. In particular it is at least $(2/\pi) \min\{Nz, 1\}$ and at most Nz .

Proof. The difference of two rank-one projections has nonzero eigenvalues $\pm s$, so its Frobenius norm is $\sqrt{2}s$. An input eigenvector maximizing one event difference gives total unhalved outcome distance at least $2 \max_x \|P_x - Q_x\|_\infty$. This proves the lower bound. For an arbitrary reference input state ρ , duality gives

$$\|\text{tr}_A(((P_x - Q_x) \otimes I)\rho)\|_1 \leq \|P_x - Q_x\|_\infty.$$

Indeed a Hermitian contraction H on the reference tests this block by $\text{tr}(((P_x - Q_x) \otimes H)\rho)$. Sum over the classical blocks and apply Cauchy–Schwarz to obtain the upper bound in (K.4).

We use here the exact measurement-discrimination theorem of Puchała–Pawela–Krawiec–Kukulski–Oszmaniec [16, Corollary 1 and Theorem 2]. If U represents the relative ordered basis, their optimized eigenvalue-arc length $\Upsilon(U)$ gives distance $2 \sin(N\Upsilon(U)/2)$ until $N\Upsilon(U) \geq \pi$, when the distance is two; their Theorem 2 allows general adaptive networks. At one use, $z < 2$ implies $\Upsilon(U) = 2 \arcsin(z/2)$. The already-perfect case is treated separately and need not be inverted. This proves (K.5). A publicly stopped tester is contained in a fixed N -slot network by performing irrelevant calls after the stop and discarding their outputs. Conversely, parallel testers are admissible here. The elementary bound $\sin t \geq 2t/\pi$ for $0 \leq t \leq \pi/2$ gives the stated lower modulus; the upper bound also follows directly by a hybrid argument. The exact multiple-use theorem is an imported result, not an inference from a finite-dimensional spectral gap. $\square$

Lemma K.3 (Measurement and programme comparisons). *For arbitrary row states,*

$$d_N(\mathcal{F}_{P,\sigma}, \mathcal{F}_{Q,\tau}) \geq d_N(M_P, M_Q).$$

When $P = Q$, the distance is at least $\max_x \|\Omega_x^{\otimes N} - \Lambda_x^{\otimes N}\|_1$, where $\Omega_x = \bigoplus_y \sigma_{xy}$ and $\Lambda_x = \bigoplus_y \tau_{xy}$. For purifications whose rowwise vector differences have lengths η_x , one has

$$(K.6) \quad d_N(\mathcal{F}_{P,\sigma}, \mathcal{F}_{Q,\tau}) \leq Nd_1(M_P, M_Q) + 2\sqrt{N \sum_x \eta_x^2}.$$

Proof. Discarding the quantum outputs and y gives the measurement channel, so every measurement tester is a tester of the displayed instruments. For a common flag, feeding a unit vector in $\text{ran } P_x$ at every call gives the x th product-state witness. For the upper bound first change P to Q while keeping σ fixed. Appending the conditional row state is a common postprocessing of the readout, and a slotwise hybrid gives the first term. Then change the row states with Q fixed. At each

slot a common processor measures Q , selects one of the supplied row programme states, retains x , and discards the unused rows. The product-purification argument of Lemma J.3 gives the second term, also with references, feedback and stopping. This is the standard common-environment method [13, Proposition 2]; retaining the readout label changes neither its positivity nor its contraction argument. The upper comparison is not asserted to equal either lower comparison. $\square$

K.2. A bounded rational atlas without column phases. There is a direct flag codec, rather than an exhaustive search over unitaries. We give its quantitative bound because it also controls singular rational descriptions whose unit vectors are not rational.

Lemma K.4 (Pivoted triangular flag charts). *Every $P \in \mathfrak{F}_d$ is obtained from an ordered basis of columns of $\Pi^{-1}L$, where Π is a permutation matrix and L is unit lower triangular with complex entries of modulus at most one below the diagonal. Its ordered rank-one projections are the successive differences of projections onto the first j columns. Thus $d!$ charts, each with $b = d^2 - d$ real coordinates, suffice. Given rational P_x , both the chart and its coordinates can be found rationally.*

Round each real and imaginary coordinate of L towards zero at grid $B \geq 1$ and let $\hat{P}$ be the decoded flag. Put

$$K_d = (2d)^{d-1}, \quad C_d = 4d^2 K_d.$$

Then every decoded flag is exactly legal and rational, and

$$(K.7) \quad d_N(M_P, M_{\hat{P}}) \leq C_d N/B.$$

There are at most $d!(2B+1)^b$ codewords. The expanded projection entries have $O_d(1 + \log B)$ bits.

Proof. Choose a nonzero column $v_x = P_x e_{k_x}$, taking the first positive diagonal position. These columns are rational whenever the input projections are rational. The columns are nonzero and mutually orthogonal; hence $A = [v_1 \cdots v_d]$ is invertible. Gaussian elimination with partial row pivoting gives $\Pi A = LR$, with L unit lower triangular and R upper triangular and invertible. At every pivot choose a residual entry of largest modulus, breaking ties by row order. Each multiplier below it has modulus at most one. Subsequent row interchanges interchange the already recorded multipliers as well. All comparisons of squared moduli and all arithmetic are rational for rational A . Right multiplication by an invertible upper triangular matrix preserves each initial column span. Thus the first j columns of $\Pi^{-1}L$ span $\text{ran}(P_1 + \cdots + P_j)$. This proves the chart assertion without choosing square roots.

Any unit lower triangular matrix in the coordinate cube $|\Re L_{ij}|, |\Im L_{ij}| \leq 1$ is invertible, even if it was not produced by elimination. If Z_j is its first j columns, put

$$E_j = Z_j(Z_j^* Z_j)^{-1} Z_j^*, \quad E_0 = 0.$$

The differences $E_j - E_{j-1}$ are mutually orthogonal rank-one projections with sum I . Permute their coordinates back by Π . The same formulas decode the rounded rational matrix exactly.

Throughout the cube, $\|L\|_2 \leq 2d$ and $\det L = 1$, so its least singular value is at least K_d^{-1} . Restriction to any first j columns preserves this lower bound. For two such matrices L, L' , write E_j, E'_j for their column-space projections. Using $(I - E'_j)Z'_j = 0$ and $E_j = Z_j Z_j^\dagger$ gives

$$\|(I - E'_j)E_j\|_F \leq K_d \|Z_j - Z'_j\|_F.$$

The analogous bound with the primes interchanged and the identity $E_j - E'_j = (I - E'_j)E_j - E'_j(I - E_j)$ imply $\|E_j - E'_j\|_F \leq 2K_d \|L - L'\|_F$. Consequently each rank-one difference is at most $4K_d \|L - L'\|_F$. The one-use block estimate used above bounds the measurement distance by the sum of these norms. Rounding changes L by at most $\sqrt{b}/B \leq d/B$, proving the one-use bound C_d/B and then (K.7) by hybridization.

Record the permutation and the b signed digits in $[-B, B]$. No phases or rational unit-vector normalizations are encoded. All inverse Gram matrices in the displayed decoding formula have fixed size and rational entries of degree bounded in terms of d in the digits and B . The determinant is nonzero. Clearing denominators and the adjugate formula bound every numerator and denominator by $B^{O_d(1)}$ times a fixed constant. This proves the bit-length claim. $\square$

Partial pivoting is used here only to select a bounded chart. The decoder itself needs no pivot test on the target and always returns a flag for any valid chart word. A target-bound verifier nevertheless checks the entire canonical encoding, not just legality.

K.3. Proof of the covering law.

Proof of Theorem K.1. For the upper bound, use the flag code at $B_f = \max\{1, \lceil 2C_d N/\delta \rceil\}$ and the conditional factor code at $B_s = \max\{1, \lceil 8\sqrt{NV}/\delta \rceil\}$. The two terms of (K.6) are each at most $\delta/2$. The factor code counts every actual pivot mask, anchor and sign. Its total cardinality is at most $C(2B_s + 1)^V$. Multiply by $d!(2B_f + 1)^b$. On the stated range this proves the upper bound in (K.2). Each factor row is trace normalized and rank nonincreasing. Formula (K.1) then gives positive Choi blocks with joint partial trace I_d ; their ranks equal those of the conditional states. Rationality and zero preservation follow exactly. Real target projectors admit the same chart selection and rounding; only the algorithmic assertion about rational targets uses finite rational comparisons.

For the lower bound, choose a small closed coordinate box around a flag in $\mathfrak{F}_d$ on which q is bi-Lipschitz to a Euclidean metric in b dimensions. Such a chart can also be checked directly from $L = I + T$: to first order the off-diagonal entries of the ordered projections recover every real and imaginary entry of strictly lower triangular T . The derivative is injective, and shrinking to a compact box gives fixed forward and inverse Lipschitz constants. Grid-pack it at q -separation greater than $A_d \delta/N$, where A_d is fixed large enough that Lemma K.2 gives distance strictly greater than 2δ . Indeed its constant lower plateau $2/\pi$ exceeds 2δ on this range. The packing has at least $c(N/\delta)^b$ points after reducing c to include the coarse-grid cases.

In every nontrivial row choose the compact maximal-rank factor patch from Theorem F.3. The product row patch has V real dimensions. Pack it in tuple Frobenius distance at more than $A_{d,n,m,r} \delta/\sqrt{N}$. Some row of two distinct grid points then differs by at least the tuple distance divided by $\sqrt{d}$. The local product-Choi test (Corollary E.2, with input dimension one) implies that its N -copy states are separated by more than 2δ . The row packing has cardinality at least $c(\sqrt{N}/\delta)^V$.

Take the Cartesian product of these two finite packings. If two points have different flags, discarding the conditional outputs gives the separating measurement witness regardless of their rows. If their flags agree, the product-state witness in a differing row separates them. These are legitimate pair-dependent testers, not incompatible tests assumed to hold in one common experiment. Hence all distinct product points have d_N -distance greater than 2δ . By the triangle inequality, a ball centred at any legal memoryless instrument covers at most one of them. Multiplying cardinalities proves the lower bound. A fixed-dimensional ceiling in the word length changes only the additive constant. Finally, $b + V = 0$ forces $d = 1$ and a singleton rank-state family, proving the last assertion. $\square$

Corollary K.5 (Readout recovered from orthogonal output supports). *Suppose the label x in (K.1) is not output, but the x th row is supported in a fixed public subspace H_x , with the H_x mutually orthogonal. Then the same coding law holds with n replaced by $n_x = \dim H_x$ in each row dimension. For public rational coordinate embeddings the upper code is rational.*

Proof. Measuring the fixed output projections onto H_x recovers x without disturbing the target's conditional state. This operation and forgetting x are fixed CPTP postprocessings inverse on

the family. Inserting them at every slot proves equality of the two metrics on target pairs, with references and adaptive feedback. The preceding upper code and product packing therefore apply. The lower packing still allows arbitrary legal centres. We do not infer a target-independent input retraction for varying flags, or a rank bound on the postprocessing of an arbitrary centre. $\square$

Remark K.6 (What can disappear when readout is unobservable). If the label x is discarded and all conditional rows equal one state tuple σ , then $\sum_x \text{tr}(P_x X) \sigma_y = \text{tr}(X) \sigma_y$ for every flag. Thus the entire basis coordinate disappears exactly. This example separates an observable readout theorem from an unjustified parameter count for arbitrary unlabelled overlapping output rows. The two proved families above do have the asserted mixed scale; neither statement classifies a general coupled coherent/support-changing instrument boundary.

APPENDIX L. UNIFORMLY SEIZABLE PROGRAMMES AND UNRESTRICTED CENTRES

Programme-state comparisons become exact when a single physical procedure can recover the programme from one use, uniformly over the whole family. The state/channel discrimination principle is classical [14, Definition 36 and Theorem 37]; see also the channel-box formulation [15, Section III.A]. We state its covering consequence with the centre classes explicit.

Let $\mathcal{S}$ be the full state space of a fixed finite direct-sum matrix algebra. Let $\mathcal{A}(\omega)$ be the instrument obtained by feeding a state $\omega \in \mathcal{S}$ into a fixed CPTP processor along with its input. Let $\mathcal{B}$ be a fixed one-use tester which returns a state in $\mathcal{S}$ from any legal instrument of the same interface. Assume

$$(L.1) \quad \mathcal{B}(\mathcal{A}(\omega)) = \omega \quad (\omega \in \mathcal{S}).$$

The tester and processor do not depend on the target, the pair of hypotheses, the tolerance or the covering centre. They may have a fixed finite quantum memory. In particular a common programme representation alone is not this hypothesis.

Theorem L.1 (Exact transfer of family covering numbers). *Under (L.1), for every $N \geq 1$,*

$$(L.2) \quad d_N(\mathcal{A}(\omega), \mathcal{A}(\tau)) = \|\omega^{\otimes N} - \tau^{\otimes N}\|_1.$$

The map $\mathcal{R} = \mathcal{A}\mathcal{B}$ on legal memoryless instrument centres is an idempotent one-use superchannel, and

$$(L.3) \quad d_N(\mathcal{A}(\omega), \mathcal{R}(K)) \leq d_N(\mathcal{A}(\omega), K).$$

For every subset $S \subseteq \mathcal{S}$ and every radius δ , its N -copy state covering number with arbitrary centres in $\mathcal{S}$ equals the d_N covering number of $\mathcal{A}(S)$ with arbitrary legal memoryless instrument centres.

In particular, for the rank-bounded state family $\mathcal{S}_r \subseteq \mathcal{S}_{m,n}$ with $v = \sum_y r_y(2n - r_y) - 1$, the latter logarithmic covering number is

$$(L.4) \quad (v/2) \log_2 N + v \log_2(1/\delta) + O(1)$$

for every fixed $0 < \delta_ < 2$, uniformly over $N \geq 1$, $0 < \delta \leq \delta_*$. If $\mathcal{A}$ sends rational programme states to rational Choi matrices, the same asymptotic law holds with rational centres and is attained by the inherited rational factor code followed by $\mathcal{A}$. The code preserves programme block ranks and zeros. No Choi-rank preservation of $\mathcal{R}$ is asserted.*

Proof. Give a common processor N independent copies of the programme, and use one fresh copy at each device call. All tester operations, including discarding the unused programmes after a stop, compose into one common channel. Data processing proves the upper bound in (L.2). Repeating the one-use seizing procedure N times and retaining its states gives the reverse bound. This is the established environment-seizing reduction with its finite-use normalization made explicit.

To implement $\mathcal{R}(K)$ on an external input, retain that input in a quantum memory, use one fresh invocation of K inside $\mathcal{B}$, and feed the resulting programme together with the retained input

to $\mathcal{A}$. This is a fixed one-use pre- and postprocessing, hence a superchannel. Equation (L.1) implies $\mathcal{R}^2 = \mathcal{R}$ and $\mathcal{R}(\mathcal{A}(\omega)) = \mathcal{A}(\omega)$. Replacing each slot of an arbitrary tester against $\mathcal{A}(\omega)$ and $\mathcal{R}(K)$ by this construction produces a common tester against $\mathcal{A}(\omega)$ and K with identical two output states. It uses no more device calls and is compatible with reference entanglement and stopping. Taking suprema proves (L.3).

A state cover $\{\tau_i\}$ gives an instrument cover $\{\mathcal{A}(\tau_i)\}$ by (L.2). Conversely, map each arbitrary instrument centre K_i to $\tau_i = \mathcal{B}(K_i)$. Equations (L.3) and (L.2) show that every target covered by K_i is covered by τ_i at the same radius in the state metric. This proves exact equality of covering cardinalities, not just a two-hypothesis equivalence. The centres τ_i need not obey the rank promises on the targets; this is why (L.1) is imposed on the whole ambient programme space.

Apply the preparation-state law of Theorem F.3, the preparation-centre retraction Theorem G.1, and its full-error extension Corollary I.4. They give (L.4) for arbitrary programme centres. Rationality-preserving $\mathcal{A}$ maps the rational rank-preserving upper code into legal rational instrument centres, while the unrestricted lower bound applies to this smaller class. If $v = 0$, zero payload suffices for the known singleton. The constants can depend on the fixed dimensions, rank bounds and error cap; they are not uniform as $\delta_* \uparrow 2$. $\square$

Corollary L.2 (Pauli-channel descriptions). *Let $\sigma_0 = I, \sigma_1 = X, \sigma_2 = Y, \sigma_3 = Z$ be the Pauli matrices and let $\Phi_p(\rho) = \sum_{j=0}^3 p_j \sigma_j \rho \sigma_j^*$. Fix a nonempty public allowed support of cardinality k for p , allowing zero probabilities on that support. With arbitrary legal qubit-channel centres, its optimal reusable description length is*

$$\frac{k-1}{2} \log_2 N + (k-1) \log_2(1/\delta) + O(1), \quad N \geq 1, \quad 0 < \delta \leq \delta_* < 2.$$

Rational upper descriptions preserve zero probabilities and never increase the Pauli-channel Choi rank of the target.

Proof. The classical programme p controls which Pauli matrix is applied. One maximally entangled input and a Bell-basis measurement recover p exactly. Both operations are fixed over the full probability simplex, so Theorem L.1 applies with $n = 1$ and one-dimensional blocks at the allowed positions. The orthogonal Bell Choi vectors show that the Choi rank equals the number of positive probabilities. The inherited scalar factor encoder keeps zeros zero and has rational output. This is a coding consequence of the standard Pauli/Bell seizing mechanism, not a new discrimination protocol. $\square$

For contrast, $U = (I + iX + iY + iZ)/2$ is unitary, so its channel has Choi rank one. Its Bell probabilities are all $1/4$, and $\mathcal{AB}$ sends it to the uniform Pauli mixture of Choi rank four. This exact example explains why arbitrary-centre retraction and rank-preserving target encoding are different claims. The processor in a programme-state comparison is quantum. Description length in (L.4) does not count its Hilbert-space dimension or give an unknown-channel learning or classical physical simulation theorem.

APPENDIX M. A NOISE-UNIFORM READOUT TRANSITION

The dependence on the readout basis can change scale when noise approaches an extremal measurement. We determine that transition in a fixed one-dimensional family, without a positive spectral margin uniform in the noise. The proof is a finite-distance argument; no conclusion is inferred merely from a Fisher-information calculation.

Let σ_1, σ_2 be the first two Pauli matrices. For $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ and a public visibility $0 \leq \lambda \leq 1$, put

$$(M.1) \quad E_{\theta,\lambda}^\pm = \frac{1}{2}(I \pm \lambda(\cos \theta \sigma_1 + \sin \theta \sigma_2)), \quad \mathcal{M}_{\theta,\lambda}^\pm(X) = \text{tr}(E_{\theta,\lambda}^\pm X).$$

Both outcome labels are retained. The output quantum system is one-dimensional. In particular, θ and $\theta + \pi$ are not identified by an outcome relabelling. Let $h(\theta, \phi) \in [0, \pi]$ be circular distance and set

$$(M.2) \quad \eta = 1 - \lambda^2, \quad K_{N,\lambda} = \lambda \sqrt{N \min\{N, \eta^{-1}\}}, \quad N \geq 1,$$

where $\eta^{-1} = +\infty$ at $\lambda = 1$. Thus $K_{N,1} = N$ and $K_{N,0} = 0$. The visibility may depend arbitrarily on N and the requested error; it is not a target coordinate charged in this theorem.

Write $\mathcal{C}_{N,\lambda}(\delta)$ for the least number of legal memoryless binary instruments of this interface whose d_N -balls of radius δ cover all targets in (M.1). Centres need not belong to the displayed family. Define $\mathcal{C}_{N,\lambda}^{\text{rat}}(\delta)$ by restricting centres to family members with rational $\cos \theta, \sin \theta$; rational visibility then gives rational effects. A constant in the following statement is independent of all three parameters N, λ, δ .

Theorem M.1 (Noise-uniform finite-use coding). *For all $N \geq 1$, $0 \leq \lambda \leq 1$, and all pairs of angles,*

$$(M.3) \quad \frac{1}{64} \min\{1, K_{N,\lambda} h(\theta, \phi)\} \leq d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min\{2, K_{N,\lambda} h(\theta, \phi)\}.$$

There are absolute $c, C > 0$ such that, for $0 < \delta \leq 2^{-10}$,

$$(M.4) \quad c \left(1 + \frac{K_{N,\lambda}}{\delta}\right) \leq \mathcal{C}_{N,\lambda}(\delta) \leq \mathcal{C}_{N,\lambda}^{\text{rat}}(\delta) \leq C \left(1 + \frac{K_{N,\lambda}}{\delta}\right).$$

Consequently the optimum fixed-length reusable payload is

$$(M.5) \quad \log_2 \left(1 + \frac{K_{N,\lambda}}{\delta}\right) + O(1).$$

The upper construction is an exact rational algorithm on rational visibility, tolerance and unit-circle input coordinates. At $\lambda = 0$ the family is a singleton and its optimum payload is zero.

The constants and the additive remainder are uniform up to the erased and projective endpoints. In contrast, they do not assert an optimal leading constant, a large-error coherent covering law, or a coding law in which an unknown visibility is also encoded.

M.1. A two-point programme with its noise dependence retained.

Lemma M.2 (An adaptive upper modulus). *Let $h = h(\theta, \phi)$ and $a = \lambda \sin(h/2)$. If $\eta > 0$, then*

$$(M.6) \quad d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min \left\{ 2, 2Na, 2\sqrt{1 - \left(\frac{\eta}{\eta + a^2}\right)^N} \right\}.$$

For $\eta = 0$ the bound $\min\{2, 2Na\}$ remains valid.

Proof. The difference of the two positive-outcome effects has operator norm a . On a bipartite input state, the two output blocks of the difference are B and $-B$, where $B = \text{tr}_A[(E_{\theta,\lambda}^+ - E_{\phi,\lambda}^+) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B\|_1 \leq a$. An eigenstate of the effect difference attains equality. Hence $d_1 = 2a$, including reference assistance. Replacing the calls one at a time and using trace-norm contraction proves $d_N \leq 2Na$ for every common adaptive tester.

Choose angle representatives with difference h and let u, v be the unit vectors along their midpoint and its oriented perpendicular in the equatorial plane. The two Bloch vectors are $m \pm av$, where $m = \lambda \cos(h/2)u$. Put $b = (1 - \|m\|^2)^{1/2} = (\eta + a^2)^{1/2}$. The two Bloch vectors $m \pm bv$ have norm one, so they define two legal binary projective instruments, say $\mathcal{A}_+$ and $\mathcal{A}_-$. The target pair is exactly the two mixtures of these same instruments with programme laws

$$p_{\pm} = ((1 \pm t)/2, (1 \mp t)/2), \quad t = a/b.$$

This construction depends on the pair, which is permissible in a metric upper bound. It is not one fixed processor for the whole circle.

For a fixed tester, first supply N independent programme bits and, at each requested call, apply the indicated $\mathcal{A}_+$ or $\mathcal{A}_-$. All subsequent operations, including the reference, feedback and a stopping rule, are a common channel of these bits. Unused bits are discarded. The root fidelity of p_+, p_- is $\sqrt{1 - t^2}$. Purifying the classical laws and using contraction gives

$$\|p_+^{\otimes N} - p_-^{\otimes N}\|_1 \leq 2\sqrt{1 - (1 - t^2)^N}.$$

This also follows directly from the classical fidelity inequality. Substitution proves (M.6). Finally $1 - (1 - t^2)^N \leq Nt^2$ and $a \leq \lambda h/2$ imply

$$d_N \leq \min\{N\lambda h, \sqrt{N/\eta} \lambda h, 2\}.$$

The limiting projective bound is supplied by the hybrid estimate, not by division by zero. $\square$

Classical simulation by common extremal channels and the effect of noise on the Heisenberg scale are established methods [7]. Environment/programme-state processing under adaptive use is also an established framework [13, 14, 15]. The point needed here is the exact two-point expression (M.6) and its uniform conversion to a family cover, rather than a new claim about the origin of those methods.

M.2. Finite entangled blocks and a binary product estimate.

Lemma M.3 (Symmetric Bernoulli products). *Let $P_\pm$ be Bernoulli laws with success probabilities $(1 \pm a)/2$, where $0 \leq a \leq 1$. For every integer $m \geq 1$,*

$$(M.7) \quad \|P_+^{\otimes m} - P_-^{\otimes m}\|_1 \geq \frac{1}{2} \min\{1, a\sqrt{m}\}.$$

Proof. Retain the largest odd number $r = 2j + 1 \leq m$ of coordinates, so $r \geq m/2$. For their majority event write $g_r(a) = P_+(\text{majority}) - P_-(\text{majority})$. Differentiating the finite binomial sum and cancelling consecutive terms gives

$$g'_r(a) = r \binom{2j}{j} 4^{-j} (1 - a^2)^j.$$

For $j \geq 1$, the bound $\binom{2j}{j} 4^{-j} \geq 1/(2\sqrt{j})$ follows by induction: the ratio of consecutive left sides is $(2j+1)/(2j+2) \geq \sqrt{j/(j+1)}$. For $0 \leq a \leq r^{-1/2}$, Bernoulli's inequality gives $(1 - a^2)^j \geq 1 - j/r \geq 1/2$. Thus $g'_r(a) \geq \sqrt{r}/(2\sqrt{2})$ on that interval. The case $r = 1$ is immediate. Integration from zero and monotonicity yield $g_r(a) \geq (2\sqrt{2})^{-1} \min\{1, a\sqrt{r}\}$. Unhalved distance is at least twice this event difference; using $r \geq m/2$ proves the stated bound. $\square$

Lemma M.4 (A noise-uniform pairwise witness). *The lower inequality in (M.3) is witnessed by nonadaptive entangled blocks, each of at most N qubits, followed by classical parity and majority processing.*

Proof. A block of k inputs is prepared in $(|0\rangle^{\otimes k} + e^{i\alpha}|1\rangle^{\otimes k})/\sqrt{2}$. Multiplying the k output signs of (M.1) has mean $\lambda^k \cos(k\theta - \alpha)$. This identity follows by applying $\bigotimes_{i=1}^k (E^+ - E^-)$ to the two off-diagonal terms of the block density matrix. For the two hypotheses choose the one common phase $\alpha = k(\theta + \phi)/2 + \pi/2$. Their parity laws then have opposite biases of magnitude

$$(M.8) \quad a_k = \lambda^k |\sin(kh/2)|.$$

Repeating $m = \lfloor N/k \rfloor$ independent blocks and ignoring remaining calls is an admissible tester. Its unhalved output distance is at least $\frac{1}{2} \min\{1, \sqrt{m} a_k\}$ by Lemma M.3. The phase and block size may depend on the pair, but never on the unknown hypothesis or a private outcome.

Assume $\lambda > 0$ and put $T = \min\{N, \eta^{-1}\}$, $k_0 = \lfloor T \rfloor$. Then $T/2 \leq k_0 \leq T$. For every $1 \leq k \leq k_0$,

$$(M.9) \quad \lambda^k \geq \lambda/2.$$

Indeed, when $0 < \eta < 1$, the inequality $\log(1 - \eta) \geq -\eta/(1 - \eta)$ and $k - 1 \leq \eta^{-1} - 1$ give $(k - 1) \log \lambda \geq -1/2$. At $\eta = 0$ the assertion is immediate.

If $h \leq 1/k_0$, choose $k = k_0$. Since $\sin(kh/2) \geq kh/\pi$ and $m \geq N/(2k)$, we obtain

$$\sqrt{m} a_k \geq \frac{\lambda \sqrt{N k_0}}{2\pi \sqrt{2}} h \geq \frac{K_{N,\lambda}}{4\pi} h.$$

This gives a lower constant $1/(8\pi)$. If $k_0 = 1$, use $k = 1$ for all $h \leq \pi$ instead. Now $T < 2$ unless $T = 1$, and $\sqrt{N} \lambda \sin(h/2) \geq K_{N,\lambda} h/(\pi \sqrt{2})$; this is stronger than required.

It remains that $k_0 \geq 2$ and $h > 1/k_0$. Here $\lambda \geq 1/\sqrt{2}$. Choose $k = \max\{1, \lfloor 1/h \rfloor\}$. Then $k \leq k_0$ and $1/2 \leq kh \leq \pi$. Equations (M.8) and (M.9) give $a_k \geq 1/(4\pi \sqrt{2})$. Even one such block has distance at least $1/(8\pi \sqrt{2}) > 1/64$. The three cases prove the claim. For $\lambda = 0$ both sides of the desired inequality vanish. $\square$

M.3. Covering and exact rational realization.

Proof of Theorem M.1. The metric assertion follows from the preceding lemmas. For a lower cover choose points on the circle with circular separation at least $256\delta/K_{N,\lambda}$ whenever that quantity is at most π . An equally spaced packing has at least a fixed constant times $K_{N,\lambda}/\delta$ points. Every two points have d_N -distance at least 4δ , because $256\delta \leq 1/4$. No radius- δ ball, even about an arbitrary legal instrument, can contain two such points. If the required angular separation exceeds π , the trivial lower bound one suffices, with a smaller absolute constant. This proves the first inequality in (M.4).

For the upper bound use the two rational semicircle charts

$$(M.10) \quad z_s(t) = s \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right), \quad s \in \{-1, 1\}, \quad -1 \leq t \leq 1.$$

Each chart has angular speed $2/(1 + t^2) \leq 2$. Round t to the nearest j/B , $-B \leq j \leq B$, with a fixed rule at ties. The circular error is at most $1/B$. Thus any integer $B \geq \max\{1, K_{N,\lambda}/\delta\}$ yields a cover with at most $4B + 2$ centres. For rational inputs, an entirely rational choice is

$$(M.11) \quad B = \max \left\{ 1, \min \left\{ \left\lceil \frac{4N\lambda}{\delta} \right\rceil, \left\lceil \sqrt{\frac{16N\lambda^2}{\eta\delta^2}} \right\rceil \right\} \right\},$$

omitting the second entry when $\eta = 0$. Integer squaring and rational comparison compute the ceiling square root exactly. Both entries are ceilings of the corresponding real bounds, so their minimum is $\lceil 4K_{N,\lambda}/\delta \rceil$. For $z = (x, y)$ on the rational unit circle choose $s = 1$ when $x \geq 0$ and $s = -1$ otherwise, and set $t = sy/(1 + sx)$. Its denominator is at least one, so it is rational and belongs to $[-1, 1]$.

The chart digit (s, j) determines a legal effect pair with eigenvalues $(1 \pm \lambda)/2$. No approximate positivity test, irrational normalization or search over real angles is needed. A fixed-length index over $4B + 2$ slots includes the chart sign. Its length gives (M.5). Redundant chart endpoints cause at most a constant overhead. At $\lambda = 0$ dispense with the chart and emit the single fair, input-erasing measurement. $\square$

Corollary M.5 (A continuous family of horizon exponents). *Fix $0 < \delta \leq 2^{-10}$. If $\lambda_N \rightarrow 1$ and $1 - \lambda_N^2 = N^{-\alpha+o(1)}$ for $\alpha \geq 0$, then*

$$\frac{\log_2 \mathcal{C}_{N,\lambda_N}(\delta)}{\log_2 N} \longrightarrow \frac{1 + \min\{\alpha, 1\}}{2}.$$

If $\lambda_N = N^{-\beta+o(1)} \rightarrow 0$, with $\beta \geq 0$, the limit is $\max\{1/2 - \beta, 0\}$. The projective and erased endpoints have coefficients one and zero, respectively.

Proof. Substitute the two regimes in (M.2) and use (M.4). The term $1 + K/\delta$ handles a vanishing or subpolynomial scale without taking the logarithm of zero. $\square$

Corollary M.6 (Noisy readout with conditional quantum states). *Fix n and ranks $1 \leq r_+, r_- \leq n$. Replace the scalar output in (M.1) by*

$$\mathcal{F}_{\theta, \lambda, \rho}^y(X) = \text{tr}(E_{\theta, \lambda}^y X) \rho_y, \quad \text{tr } \rho_y = 1, \quad \text{rank } \rho_y \leq r_y,$$

retaining the outcome y . Put $V = \sum_{y \in \{+, -\}} [r_y(2n - r_y) - 1]$. For some fixed $\delta_0 > 0$, the optimum covering length is

$$(M.12) \quad \log_2 \left(1 + \frac{K_{N, \lambda}}{\delta} \right) + \frac{V}{2} \log_2 N + V \log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The constants are uniform in $N \geq 1$ and $\lambda \in [0, 1]$, at fixed n, r_+, r_- . For rational visibility, upper centres can be rational and never increase either conditional state rank; lower centres may be arbitrary legal memoryless instruments of the same interface.

Proof. Discarding the quantum output leaves $\mathcal{M}_{\theta, \lambda}$, so the readout lower bound is independent of the conditional states. Conversely, using $I/2$ independently at each call produces exactly $\Omega_\rho^{\otimes N}$, where $\Omega_\rho = \frac{1}{2}\rho_+ \oplus \frac{1}{2}\rho_-$, independently of both θ and λ . Choose compact smooth patches of the two state rank strata. Their product has dimension V and is locally bi-Lipschitz in the Hilbert–Schmidt metric on Ω_ρ . The local binary-test proof of Corollary E.2, applied to these states, supplies a packing of size $c(\sqrt{N}/\delta)^V$ at output distance greater than 2δ for all sufficiently small fixed δ_0 . Cross it with the angular packing used above. Two differing angles are separated by the readout witness regardless of their states; pairs with the same angle are separated by the maximally mixed-input witness. Triangle inequality therefore proves the product lower bound for arbitrary centres.

For the upper cover first encode the angle to error at most $\delta/2$. For fixed conditional states their preparation is a common operation following the readout, so it cannot enlarge this error, including within an adaptive tester. Encode each trace-one state separately by Theorem F.2, with preparation N -copy error at most $\delta/4$. At each call a common processor can consume one copy of each row state and select the state indexed by the measurement outcome. Providing all $2N$ row programmes in advance controls every tester. Triangle inequality on the two tensor products bounds their contribution by $\delta/2$. The product of the two rank-aware factor codebooks has size $C(\sqrt{N}/\delta)^V$. Rational state factors and the rational angle charts give rational Choi blocks $(E_{\theta, \lambda}^y)^\top \otimes \rho_y$ when λ is rational. Each encoded state is trace one and rank nonincreasing, which proves the asserted legality and payload bound. $\square$

Remark M.7 (The retained flag and the scope of the transition). For $n = 1$, forgetting y yields the trace map for every θ and λ ; its covering number is one. For conditional states, forgetting the flag likewise erases the readout whenever $\rho_+ = \rho_-$. Thus the observable interface is essential. At $0 < \lambda < 1$ the measurement Choi blocks have full rank, whereas they have rank one at $\lambda = 1$. Their approach to that boundary is controlled explicitly by $N(1 - \lambda^2)$; a fixed-interior theorem alone would leave its constants undetermined. The transition is not a general classification of all coupled Choi tangent directions. Nor does a pairwise GHZ witness supply a single globally successful estimator or extend this coherent covering theorem to all fixed radii below two. Those are different statements from (M.4).

Remark M.8 (Rational-circle convention). The rational upper-centre notation in this section specifies rational circle coordinates. Decoded effects are fully rational when the public visibility

is rational; for irrational visibility no such matrix rationality follows. Section N gives a different joint code for rational Cartesian targets, in which visibility is encoded and every decoded matrix is rational.

APPENDIX N. COUPLED GEOMETRY OF THE BINARY MEASUREMENT BALL

We now encode the visibility as well as the direction. More generally, we give a local geometric criterion for arbitrary regular subfamilies of the unbiased binary measurement ball. The parameter space in this criterion is the identifiable image of the family, not a redundant choice of angular coordinates.

Let $\mathbb{B} = \{x \in \mathbb{R}^3 : |x| \leq 1\}$. With ordered outcomes put

$$(N.1) \quad E_x^\pm = \frac{1}{2}(I \pm x \cdot \sigma), \quad \mathcal{M}_x^\pm(X) = \text{tr}(E_x^\pm X), \quad x \in \mathbb{B}.$$

Here $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ is the Pauli triple. We use d_N with its unhalved trace-norm convention and all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping. The channel outputs the ordered classical outcome and no residual quantum system. Define

$$(N.2) \quad a_N(r) = \sqrt{\frac{N}{1 - r^2 + N^{-1}}}, \quad w_N(x) = a_N(|x|), \quad N \geq 1.$$

The cutoff N^{-1} is part of a finite-use estimate. It is not a positive-noise assumption on the target.

Theorem N.1 (Finite-use metric on the measurement ball). *For $x, y \in \mathbb{B}$, let $s = \min(|x|, |y|)$ and $D_N(x, y) = a_N(s)|x - y|$. Then*

$$(N.3) \quad \frac{1}{256} \min\{1, D_N(x, y)\} \leq d_N(\mathcal{M}_x, \mathcal{M}_y) \leq \min\{2, 6D_N(x, y)\}.$$

In particular $d_1(\mathcal{M}_x, \mathcal{M}_y) = |x - y|$. The constants in (N.3) are independent of the visibility, direction, horizon and pair separation. The lower bound is attained up to these constants by pair-dependent nonadaptive tests; it does not assert the existence of one estimator for the whole ball. The comparison constants are not optimized.

This supplies a finite-distance local comparison even at erasure and at projective effects. The comparison is not obtained by integrating a Fisher metric through a singular boundary. We first prove its radial part, for which there is an exact classical reduction.

Lemma N.2 (Binomial distance with a support cutoff). *For $0 \leq s \leq r \leq 1$, a unit u , and $P_t = \text{Bern}((1 - t)/2)$,*

$$(N.4) \quad d_N(\mathcal{M}_{ru}, \mathcal{M}_{su}) = \|P_r^{\otimes N} - P_s^{\otimes N}\|_1.$$

Writing $R = a_N(s)(r - s)$, this distance lies between $\frac{1}{64} \min\{1, R\}$ and $\min\{2, 3R\}$.

Proof. First perform the projective measurement along u and then flip its output with probability $(1 - t)/2$. For any fixed adaptive tester, all its outputs are a common channel of N independent flip bits. Supply all bits in advance and discard any unused suffix after public stopping. This proves the upper inequality in (N.4). Repeated input of the $+u$ eigenstate gives exactly these bits and proves equality.

Put $p = (1 - r)/2$, $q = (1 - s)/2$, and $\Delta = q - p$. If $q = 0$, the ordering forces $s = r = 1$, so the two projective measurements coincide. More generally, the case $\Delta = 0$ is immediate. Otherwise $0 < q \leq 1/2$. Let

$$k = \min\{N, \lfloor (2q)^{-1} \rfloor\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, q^{-1}\}$ and $m \geq N/(2k)$. In a block of k trials the no-success event has probabilities $e_p = (1-p)^k$ and $e_q = (1-q)^k$. Their difference obeys

$$e_p - e_q \geq k\Delta(1-q)^{k-1} \geq k\Delta/2.$$

The second inequality follows from $(k-1)q \leq 1/2$ and Bernoulli's inequality. A common randomized processing maps a Bernoulli parameter e to $b + ce$, where

$$c = \frac{1}{\max\{e_p + e_q, 2 - e_p - e_q\}} \geq \frac{1}{2}, \quad b = \frac{1}{2}(1 - c(e_p + e_q)).$$

Indeed $b \geq 0$, $b+c \leq 1$, and the processed pair has parameters $(1 \pm c(e_p - e_q))/2$. Apply Lemma M.3 to the m independent processed blocks. Contraction under processing gives

$$\|P_r^{\otimes N} - P_s^{\otimes N}\|_1 \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, q^{-1})}\}.$$

Since $1 - s^2 = 2q(1+s)$, we have $a_N(s) \leq \sqrt{N \min(N, q^{-1})}$. Also $\Delta = (r-s)/2$. The asserted lower constant $1/64$ follows.

For the upper bound a one-bit coupling gives $N(r-s)$. Let $F = \sqrt{pq} + \sqrt{(1-p)(1-q)}$. Directly,

$$1 - F = \frac{1}{2}[(\sqrt{p} - \sqrt{q})^2 + (\sqrt{1-p} - \sqrt{1-q})^2] \leq \frac{\Delta^2}{2q(1-q)} = \frac{(r-s)^2}{2(1-s^2)}.$$

The no-success statistic above is a lower-bound proof device; its optimality is not asserted. The product-fidelity inequality gives the second bound $2\sqrt{N}(r-s)/\sqrt{1-s^2}$ when $s < 1$. For $z = N(1-s^2) \geq 0$,

$$\min\{1, 2/\sqrt{z}\}\sqrt{z+1} \leq \sqrt{5} < 3.$$

Combining the two upper estimates proves the claim, without division by zero at the identical endpoint $r = s = 1$. $\square$

Proof of Theorem N.1. Assume $x = ru$, $y = sv$, $r \geq s > 0$, and write $h \in [0, \pi]$ for the angle between u, v . Any two directions lie on a great circle; common unitary conjugation of all input systems preserves d_N . Thus the fixed-visibility theorem M.1 applies with $K = K_{N,s}$. Notice

$$(N.5) \quad sa_N(s) \leq K \leq \sqrt{2} sa_N(s).$$

Let $d = d_N(\mathcal{M}_{ru}, \mathcal{M}_{sv})$ and $d_r = d_N(\mathcal{M}_{ru}, \mathcal{M}_{su})$. Inputting the $+u$ eigenstate gives Bernoulli parameters $p = (1-r)/2$ and $q' = (1-s \cos h)/2 \geq q = (1-s)/2$. For fixed p , the total variation of binomial products increases with the larger parameter. To see this without a differentiability assumption, the binomial likelihood ratio is monotone in the success count; its optimal event is a lower count interval. Each such event has probability decreasing in the other parameter. Endpoint cases follow by continuity. Consequently (N.4) gives $d_r \leq d$. The triangle inequality now yields

$$d_N(\mathcal{M}_{su}, \mathcal{M}_{sv}) \leq d_r + d \leq 2d.$$

Set $A = a_N(s)(r-s)$ and $B = sa_N(s)h$. The radial lemma and the fixed-visibility lower bound imply

$$d \geq \frac{1}{64} \min\{1, A\}, \quad d \geq \frac{1}{128} \min\{1, B\}.$$

Hence $d \geq \frac{1}{256} \min\{1, A+B\}$. Since $|ru - sv| \leq (r-s) + sh$, this proves the required lower bound. For the upper bound the radial and angular estimates give $d \leq 3A + \sqrt{2}B$. Moreover $r-s \leq |ru - sv|$ and $sh \leq (\pi/2)|ru - sv|$, by $|ru - sv|^2 = (r-s)^2 + 4rs \sin^2(h/2)$. Thus $d \leq (3 + \pi/\sqrt{2})D_N < 6D_N$. When $s = 0$, the radial lemma proves both bounds, independently of the undefined direction at the origin. The one-use identity follows, as in Lemma M.2, from the eigenvalues $\pm|x-y|/2$ of $E_x^+ - E_y^+$ and reference trace-norm duality. $\square$

N.1. A local volume criterion and boundary contact. For compact $X \subset \mathbb{B}$ let $\mathcal{C}_N(X, \delta)$ allow arbitrary legal memoryless centres of the binary qubit interface, and let $\mathcal{C}_N^X(X, \delta)$ require centres in $\{\mathcal{M}_x : x \in X\}$. The measure in the next theorem is ordinary Euclidean local geometry on the identifiable set X .

Theorem N.3 (Weighted local covering criterion). *Suppose a finite Borel measure μ on a nonempty compact $X \subset \mathbb{B}$ satisfies, for some $k > 0$, $r_0 > 0$ and $c_0, C_0 > 0$,*

$$(N.6) \quad c_0 t^k \leq \mu(X \cap B(x, t)) \leq C_0 t^k \quad (x \in X, 0 < t \leq r_0).$$

There are $c_X, C_X, \delta_X > 0$, depending only on these regularity constants, such that, for every $N \geq 1$ and $0 < \delta \leq \delta_X$,

$$(N.7) \quad c_X \delta^{-k} \int_X w_N(x)^k d\mu(x) \leq \mathcal{C}_N(X, \delta) \leq \mathcal{C}_N^X(X, \delta) \leq C_X \delta^{-k} \int_X w_N(x)^k d\mu(x).$$

No map retracting arbitrary centres onto X is assumed. In a parametrized family the set X is its image under (N.1); indistinguishable parameter values are not charged twice.

Proof. Put $e_N(x) = 1 - |x|^2 + N^{-1}$, so $e_N \geq N^{-1}$ and $|e_N(x) - e_N(y)| \leq 2|x - y|$. If $|x - y| \leq t/w_N(x)$ and $t \leq 1/4$, then

$$|e_N(x) - e_N(y)| \leq 2t\sqrt{e_N(x)/N} \leq 2te_N(x).$$

Thus the weights are comparable on these small Euclidean balls. Conversely, if $d_N(\mathcal{M}_x, \mathcal{M}_y) \leq t < 1/256$, Theorem N.1 implies $|x - y| \leq 256t/a_N(s)$, $s = \min(|x|, |y|)$. Applying the same estimate first at the point of smaller norm shows comparability to $w_N(x)$ when, for example, $t \leq 2^{-14}$. Consequently absolute $c, C > 0$ satisfy the two local inclusions

$$X \cap B(x, ct/w_N(x)) \subseteq B_{d_N}(x, t) \cap X \subseteq X \cap B(x, Ct/w_N(x)).$$

Reduce t by a fixed multiple of r_0 if necessary. Since $w_N(x) \geq 1/\sqrt{2}$, all Euclidean radii used here lie below r_0 . The measure ν_N is auxiliary geometric volume, rather than a statistical prior or a hidden-register entropy. The weighted measure $d\nu_N = w_N^k d\mu$ of either operational ball is bounded above and below by fixed multiples of t^k , by (N.6) and weight comparability.

A maximal δ -separated set in the compact metric space (X, d_N) covers X by radius- δ balls. Its radius- $\delta/3$ balls are disjoint, proving the upper bound in (N.7). A maximal 3δ -separated set covers X by radius- 3δ balls, so it has at least $c\nu_N(X)\delta^{-k}$ points. Any radius- δ ball about any legal centre contains at most one of these points, by the triangle inequality. This gives the lower bound for unrestricted centres. Compactness and the one-use identity ensure that these are genuine metrics on X ; the finite hybrid bound gives the required continuity. $\square$

Corollary N.4 (Boundary-contact law). *In Theorem N.3, put $F(t) = \mu\{x \in X : 1 - |x|^2 \leq t\}$. Suppose $F(0) = 0$ and $ct^\alpha \leq F(t) \leq Ct^\alpha$ for $0 < t \leq t_0$, with $\alpha > 0$. Uniformly on the same small-error range,*

$$(N.8) \quad \mathcal{C}_N(X, \delta) \asymp_X \delta^{-k} \begin{cases} N^{k/2}, & \alpha > k/2, \\ N^{k/2} \log(N+2), & \alpha = k/2, \\ N^{k-\alpha}, & 0 < \alpha < k/2. \end{cases}$$

Here $F(0) = \mu(X \cap S^2)$ is boundary mass of the target measure, not ambient surface measure. If instead $F(0) > 0$, the law is $\asymp_X \delta^{-k} N^k$.

Proof. Write $b = k/2$ and $\tau = N^{-1}$. Stieltjes integration by parts gives

$$\int_{[0, t_0]} (t + \tau)^{-b} dF(t) = (t_0 + \tau)^{-b} F(t_0) + b \int_0^{t_0} \frac{F(t)}{(t + \tau)^{b+1}} dt$$

when $F(0) = 0$. Substitute the two bounds on F and split the last integral at τ . It is bounded, logarithmic, or comparable to $\tau^{\alpha-b}$ according as $\alpha > b$, $\alpha = b$, or $\alpha < b$. The part beyond t_0 is bounded, and multiplication by N^b proves (N.8). Bounded N is absorbed into the constants. An atom at depth zero contributes exactly $N^k F(0)$ to the weighted volume; $w_N \leq N$ supplies the matching upper bound. $\square$

For example, in a regular k -dimensional chart with a boundary subset of codimension ℓ , a two-sided relation between the support deficit $1 - |x|^2$ and the m th power of transverse distance yields $\alpha = \ell/m$, provided the transverse volume has order t^ℓ . This is a geometric condition to check, not an automatic regularity assertion. It distinguishes ordinary interior accumulation, critical logarithmic enhancement and boundary-dominated coherent accumulation.

Corollary N.5 (Joint visibility and direction). *Let $\mathbb{B}_d$ be the closed unit ball in a fixed d -dimensional linear subspace of $\mathbb{R}^3$, $d = 1, 2, 3$. For an absolute small-error range,*

$$(N.9) \quad C_N(\mathbb{B}_d, \delta) \asymp \begin{cases} \sqrt{N} \delta^{-1}, & d = 1, \\ N \log(N+2) \delta^{-2}, & d = 2, \\ N^2 \delta^{-3}, & d = 3. \end{cases}$$

For the coordinate disk and full ball, rational target data admit exact rational family-centred codes attaining these orders. In particular, the optimal fixed-length payloads for $d = 2, 3$ are respectively

$$\log_2 N + \log_2 \log(N+2) + 2 \log_2(1/\delta) + O(1), \quad 2 \log_2 N + 3 \log_2(1/\delta) + O(1).$$

Visibility is a charged target coordinate in these statements.

Proof. Lebesgue measure on $\mathbb{B}_d$ satisfies (N.6) with $k = d$, and its boundary-depth distribution has exponent $\alpha = 1$. For the disk $(k, \alpha) = (2, 1)$ is critical: its weighted volume contains $N \int_0^1 (t + N^{-1})^{-1} dt \asymp N \log(N+2)$. For the ball $(k, \alpha) = (3, 1)$, the boundary layer instead dominates and gives N^2 . These factors are geometric and are unchanged in order by replacing the Ahlfors measure by an equivalent one, with constants depending on the regularity data. Apply Corollary N.4. The exact rational upper construction follows from Theorem N.6 below. In a coordinate line the direct scalar grid in the same construction gives the stated one-dimensional order. $\square$

N.2. Exact joint coding without irrational normalization. We give a rational construction for $d = 2, 3$ that takes the Cartesian Bloch vector as input. Its norm need not be rational. Fix public $N \geq 1$ and rational $0 < \delta \leq 1/4$. Set

$$(N.10) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad t_i = i/B, \quad r_i = \frac{2t_i}{1+t_i^2}, \quad A_i = \max \left\{ 1, \left\lceil \sqrt{64(d-1)K_{N,r_i}^2/\delta^2} \right\rceil \right\} \quad (i > 0).$$

The radial layer $i = 0$ has one codeword. At $i > 0$ use $2d$ signed stereographic charts and the grid $\{-A_i, \dots, A_i\}^{d-1}/A_i$. For chart axis j and sign s , the map is

$$(N.11) \quad u_j = s \frac{1 - |z|^2}{1 + |z|^2}, \quad u_\ell = \frac{2z_\ell}{1 + |z|^2} \quad (\ell \neq j).$$

Every decoded centre is $r_i u \in \mathbb{B}_d$ with rational entries. The exact capacity, including possibly redundant chart words, is

$$(N.12) \quad L_{N,\delta,d} = 1 + \sum_{i=1}^B 2d(2A_i + 1)^{d-1}.$$

A single fixed-length integer index in this union of layers charges the visibility digit as well as the direction digits.

Theorem N.6 (Rational joint codec). *For rational $x \in \mathbb{B}_d$, $d = 2, 3$, an exact rational algorithm returns an index in (N.12). Its decoded measurement is within δ in d_N , and every index decodes to a legal rational measurement. There are absolute constants such that*

$$L_{N,\delta,2} \leq CN \log(N+2)\delta^{-2}, \quad L_{N,\delta,3} \leq CN^2\delta^{-3}.$$

The stored payload is $\lceil \log_2 L_{N,\delta,d} \rceil$ bits. The JSON envelope, encoder workspace, public d, N, δ and expanded matrices are not this payload. No input visibility is public, and no floating-point operation decides a chart or grid digit.

Proof. Write $r = |x|$, $\beta = \arcsin r$, $t = \tan(\beta/2) \in [0, 1]$. Choose a nearest i/B to t , resolving ties towards zero. Then $|\beta - 2\arctan(i/B)| \leq 1/B$. The radial flip programmes have root fidelity $\cos((\beta - \beta_i)/2)$, and hence their N -copy distance is at most $\sqrt{N}|\beta - \beta_i| \leq \delta/8$. When $i = 0$, decode the unique erased measurement and stop. Otherwise choose a maximal-absolute-coordinate axis of x , with its sign. The exact chart parameters of x/r are

$$z_\ell = \frac{x_\ell}{r + |x_j|}, \quad \ell \neq j.$$

They lie in $[-1, 1]$. Round each to a nearest grid point at denominator A_i . The differential norm of (N.11), measured in spherical length, is $2/(1 + |z|^2) \leq 2$. Thus the angular error is at most $\sqrt{d-1}/A_i$ and its operational cost is at most $\delta/8$, by the fixed-visibility upper bound and (N.10). The triangle inequality leaves at least three quarters of the requested budget unused.

All comparisons above can be made in the rationals. If $Q = \sum x_j^2$, comparison of t with a rational $v \in [0, 1]$ is comparison of Q with $[2v/(1+v^2)]^2$. For a direction digit compare $b/(\sqrt{Q} + a)$ with rational $v \geq 0$, where $a = |x_j|$ and $b = |x_\ell|$. First test the sign of $b - va$. If it is negative, the comparison is negative; otherwise compare $(b - va)^2$ and v^2Q . The sign test is essential before squaring. Binary integer searches therefore determine every nearest grid digit, including ties, without computing an irrational normalization. Integer ceiling square roots suffice for both grids.

It remains to bound the union capacity. Here $B \asymp \sqrt{N}/\delta$. For $i \leq B/2$, $K_{N,r_i} \leq C\sqrt{N}$, which contributes at most $CN^{d/2}\delta^{-d}$. For $i > B/2$, put $j = B - i$. Since $1 - r_i^2 = (1 - t_i^2)/(1 + t_i^2)^2$, we have

$$K_{N,r_i} \leq C \min\{N, \sqrt{N}B/(j+1)\},$$

with an adjusted absolute constant also at $j = 0$. Split the sum at $j+1 = B/\sqrt{N} \asymp \delta^{-1}$. For $d = 2$, the remaining sum is harmonic and contributes at most $CN\delta^{-2} \log(N+2)$. For $d = 3$, the sum of $(j+1)^{-2}$ contributes at most $CN^2\delta^{-3}$. The near-boundary layers have these same or smaller orders. The added constants in A_i and the single erased word are absorbed into these bounds. This proves the capacity and payload assertions. Counting layer sizes and ranking or unranking the integer grid are finite exact algorithms, with cost polynomial in the numerical radial grid and operand lengths; no polynomial bound in the encoded length of N or $1/\delta$ is asserted. $\square$

Remark N.7 (Priority and interface). Sedláč-Ziman [17, Section VI, Example 4, equation (37)] already treat distinct white-noise levels in single-shot qubit measurement discrimination. Puchała-Pawela-Krawiec-Kukulski [18, Theorem 1] identify the single-shot diamond distance of projective measurements with an optimized unitary-channel distance. Neither result is claimed anew here. Our conclusion is the uniform finite-use metric comparison (N.3), its weighted covering criterion, and the resulting contact and exact rational joint-description laws. The classical programme and entangled-block mechanisms retain their earlier attribution. The geometric criterion covers regular identifiable subsets of (N.1), not arbitrary biased POVMs or general disturbing instruments. The supplied vector is encoded; no unknown measurement is learned. Pairwise testing, a common estimator, a global environment seizer and a physical classical simulation of unknown quantum inputs are different tasks.

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